



A review of the application of deep eutectic solvents for metal recovery from diverse secondary sources

Fereshteh Moradi, Francis Bougie *

Department of Chemical Engineering, Laval University, 2325 University Street, G1V 0A6 Quebec, Canada

ARTICLE INFO

Keywords:

Deep eutectic solvents
Solvometallurgy
Leaching
Metal-associated wastes
Liquid–liquid extraction
Green chemistry
Circular economy

ABSTRACT

In recent years, the growing demand for critical, strategic, and precious metals across various industries has led to a worldwide shortage. The increasing disposal of metal-containing waste poses significant environmental and health risks while also accelerating the depletion of natural resources. Given the supply risks, environmental hazards, and challenges associated with extracting these metals from primary ores, it is vital to develop sustainable methods for recovering them from secondary sources. Metal recovery has garnered research interest due to its essential role in promoting a circular economy. Metallurgical processes, such as pyrometallurgy, bio-leaching, and hydrometallurgy, often lead to secondary pollution and economic challenges. Methods based on solvometallurgy and green chemistry, such as the use of deep eutectic solvents (DESs), offer potential for separating metals from waste. DESs provide a sustainable approach for selective recovery, potentially replacing mineral acids while minimizing wastewater production. This paper provides a literature survey on the extraction of metals from sources including electrical and industrial waste, minerals, biological materials, and environmental samples using DESs. It discusses concepts, specifications, and characterization methods related to DESs.

Additionally, the article classifies the techniques used for metal separation and recovery with DESs, along with key parameters that influence efficiency. The review emphasizes the selective and high-efficiency recovery of various metals using DESs. It highlights the necessity of designing DESs with low viscosity, strong coordination, and high reducibility for optimal separation. We hope that the challenges and opportunities for advancing metal recovery through DESs are clearly outlined, providing a roadmap for new applications of DES technology.

1. Introduction

The rapid growth of modern technologies has intensified global demand for critical metals, raising concerns about resource depletion, supply chain vulnerability, and environmental sustainability [1–4]. Primary mining alone cannot meet this demand, and several essential metals are projected to face depletion within the next few decades. Simultaneously, substantial quantities of valuable metals are discarded in industrial residues and end-of-life (EoL) products, including batteries, magnets, catalysts, and electronic devices. To address these challenges, recovering secondary resources is vital to reducing dependence on primary ores, ensuring sustainable metal supplies, and mitigating environmental impacts [2,5,6]. Within this framework, the circular economy and urban mining have emerged as key strategies for addressing these challenges. By reusing and recycling discarded products, they not only lower CO₂ emissions and environmental risks but also provide higher metal concentrations than many natural deposits. For instance, the

gallium content of red LEDs is significantly greater than that found in typical ores [7–9]. Such approaches highlight the importance of developing efficient and sustainable processes for extracting metals from diverse secondary sources [6].

Various processes have been developed to recycle metals from waste streams, typically starting with physical pre-treatments such as manual disassembly and sorting into metallic and non-metallic fractions. The metallic fraction is then processed through pyrometallurgy, hydrometallurgy, or biometallurgy [5,10]. Pyrometallurgy operates at high temperatures (>1200 K) to produce high-purity alloys, but it consumes significant energy, emits toxic gases, and lacks selectivity [11,12]. Biometallurgy uses acidophilic sulfur- and iron-oxidizing bacteria to solubilize metals such as gallium, copper, zinc, and nickel. It offers low cost and reduced energy demand but suffers from slow kinetics [5,13]. Hydrometallurgy presents a more energy-efficient processing method, primarily due to its low-temperature operations [14]. However, recovery times are often longer, especially during the separation and purification of metals [15]. This technique comprises two critical steps,

* Corresponding author.

E-mail address: francis.bougie@gch.ulaval.ca (F. Bougie).

<https://doi.org/10.1016/j.molliq.2025.128903>

Received 29 July 2025; Received in revised form 12 October 2025; Accepted 6 November 2025

Available online 17 November 2025

0167-7322/© 2025 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

List of Key Acronyms

AALLME	Air-Assisted Liquid-Liquid Microextraction	MCA	Monochloroacetic Acid
AGU	1-Aminoguanidine Hydrochloride	MSDES	Metal Salts-Based Deep Eutectic Solvent
BEA	Bis(2-ethylhexyl) Amine	MTPB	Methyltriphenylphosphonium Bromide
BtCl	Betaine Hydrochloride	NADES	Natural Deep Eutectic Solvent
BTPC	Benzyltriphenylphosphonium Chloride	NMC	$\text{LiNi}_{1/3}\text{Mn}_{1/3}\text{Co}_{1/3}\text{O}_2$
CAM	Cathode Active Material	NRTL	Non-Random Two-Liquid
ChCl	Choline Chloride	OC	Octanoic Acid
COSMO-RS	Conductor-like Screening Model for Real Solvent	PA	Propionic Acid
CRM	Critical Raw Material	PCB	Printed Circuit Board
D ₂ EHPA	Di(2-ethylhexyl) Phosphoric Acid	PEG	Polyethylene Glycol
DA	Decan-1-ol	PGM	Platinum Group Metal
DAC	N,N-diethylethanolammonium Chloride	PTSA	ρ -Toluenesulfonic Acid
DAG	1,3-Diaminoguanidine Hydrochloride	PV	Photovoltaic
DES	Deep Eutectic Solvent	QSPR	Quantitative Structure-Property Relationship
DFT	Density Functional Theory	REE	Rare Earth Element
DLLME	Dispersive Liquid-Liquid Microextraction	REO	Rare Earth Oxide
DLLME-SF	DLLME Solidifying Floating	RLLE	Reactive Liquid-Liquid Extraction
DSC	Differential Scanning Calorimetry	SALLME	Shaker-Assisted Liquid-Liquid Microextraction
EADLLME	Effervescence-Assisted DLLME	SLE	Solid-Liquid Equilibrium
EG	Ethylene Glycol	SME	Solvent Microextraction
ELME	Emulsification Liquid-Liquid Microextraction	TAC	Trioctylmethylammonium Chloride
EoL	End-of-Life	TADLLME	Temperature-Assisted DLLME
E-waste	Electronic Waste	TBAB	Tetrabutylammonium Bromide
FL	Fluorescent Lamp	TBAC	Tetrabutylammonium Chloride
FTIR	Fourier Transform Infrared Spectroscopy	TCA	Trichloroacetic Acid
GUC	Guanidine Hydrochloride	TEA	Techno-Economic Assessment
HBA	Hydrogen Bond Acceptor	TfOH	Trifluoromethanesulfonic Acid
HBD	Hydrogen Bond Donor	Tg	Glass Transition Temperature
HBTa	3-Benzoyl-1,1,1-Trifluoroacetone	TGA	Thermogravimetric Analysis
HDEHP	Bis(2-ethylhexyl) Phosphoric Acid	THEDES	Therapeutic Deep Eutectic Solvent
HDES	Hydrophobic Deep Eutectic Solvent	Tm	Melting Temperature
HNMR	Proton Nuclear Magnetic Resonance	TOAB	Tetraoctylammonium Bromide
HREE	Heavy REE	TOAC	Trioctylammonium Chloride
HTTA	Thenoyltrifluoroacetone	TOMAC	Trialkylmethylammonium Chloride
IL	Ionic Liquid	TOPO	Trioctylphosphine Oxide
LA	Lauryl Alcohol	TPAB	Tetrapentylammonium Bromide
LAB	Lead Acid Battery	TPP	Triphenyl Phosphate
LCO	LiCoO_2	UADLLME	Ultrasonic-Assisted DLLME
LIB	Lithium-Ion Battery	UNIFAC	Universal Quasichemical Functional Group Activity Coefficient
LLME	Liquid-Liquid Microextraction	UV-vis	Ultraviolet-Visible
LMO	LiMn_2O_4	VALLME	Vortex-Assisted Liquid-Liquid Microextraction
LNCO	$\text{LiNi}_{0.8}\text{Co}_{0.2}\text{O}_2$	VFT	Vogel-Fullcher-Tamman
LNMO	$\text{LiNi}_{0.5}\text{Mn}_{1.5}\text{O}_4$	VOC	Volatile Organic Compound
LPME	Liquid-Phase Microextraction	[BTBA][Cl]	Benzyltributylammonium Chloride
LREE	Light REE	[N ₈₈₈₁][Cl]	Methyltrioctylammonium Chloride
LTTM	Low Transition Temperature Mixture	[N ₈₈₈][Br]	Tetraoctylammonium Bromide

namely leaching and extraction, where in the first one metal oxides are transformed into soluble ions using aqueous inorganic or organic acids and alkaline solutions [10,16]. Following leaching, the resulting solution undergoes purification techniques, including electrodeposition, ion exchange, adsorption, and solvent extraction [5]. While effective, hydrometallurgy faces significant drawbacks, including acid toxicity, heavy metal pollution, sludge generation, and high water and reagent consumption, which result in substantial wastewater and create both environmental and economic challenges. Nevertheless, it remains widely applied because it offers reliable operation, reasonable process control, and the ability to obtain high-purity metals [11,16,17]. As most current recycling strategies for metal-bearing wastes continue to depend on acid- or alkali-based hydrometallurgical processes, increasing attention is being directed toward safer and more sustainable

approaches that circumvent the inherent drawbacks of aqueous media [18,19]. Consequently, this has given rise to solvometallurgy as a new area of research focused on developing cleaner and more efficient alternatives to conventional leaching and solvent extraction methods. In contrast to hydrometallurgy, solvometallurgy employs non-aqueous solvents, which allows for increased selectivity [18,19]. Traditional organic solvents are volatile, flammable, and toxic, resulting in volatile organic compound (VOC) emissions, handling hazards, and waste-management challenges. Supercritical CO₂ offers tunable solvating ability and easy separation but requires high pressures and specialized equipment, resulting in high energy consumption. Ionic liquids (ILs), despite their negligible vapor pressure, strong solvating ability, and high thermal stability, are limited by high synthesis costs and complexity, poor biodegradability, elevated viscosity, and potential toxicity. These

limitations underscore the urgent need for safer, cost-effective, and environmentally benign alternatives [2].

Deep eutectic solvents (DESs) are a promising option for developing cleaner processes, thanks to their high thermal and chemical stability, simple preparation methods, cost-effectiveness, low vapor pressure, and minimal or negligible toxicity [20]. They remain liquid at room temperature and, because of their unique and tunable properties, can potentially replace water in extractive metallurgy and metal processing [21]. Additionally, leaching processes can be performed without the need for extra reagents, reducing agents, or complicated procedures, making them a more efficient and environmentally friendly option [15]. DESs promote the dissolution of metal oxides through proton attack and demonstrate selective leaching of specific oxides by forming distinct complexes [5]. Furthermore, DESs can be reused multiple times while maintaining their original leaching efficiency [15]. Recently, there has been growing interest in using DESs for metal recovery [5]. The selected approach uses solvometallurgical methods based on DESs, which can rapidly, cost-effectively, and selectively recover multiple metals from EoL products with minimal water consumption and near-zero wastewater generation [19]. Moreover, it streamlines the process by integrating leaching and solvent extraction, resulting in more selective extraction, decreased acid consumption, and fewer purification steps [22,23].

Given the urgent demand for sustainable metal recovery, this review provides a timely and comprehensive synthesis of the latest advances in applying both hydrophilic and hydrophobic DESs for extracting metals from a wide range of secondary resources. A bibliometric analysis of the literature (Fig. 1), based on Web of Science data from 2015 to 2025, highlights the rapid growth of publications and citations in this area, underscoring the importance of an updated and critical review. The left vertical axis (blue bars) represents the annual number of publications. In contrast, the right vertical axis (black line) tracks citation counts, revealing not only an increase in research activity but also a rise in scientific influence. Notably, citation growth has outpaced publication growth in recent years, reflecting the substantial impact of DES research on the field of sustainable metallurgy.

Unlike earlier reviews that concentrated on a narrow subset of waste streams or a single class of solvents, the present work integrates all relevant secondary sources, including electronic waste (lithium-ion batteries (LIBs), magnets, printed circuit boards (PCBs), fluorescent lamps (FLs)), industrial residues, mineral by-products, and biological or environmental matrices, into a unified framework. Table 1 systematically maps prior reviews against these sources and metal-bearing

matrices, revealing both the fragmented coverage of the literature and persistent knowledge gaps, such as those related to photovoltaic modules and spent catalysts. By encompassing all secondary sources in a single study, this review offers a more comprehensive and inclusive perspective than existing works. The table is intended as a scoping overview rather than a detailed comparative analysis; readers are referred to the cited works for in-depth conclusions regarding individual studies.

A central emphasis of this review is establishing mechanistic property–performance relationships, rather than providing only case-by-case descriptions. Specifically, we examine how viscosity (mass-transfer limitations), acidity and proton activity (oxide dissolution), coordination/ligand strength (metal selectivity and complex stability), and hydrophilicity/hydrophobicity (phase behavior and downstream separations) govern the outcomes of leaching and liquid–liquid extraction. Because these physicochemical parameters are tunable through careful selection and preparation of hydrogen bond acceptors (HBAs) and hydrogen bond donors (HBDs), DESs constitute a customizable platform for targeted applications. Accordingly, this paper is designed to serve as a functional reference that can act as a database for identifying the best systems for recovering specific elements by linking solvent design parameters directly to metal-specific recovery performance across the whole processing chain (leaching, extraction, metal recovery, and solvent regeneration).

A further novel contribution is the explicit treatment of hydrophilic–hydrophobic DES pairings as complementary and potentially synergistic systems, for instance, employing hydrophilic DESs for rapid oxide dissolution followed by hydrophobic DESs (HDESs) for selective extraction and minimization of aqueous waste. This DES–DES concept, which has received little attention in prior reviews, offers a practical pathway toward higher efficiency, selectivity, and sustainability across the complete recovery chain. Finally, the review remains entirely up to date (2023–2025) and is written with industrial implementation in mind. This review consolidates fundamental concepts and recent advances, while also critically examining barriers, stability, and redox robustness, as well as viscosity management and handling, impurity tolerance, solvent regeneration, and scale-up. These factors are analyzed in terms of their impact on the feasibility of closed-loop, pilot-ready flowsheets. The paper's structure (Fig. 2) guides the reader from fundamentals to applications and industrialization, while Table 1 provides a clear overview of prior coverage.

2. Methodology of this review

This review was conducted using a structured and systematic approach to ensure transparency, reproducibility, and scientific rigor. Relevant publications were collected from leading databases using combinations of keywords such as “deep eutectic solvents”, “metal recovery”, “waste management”, “resource recovery”, “recycling metals”, “rare earth elements”, “circular economy”, “green solvents” and “hydrophilic and hydrophobic deep eutectic solvents”. The search focused on publications, with particular emphasis on works published between 2023 and 2025, to capture the most recent advances in the field. To maintain scientific quality, only peer-reviewed journal articles, review papers, and book chapters were included, while non-peer-reviewed reports, patents, and unrelated studies were excluded.

The reviewed studies were systematically categorized according to secondary resource types (e-waste, industrial residues, mineral wastes, environmental and biological matrices) and process stages (leaching, extraction, purification, and solvent regeneration). In addition, essential physicochemical parameters, including viscosity, acidity, coordination/ligand strength, and hydrophilicity/hydrophobicity, were extracted to assess performance correlations and trends. Comparative tables were constructed to map the scope of existing reviews, allowing us to highlight underexplored resources such as photovoltaic modules and spent catalysts. This organization not only contextualizes earlier works but

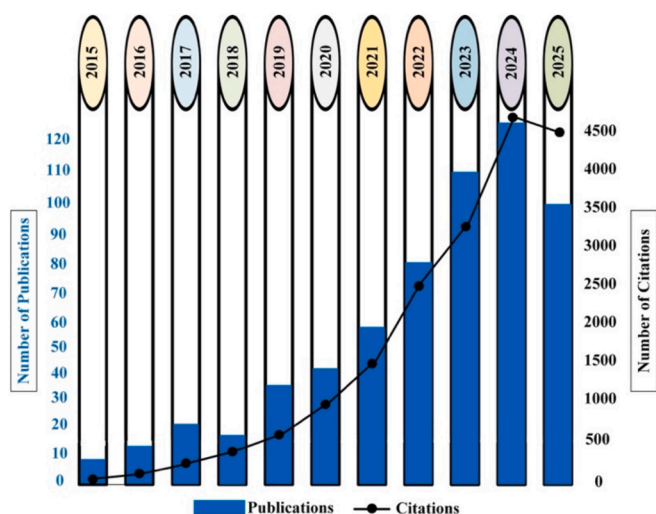


Fig. 1. Publications and citations on metal recovery using DESs over the past decade based on data extracted from the Web of Science database.

Table 1
Review papers on the metal recovery application of DESs from various sources.

Secondary Source																	Refs.
Biological/Organic Materials/Food Product			Environmental Analysis		Industrial				Mineral			E-waste					
Biological	Waste Oil	Food	Soil	Water	Industrial Dust	Spent Catalyst	Waste Water	Fly Ash	REE	Metals Oxide	Ores	Lamp	Magnet	Solar Panel	PCB	LIB	
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x**	✓*	[24]
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	✓	[15]
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	✓	[25]
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	✓	[26]
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	✓	[27]
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	✓	[28]
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	✓	[29]
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	✓	[30]
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	✓	[31]
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	✓	[32]
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	✓	[33]
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	✓	[34]
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	✓	[35]
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	✓	x	[36]
x	x	x	x	x	x	x	x	x	✓	✓	x	x	x	x	x	x	[37]
x	x	x	x	x	x	x	✓	x	x	x	x	x	x	x	x	x	[38]
x	✓	✓	✓	✓	✓	✓	x	x	✓	✓	x	x	x	x	x	✓	[39]
x	x	x	✓	✓	✓	x	x	x	✓	✓	✓	x	✓	x	✓	✓	[40]
x	x	x	x	x	x	✓	x	x	x	✓	x	✓	✓	x	x	✓	[2]
✓	x	✓	x	✓	x	x	x	x	x	x	x	✓	✓	x	x	✓	[3]
x	x	x	x	x	x	x	x	x	x	x	x	✓	✓	✓	✓	✓	[41]
x	x	x	x	x	x	x	x	x	✓	x	x	x	x	x	x	x	[42]
x	x	x	x	x	x	x	x	x	✓	x	✓	x	x	x	x	x	[43]
x	x	x	x	✓	x	x	x	x	x	x	x	x	x	x	x	x	[44]
x	x	x	x	✓	x	x	✓	x	x	x	x	x	x	x	x	x	[45]
x	x	x	x	✓	x	x	x	x	x	x	x	x	x	x	x	x	[46]
x	x	x	x	✓	x	x	x	x	x	x	x	x	x	x	x	x	[47]
x	x	x	x	✓	x	x	x	x	x	x	x	x	x	x	x	x	[48]
✓	x	✓	✓	✓	x	x	x	x	x	x	x	x	x	x	x	x	[49]
x	x	✓	x	x	x	x	x	x	x	x	x	x	x	x	x	x	[50]
x	x	x	✓	x	x	x	x	x	x	x	x	x	x	x	x	x	[51]
✓	✓	✓	x	✓	x	x	x	x	x	x	x	x	x	x	x	x	[52]
✓	✓	✓	x	✓	x	x	x	x	x	x	x	x	x	x	x	x	[53]
x	✓	x	x	✓	x	x	x	x	x	x	x	x	x	x	x	x	[54]
x	x	x	x	✓	✓	x	x	x	✓	x	x	x	x	x	x	✓	[55]
x	x	x	x	✓	✓	x	x	x	✓	✓	✓	x	✓	x	x	✓	[20]
x	✓	✓	✓	✓	✓	x	x	✓	✓	✓	✓	✓	✓	x	✓	✓	[4]
✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	[5]
✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	This Paper

* Investigated

** Not investigated

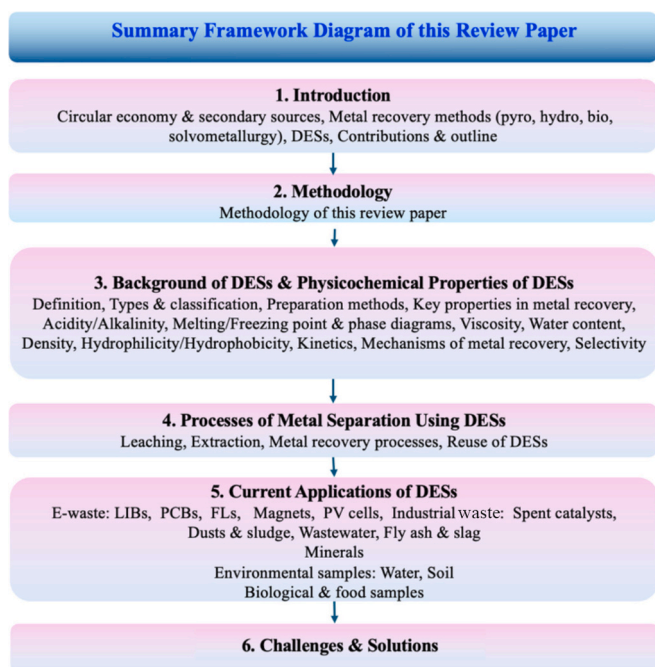


Fig. 2. Organization of this article.

also demonstrates how the present study extends beyond them by integrating all relevant secondary sources into a single framework.

To complement the systematic literature review, a bibliometric analysis was conducted using the Web of Science database, covering the period from 2015 to 2025. Data were processed to remove duplicates, validate citations, and standardize keywords. The cleaned dataset was analyzed with VOSviewer software, which generated a keyword co-occurrence network. As shown in Fig. 3, four dominant thematic clusters emerged: DESs and solvent properties (red), circular economy and waste management (green), recycling and e-waste (blue), and rare earth

and critical metals (yellow). In this visualization, node size reflects keyword frequency, link thickness indicates the strength of co-occurrence, and colors distinguish research domains. The map highlights the interdisciplinary character of DES research and its increasing integration with sustainable metallurgy and circular economy strategies. By combining systematic literature collection with bibliometric mapping, this review ensures both qualitative depth and quantitative insight.

3. Background of DESs

As environmental awareness grows, the development of green solvents has become increasingly important. Since the introduction of green chemistry in 1992, efforts have focused on minimizing the generation of harmful chemical waste. A key advancement has been the use of low-toxicity solvents, such as DES, which support cleaner and more sustainable scientific processes [56]. The word “eutectic” was first introduced in 1884 to describe metal alloys that melt at temperatures lower than those of their individual components [57]. It comes from the Ancient Greek term εὐτηκτος (eûtēktos), meaning “easily melted.” In a phase diagram, the eutectic point represents the precise composition and temperature at which a mixture of two solid substances melts completely at the lowest possible temperature relative to each component [58]. A eutectic mixture is typically a stable, non-reactive, and isothermal combination of different substances that behaves reversibly [59]. When a multi-component system reaches its minimum freezing point upon solidification, it is identified as a eutectic mixture [2]. DESs are formed by mixing two or more compounds, typically an HBD and an HBA. This combination results in a stable liquid eutectic system with a melting point significantly lower than that of the original components, primarily due to the strong hydrogen bonding between them. The presence of non-symmetrical ions with low lattice energy further reduces the melting point, even lower than that of an ideal mixture. This pronounced drop in temperature is why the term “Deep” is used [58,60]. Despite their growing use, the precise definition of DESs remains a subject of debate. Many solid compound mixtures exhibit eutectic behavior or hydrogen bonding, which alone does not qualify them as

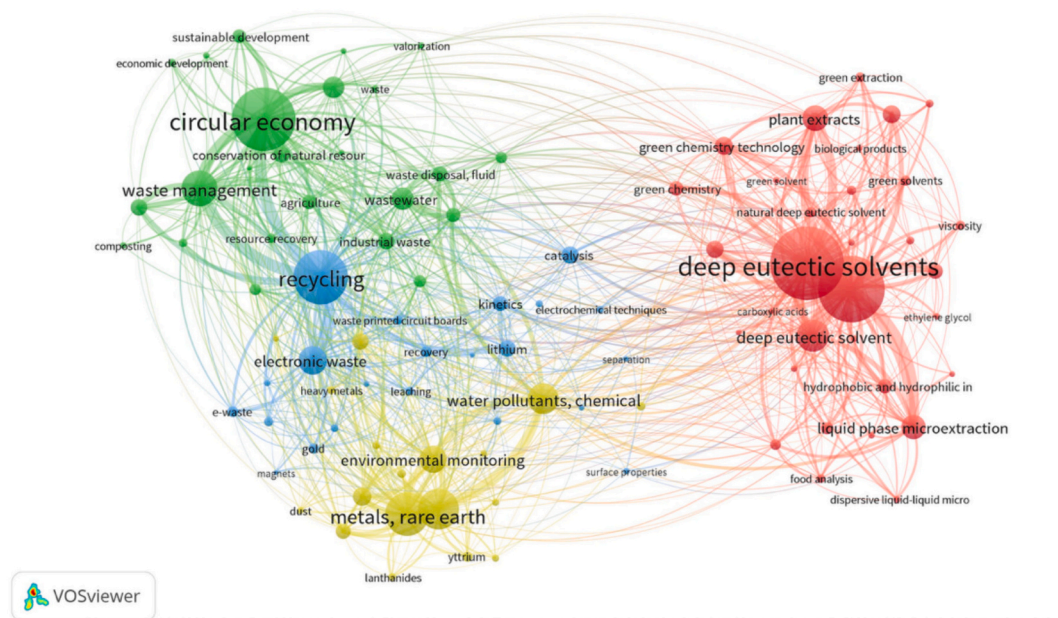


Fig. 3. Keyword co-occurrence network of publications on DESs for metal recovery, based on data extracted from the Web of Science (2015–2025) and analyzed using VOSviewer. Node size represents keyword frequency, link thickness indicates co-occurrence strength, and colors distinguish thematic clusters: red (DESs and solvent properties), green (circular economy and waste management), blue (recycling and e-waste), and yellow (rare earth and critical metals). The network highlights the interdisciplinary nature of DES research and its integration into sustainable metallurgy and circular economy strategies. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

DESS. To distinguish DESs from other systems, they are best described as mixtures of two or more pure substances that display a eutectic point markedly lower than that of an ideal mixture, characterized by a significant negative deviation from ideal behavior ($\Delta T > 0$), where ΔT represents the temperature difference between the expected and actual eutectic point, as illustrated in Fig. 4 [58].

The low melting point of DESs is primarily attributed to the strong hydrogen bonding between their components, particularly between halide salts (HBAs) and HBDs. While weaker forces such as van der Waals and electrostatic interactions may also occur, hydrogen bonds dominate, creating a robust intermolecular network. This bonding leads to charge delocalization, which significantly lowers the melting point, typically below 400 K, compared to the individual components [61,62]. The eutectic temperature acts like a triple point with zero degrees of freedom [59]. DESs that are stable and liquid at room temperature, customizable, highly effective in solubilizing target compounds, have low toxicity, and are composed of inexpensive components are highly desirable for a wide range of applications. However, these properties can be achieved simply by selecting the proper constituent components [56,58]. DESs can be classified either by their overall composition or by the characteristics of their individual components. More specifically, they are divided into five main types based on the complexing agents involved, such as organic salts, non-ionic compounds, metal halides, and their hydrates, which interact to form a eutectic system [62,63], as shown in Fig. 5.

DESs are formed through the combination of Lewis or Brønsted acids and bases. The general formula commonly used to describe DESs is $\text{Cat}^+\text{X}^-z\text{Y}$. In this formula, Cat^+ refers to the cationic component, which typically comes from salts of ammonium, sulfonium, or phosphonium. The symbol X denotes a Lewis base, usually the halide anion derived from the salt. The symbol Y represents a specific Lewis or Brønsted acid. At the same time, z indicates the ratio of Y molecules to the anion X, showing how many Y molecules interact with the anion [64]. DESs are mainly classified according to the type of complexing agent involved. The classification of a base as either Lewis or Brønsted is influenced by the value of z [2]. This paper examines the classification framework and various types of DESs, with a particular emphasis on two key categories: hydrophilic and hydrophobic DESs. It also outlines the commonly used HBAs and HBDs associated with each type, as shown in Fig. 5.

The first type of DES is formed by combining organic salts with metal chlorides, such as CuCl_2 , FeCl_2 , SnCl_2 , and ZnCl_2 . Type II DESs consist of quaternary ammonium salts mixed with hydrated metal halides, including chromium, cobalt, copper, nickel, and iron. These hydrated salts are not only inexpensive and stable in air and moisture, but also more versatile than the anhydrous halides used in Type I. This versatility

makes Type II DESs highly suitable for industrial applications. Furthermore, the presence of crystalline water in Type II DESs typically leads to lower melting points and viscosity, as hydration reduces the overall network energy of the mixture [15,20,65]. Type III DESs primarily consist of HBAs such as quaternary ammonium, phosphonium, and imidazolium halides, paired with a variety of HBDs, including amides, carboxylic acids, polyols, and sugars [64]. These DESs are effective at solvating metal halides and transition elements, making them versatile and highly valuable. They are biodegradable, affordable, and customizable based on the choice of HBDs, which in turn influence their physical properties. The ability of Type III DESs to dissolve a wide range of transition metal salts and oxides has led to significant interest and numerous studies on their applications [63,64,66]. Type IV DESs are made by mixing an HBD with metal chlorides or chloride-based compounds. These DESs feature metal chlorides as the HBA, paired with HBDs such as urea, glycol, or glycerol. Type IV DESs can also include actinides as the HBA, such as in a uranium-based DES synthesized from uranyl nitrate hexahydrate and urea, which offers exceptional thermal stability (from 267.8 K to 573 K). The broad variety of potential DES combinations in this category provides significant benefits, making them suitable for a diverse range of industrial applications. While their components may resemble those in Type I DESs, the key distinction lies in the use of transition metal salts as the cation in Type IV DESs [2,57,66,67]. A new classification, Type V DESs, has recently been introduced, characterized by non-ionic bonding. These DESs are composed entirely of non-ionic HBAs and HBDs, such as thymol and menthol mixtures. They are notable for having lower viscosity than their ionic counterparts and do not contain chlorides. Type V DESs are considered environmentally friendly due to their use of non-ionic components, which offer flexibility in formulation with various levels of hydrophobicity. Despite lacking ionic contributions, they still exhibit melting point behavior similar to that of typical DESs, likely due to the strong hydrogen bonding in these mixtures [58,62,66]. The diverse materials used in DESs, including salts, acids, alcohols, and others, make them highly customizable, allowing their physical and chemical properties to be tailored for specific applications [20].

DESs are classified as either hydrophilic or hydrophobic based on their interaction with water. Types I and II are completely water-miscible, while Types III, IV, and V may or may not mix with water, depending on their components [39,57]. Many DESs are hydrophilic due to their ability to form hydrogen bonds with water molecules. These hydrophilic DESs are particularly effective in extracting bioactive compounds such as flavonoids, alkaloids, and essential oils, and are also helpful for removing sulfur compounds from fuels [64]. However, their high polarity limits their usefulness in non-polar environments, especially for extraction tasks in aqueous systems [20]. In a two-phase extraction system, solvents that do not mix with water play a crucial role in extracting solutes by creating a biphasic environment. HDESSs are especially important because they are immiscible with water and demonstrate high extraction efficiency for non-polar analytes [68]. These solvents offer a promising alternative to conventional hazardous organic solvents and costly hydrophobic ILs, particularly in addressing the stability issues that arise upon contact with water [64,68]. HDESSs are typically created using quaternary ammonium salts with bulky alkyl groups, which make them effective for various applications [59]. They are commonly used in liquid-liquid extraction techniques to remove alkalis and transition metal ions from water [64]. Although HDESSs are still relatively new compared to other classes of DESs, the number of publications focusing on them is steadily increasing. This emerging class of solvents is expected to lead to a revolutionary shift toward green solvents in various fields, including biotechnology, bioengineering, nanotechnology, biomedicine, and separation and extraction technology [60].

Despite extensive research on DESs over the past decade, the fundamental mechanisms behind their formation and properties remain largely poorly understood. Advancing this fundamental knowledge is

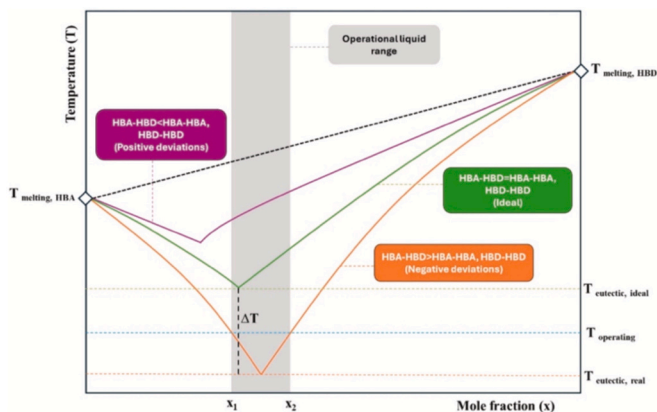


Fig. 4. Solid-liquid equilibrium phase diagrams illustrating a theoretical binary ideal eutectic system (represented in green) and a DES system (shown in orange) [37]. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

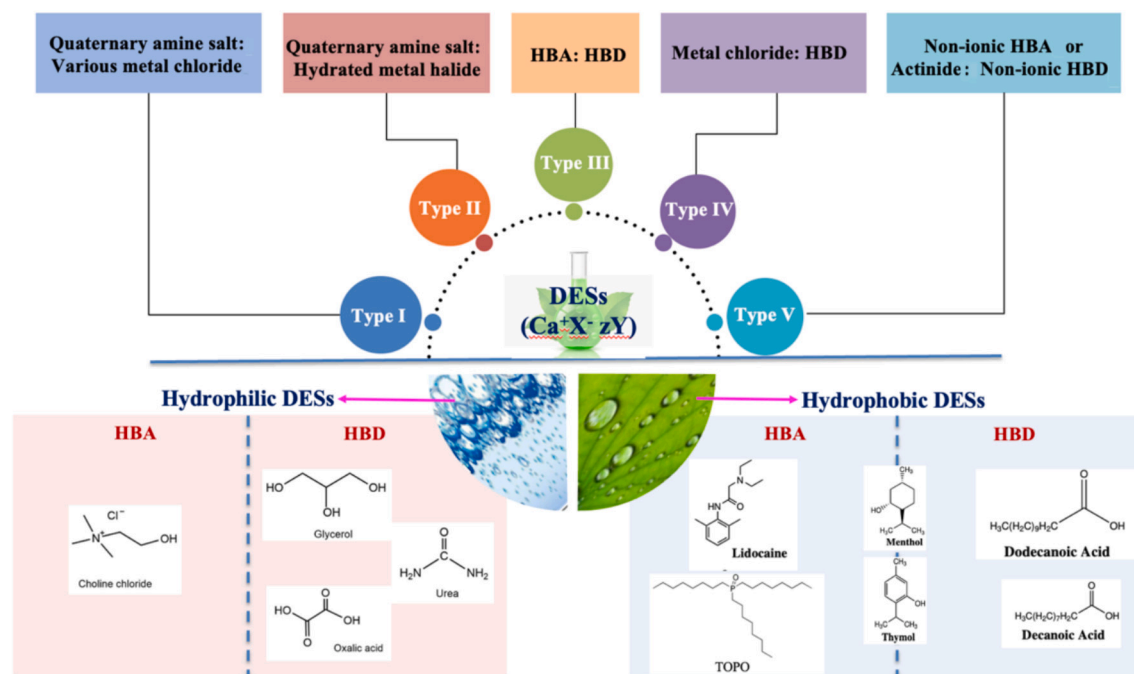


Fig. 5. Classification of DESs based on their composition and nature of the individual constituents.

crucial for developing task-specific, safe, and efficient alternatives to traditional organic solvents. However, significant challenges still exist in scaling up these applications and ensuring their economic viability [58,68]. The following sections will explore current research efforts, outlining both their strengths and limitations.

3.1. Preparation of DESs

DESs are typically prepared by combining two components, an HBA and an HBD, that interact through hydrogen bonding [68]. The process is simple and does not involve multi-step reactions or require the use of organic solvents [20]. Unlike typical chemical syntheses, which involve reactions between reagents, DESs form through physical mixing, making “preparation” a more accurate term than “synthesis” [20]. The main methods for preparing DESs are:

Heating and Stirring Method: This widely used method involves mixing the HBA and HBD, sometimes with water, and heating the mixture in a water or oil bath (323–403 K) while stirring continuously.

The process, often performed under an inert atmosphere, continues until a clear, homogeneous liquid is formed. Despite being simple, cost-effective, and safe, this method can be time-consuming and may lead to degradation through esterification if the temperature is too high. Thus, optimizing temperature and duration is essential [20,69,70].

Grinding Method: This approach, often carried out under a nitrogen atmosphere or in a glovebox, involves grinding the HBA and HBD together in a mortar or mill until a uniform liquid forms. Unlike the heating method, this process avoids esterification, resulting in purer DESs [20,57,59,70].

Twin-Screw Extrusion: A scalable mechanochemical technique, twin-screw extrusion involves feeding the HBA and HBD into a pre-heated twin-screw extruder. The screws knead and compress the materials, forming DESs with minimal heat exposure, thereby reducing the risk of thermal degradation. This method supports continuous production, high efficiency, and easy collection [20,70].

Microwave-Assisted Preparation: This rapid and energy-efficient method involves mixing the DES components in glass containers and exposing them to microwave radiation for a few seconds. It significantly reduces synthesis time (as short as 30 s) and lowers energy consumption

by a factor of up to 650 compared to conventional methods. However, careful control of microwave power, exposure time, and reactant choice is required [20,69,70].

Ultrasound-Assisted Method: In this method, HBA and HBD are sealed in a vial and placed in an ultrasonic bath (room temperature to 333 K) for 1 to 5 h. Afterward, the mixture is left at ambient conditions for 24 h to ensure uniformity. This approach yields stable DESs [52,70].

Vacuum Evaporation Method: In this method, the DES components are first dissolved in water. The water is then removed under vacuum at 323 K using a rotary evaporator. Although effective, this method can be slow and challenging due to the difficulty in entirely removing water. The final product is stored in a desiccator until its weight stabilizes, a process that is particularly useful for high-melting-point compounds [62,69,70].

Freeze-Drying (Lyophilization): This process involves dissolving the HBA and HBD in distilled water (5 % solution), freezing the mixture at temperatures ranging from 77 to 253 K, and then freeze-drying it. The water sublimates, leaving behind a clear, viscous DES. This method ensures high purity by removing moisture and can be applied either to DES mixtures or individual components [20,70].

Despite the availability of various preparation techniques, discrepancies in reported physicochemical properties persist across studies. Establishing a standardized method could improve reproducibility and ensure more consistent results [67]. Additionally, since most DESs are hygroscopic, even small amounts of absorbed water can significantly influence their properties. As such, preparation and storage in moisture-controlled environments, such as airtight vessels or desiccators, are essential [62]. Importantly, these methods typically do not require solvents, purification steps, or produce by-products, reinforcing DESs’ potential as cost-effective and eco-friendly alternatives to conventional organic solvents and ILs [20,67].

3.2. Main physicochemical properties of DESs in metal recovery

The performance of DESs in metal recovery is strongly influenced by their physicochemical properties. Key parameters include reduction potential, coordination ability, and proton activity, which directly affect the efficiency of metal dissolution and complexation. Other important

factors, including density, viscosity, water content, hydrophilicity/hydrophobicity, and acidity, significantly influence the selectivity and practicality of these solvents in separation and recovery processes. Thermal and chemical stability are also critical for ensuring reliable application under industrial conditions. Controlling the decomposition temperature is particularly important, as it governs the stability, performance, and reusability of the solvent during intensive operations [37].

3.2.1. Acidity/alkalinity

The acidity and basicity of DESs are among their most crucial characteristics, underpinning their broad applicability across various industrial sectors [68]. These properties also affect practical considerations in process design, such as the choice of materials for pipelines, reactors, and storage tanks. In hydrometallurgical processes, particularly leaching, the hydrogen ion concentration, or pH, plays a vital role in selecting suitable DESs. pH is a measure of hydrogen ion concentration in an aqueous solution, which may consist of pure water or water containing small amounts of dissolved substances. The relationship between pH values across different media is governed by the Gibbs free energy associated with proton transfer between solvents. However, comparing pH values across non-aqueous and aqueous systems remains theoretically problematic. Although pH represents the degree of acidity or basicity of a solution, it differs from acidity and alkalinity, which quantify the buffering capacity of an aqueous system to resist changes in pH. These are quantified by titration using acids or bases of known concentration to determine the amount required for complete neutralization.

DESs are formed through combinations of Lewis or Brønsted acids and bases, and their acidity or basicity is determined by the nature of the individual components (HBAs and HBDs) as well as their molar ratios. Notably, the acid–base behavior of DESs is strongly influenced by the chemical properties of the HBD component [20]. Temperature is another influential factor, with studies showing that increasing temperature generally makes DESs more acidic in a linear manner [68]. For extracting highly polar molecules or metal oxides, which typically require an acidic medium, DESs based on organic acids or certain natural components are often preferred due to their stronger acidity. The acidity of DESs is primarily dictated by the intrinsic acidity of their HBDs. Studies have also shown that proton activity (H^+ concentration) is a critical factor in dissolving metal oxides within DESs, particularly when poorly coordinating HBDs are used. In such cases, protons interact with oxygen atoms in the metal–oxide matrix, facilitating the disruption of metal–oxygen bonds and promoting oxide dissolution [5,71]. The extraction of metals using DESs is largely pH-dependent, as pH governs both the metal species present and the ionization states of functional groups in different phases of the solvent system [54]. Furthermore, increasing the alkyl chain length of HBA or HBD components reduces the solvent's proton-donating capacity [72]. The Brønsted acidity of DESs can be assessed by directly measuring pH or using the Hammett acidity function, while Lewis acidity is typically evaluated by FTIR spectroscopy [61]. The Hammett function is especially suited for highly acidic or non-aqueous systems and is determined through UV-vis spectroscopy using basic indicators, which measure the absorbance ratio between the protonated and deprotonated forms of the indicator [29]. Several studies have applied acidity and alkalinity measurements to DESs. For example, Cui et al. prepared a series of ChCl-based DESs with different HBDs, such as PTSA, trichloroacetic acid (TCA), monochloroacetic acid (MCA), and propionic acid (PA), and determined their Hammett acidity functions using 4-nitroaniline as an indicator. The results showed a clear acidity sequence: [ChCl: 2PTSA] ($H_0 = 0.87$) > [ChCl: 2TCA] ($H_0 = 0.96$) > [ChCl: 2MCA] ($H_0 = 1.46$) > [ChCl: 2PA] ($H_0 = 2.23$), reflecting the intrinsic proton-donating ability of the acid components and their hydrogen-bonding interactions. The UV-vis absorbance spectra (Fig. 6) further confirmed these results: the maximal absorbance of unprotonated 4-nitroaniline at 380 nm decreased progressively in the presence

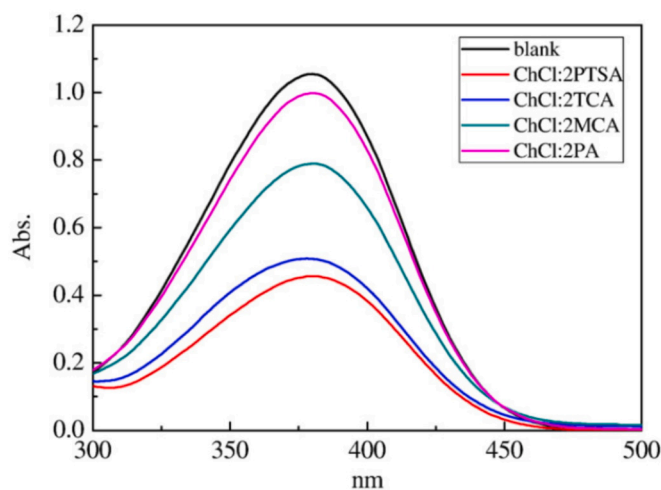


Fig. 6. UV-vis absorption spectra of 4-nitroaniline measured in aqueous solutions of various DESs (40.0 mmol/L, 293.15 K) [73].

of increasingly acidic DESs, indicating a higher fraction of protonated species. This spectral shift underpins the calculation of H_0 values and demonstrates how stronger hydrogen-bond donor acids enhance the acidity of DES systems. Together, these findings highlight that acidity in DESs can be systematically tuned by careful selection of components, with direct implications for optimizing leaching efficiency in metal recovery applications [73]. Beyond these indicator-based methods, more advanced spectroscopic approaches have been developed. NMR and IR spectroscopy have emerged as powerful tools for directly characterizing the acidity and basicity of DESs at the molecular level. For example, Shi et al. (2021) [74] characterized the acidity and basicity of green solvents, including DESs, using ^{31}P -TMPO and 1H -pyrrole NMR, enabling direct molecular-level evaluation of proton donor/acceptor strengths. Likewise, Qin et al. (2021) [75] quantified the alkalinity of DESs by combining Hammett functions with 1H NMR, revealing strong correlations between solvent composition, structure, basicity, and functional performance such as CO_2 capture capacity. These NMR- and IR-based acidity scales overcome the limitations of aqueous pH concepts, providing a more rigorous and universally comparable framework for assessing Brønsted–Lewis acidity in DESs.

Together, Hammett acidity functions and advanced spectroscopic scales offer complementary insights, enhancing the design of DESs with tailored reactivity for targeted industrial applications.

3.2.2. Melting/freezing point

DESs with melting points close to room temperature are especially attractive for practical applications because they reduce energy consumption compared to high-temperature systems [76]. Lower melting points are advantageous in extraction and separation methods, as they facilitate more efficient mass transfer of target substances [52]. Therefore, selecting DESs with low melting temperatures is a key consideration for such applications. DESs are typically identified by a noticeable decrease in their melting temperature (T_m) [20] (see Fig. 4). Investigating the structural features and thermophysical behavior of DESs, particularly through their solid–liquid equilibrium (SLE) phase diagrams across the full range of mole fractions, is crucial. These diagrams illustrate how melting temperatures vary with composition, which helps determine ideal solubility curves and the melting behavior of pure components [58]. Since different compositions can significantly influence the properties and performance of DESs, studying phase behavior beyond the eutectic point is essential. Despite this importance, available data on eutectic compositions and binary phase diagrams remain limited. Comprehensive phase diagram data are critical for the effective design and optimization of DESs for specific industrial applications.

Table 2

Summary of SLE phase diagram data for DESs reported in the literature.

DESs (HBA: HBD)	Phase Diagram Data (Eutectic Point)		Measured Method	Refs.
	Molar Fraction (HBD)	Temperature (K)		
TBAB: Boric Acid	0.43–0.60	226	DSC and NRTL model	[83]
TBAB: Benzilic Acid	0.43–0.50	244		
TBAB: Azelaic Acid	0.33–0.5	219		
Acetamide: TFOH	0.66	183	DSC	[84]
TBAB:Nonanoic Acid	0.65	260	DSC and COSMO-RS model	[77]
DL-menthol: Borneol	0.3	240	DSC	[78]
DL-menthol: Camphor	0.4	230		
Thymol: Borneol	0.4	250		
Thymol: Camphor	0.5	225	COSMO-RS and UNIFAC models	
Capric: Lauric Acids	0.25	295		
Capric: Oleic Acids	0.6	272		
TBAB: Lauric Acids	0.55	265		
TBAB: Oleic Acids	0.7	265	TGA	[86]
Glycolic Acid: H ₂ O	0.8	250		
Trimethylglycine: H ₂ O	0.8	240		
Thymol: Phenylacetic Acid	0.25	272		[87]
Thymol: Urea	0.2	265		
Thymol: Water	0.15	265		
Mannitol: Meso-Erythritol	0.8	390	Automatic Glass Capillary Device (M-565)	[88]
Mannitol: Xylitol	0.9	360		
Mannitol: Sorbitol	0.9	420		
Maltitol: Mannitol	0.3	415		
ChCl: Mannitol	0.5	380		
ChCl: Maltitol	0.6	310	–	[89]
Dodecanoic Acid: Octanoic Acid	0.75	283		
Dodecanoic Acid: Nonanoic Acid	0.75	278		
Dodecanoic Acid: Decanoic Acid	0.65	293	Cooling Curve Method	[90]
Glycolic Acid: L-Proline	0.7	248		
Trioctylphosphine oxide (TOPO): HBTA	0.6	295	DSC and TGA	[91]
EG: Oxalic Acid Dihydrate	0.1	153	DSC	[92]
Ibuprofen: Lidocaine	0.6	320	DSC	[93]
Ibuprofen: Proton Sponge	0.7	310		
Ibuprofen: DL-Menthol	0.75	290		
Ibuprofen: Methyl Anthranilate	0.8	280	COSMO-RS	[79]
L-Menthol: Thymol	0.45	200		
L-Menthol: TOPO	0.2	240		
TOPO: Phenol	0.7	243	DSC	[94]
Thymol: TOPO	0.7	255	–	[95]
TOPO: Capric Acid	0.6	260		
Hydrocinnamic Acid: Capric Acid	0.5	290		
L(–)-menthol: Caprylic Acid	0.45	265	Predicted by PC-SAFT	[96]
L(–)-menthol: Capric Acid	0.35	280		
L(–)-menthol: Lauric Acid	0.3	291		
L(–)-menthol: Myristic Acid	0.2	300		
L(–)-menthol: Palmitic Acid	0.15	305		
L(–)-menthol: Stearic Acids	0.15	310		
Thymol: Caprylic Acid	0.7	270		
Thymol: Capric Acid	0.58	285		
Thymol: Lauric Acid	0.43	297		
Thymol: Myristic Acid	0.35	307		
Thymol: Palmitic Acid	0.25	312		
Thymol: Stearic Acids	0.2	314		
EG: Sulfosalicylic Acid Dihydrate	0.1	148	DSC	[97]
ChCl: Phenylpropionic Acid	0.65	293	–	[98]
ChCl: Phenylacetic Acid	0.65	298		
ChCl: Oxalic Acid	0.5	298		
ChCl: Malonic Acid	0.5	278		
ChCl: Succinic Acid	0.5	348		
HTTA: TOPO	0.35	290	COSMO-RS	[80]
HTTA: TPP	0.35	290		
Aliquat336: L-menthol	0.4	253	DSC	[99]
TOPO: Malonic Acid	0.15	323	SLC and DSC	[100]
EG: Malonic Acid	0.2	173	DSC	[101]
EG: Succinic Acid	0.2	198		
EG: Citric Acid	0.2	193		
EG: Maleic Acid	0.2	173		
1,2-Propanediol: Maleic Acid	0.2	193		
Glycerol: Maleic Acid	0.3	213		
1,4-Butanediol: Maleic Acid	0.2	183	COSMO-RS	[102]
Sucrose: Glucose	0.6	400		
Sucrose: Fructose	0.7	370		
Glucose: Fructose	0.65	370	DSC	[103]
Guanidine Hydrochloride (GUC): Lactic Acid	0.65–0.9	198		

(continued on next page)

Table 2 (continued)

DESs (HBA: HBD)	Phase Diagram Data (Eutectic Point)		Measured Method	Refs.
	Molar Fraction (HBD)	Temperature (K)		
Biphenyl: Bibenzyl	0.55	305	Capillary Melting Point Apparatus	[104]
ChCl: Urea	0.65	300		
TPABr: Erythritol	0.45	350		
TPABr: Succinic Acid	0.35	350		
1,10-phenanthroline: Thymol	0.8	273	Automatic Glass Capillary Device (M-565)	[82]
1,10-phenanthroline: Chlorothymol	0.5	293		
1,10-phenanthroline: BHT	0.2	343		
Tetrapentylammonium Bromide (Pe4NBr): Erythritol	0.4	358	Orthogonal Redlich–Kister Polynomial	[105]
Thymol: TOPO	0.3	260	COSMO-RS	[81]
Menthol: TOPO	0.2	285		
Dodecanoic Acid: TOPO	0.45	260		
Dihexylthiourea: TOPO	0.45	285		
Thymol: Hexadecanol	0.4	302		
Menthol: Hexadecanol	0.2	300		
Thymol: Camphor	0.5	200		
Menthol: Camphor	0.5	280		

Many studies rely on presumed eutectic mixtures without confirming their compositions through binary phase diagrams, making the inclusion of such diagrams vital for proper validation [58,62]. Several studies are summarized in Table 2. Because including all phase diagrams would make the paper excessively long, readers are referred to the cited references for full details. Instead, Table 2 highlights the two most important parameters from the phase diagrams: composition and eutectic temperature.

Several factors influence the melting point of DESs, including the molar ratio of components, the alkyl chain length of HBAs and HBDs, the overall chemical composition, the strength of intermolecular interactions, the structural arrangement of components, and the degree of ionic symmetry [60,68]. Their typically low melting points are primarily due to reduced lattice energy and strong hydrogen bonding between HBAs and HBDs, which prevents crystallization. Additionally, entropy changes occur during the liquid formation process, and the melting points of the individual components influence the overall melting behavior of the mixture [52,62,63]. In general, the freezing points of DESs range from -342 K to 422 K [58]. Moreover, larger ions with smaller charges require less energy to dissociate, resulting in lower freezing points. HBDs often increase the effective size of anions by interacting with them, further lowering the melting temperature [62]. Carboxylic acids with shorter carbon chains generally have lower molecular weights, which leads to a greater freezing point depression. This trend is illustrated by the ChCl: oxalic acid [68] (307 K) compared with ChCl: adipic acid [68] (358 K) [52,64]. Cation symmetry also influences melting behavior; less symmetric cations typically result in lower melting points [52,64]. When comparing Type I and Type II DESs, a further decrease in melting point is observed when anhydrous metal chlorides are replaced with their hydrated counterparts. This is attributed to the presence of water, which disrupts the crystalline structure and reduces lattice energy, thereby facilitating the melting process [20]. In Type III DESs, where carboxylic acids act as HBDs, the coordination of halide anions (e.g., Cl^-) often involves at least two hydroxyl groups. This multipoint hydrogen bonding stabilizes the anion and plays a crucial role in disrupting the crystalline structure of the salt, facilitating eutectic formation. The structure and type of HBD have a significant impact on the freezing point of the resulting DES. However, the literature currently lacks a clear correlation between the freezing points of DESs and the melting points of their individual components [76].

Unlike hydrophilic DESs, hydrophobic DESs can exhibit either shallow or deep depressions in their melting points. Most HDESs have melting temperatures below 273 K, which allows for their potential use as reaction media or solvents at room temperature, indicating that they

remain stable in the liquid state under these conditions [60]. Additionally, most ionic and non-ionic HDESs melt at progressively higher temperatures as the alkyl chain length of the fatty acid HBD increases. Moreover, the melting point of ionic HDESs is significantly affected by both the anions and cations present in the ionic components [57]. It is essential to note that many non-ionic DESs exhibit only a glass transition temperature (T_g) rather than a sharp melting point, indicating a transition from a rigid, glassy state to a viscous or amorphous liquid state. As such, they are classified as low-transition-temperature mixtures (LTTMs) rather than true solids [64].

The melting or freezing point of a DES mixture can be determined using various thermal analysis techniques, such as a melting point apparatus, DSC, X-ray scattering, visual observation, and TGA. These methods are widely employed to characterize the fundamental thermal properties of DESs. Instruments from leading manufacturers, including TA Instruments, Netzsch, Mettler-Toledo, PerkinElmer, and Shimadzu, are commonly used for such analyses. These techniques provide insights into phase transitions, thermal stability, and the proportion of volatile components present [56,62,67]. In addition to experimental approaches, predictive models such as the conductor-like screening model for real solvents (COSMO-RS) have proven particularly effective for estimating solid–liquid equilibrium (SLE) phase diagrams of DESs, with several studies successfully applying this method [77–80]. The COSMO-RS model evaluates the chemical potential and activity coefficients of mixture components by analyzing interactions between molecular surface charge segments (σ -surfaces). This predictive capability supports rational experimental design and enables accurate estimation of phase behavior. SLE phase diagrams are particularly important for identifying the degree of non-ideality and the liquid range of DESs, while also preventing misinterpretations of melting point depression. For instance, some systems may form cocrystals at specific ratios (1: 1) but still exhibit significant melting point depression near pure component compositions, resulting in liquid states. Beyond phase equilibria, COSMO-RS has also proven valuable for predicting solute solubility in DESs.

Although computational approaches, such as molecular simulations and machine learning, are gaining increasing attention, their reliability is still constrained by the limited availability of experimental data [81,82]. For example, Abdallah et al. (2021) generated SLE phase diagrams from experimental data for ideal systems and modeled them using both the semi-predictive UNIFAC and COSMO-RS models [78]. Likewise, Kianpoor et al. (2024) employed activity coefficient models, including the Wilson, NRTL, and regular solution models, to correlate their experimentally measured SLE data [78]. Likewise, Kianpoor et al. (2024) employed activity coefficient models, including the Wilson,

NRTL, and regular solution models, to correlate their experimentally measured SLE data [83].

3.2.3. Viscosity

Viscosity is a key property of DESs, as low viscosity promotes more efficient mass transfer, enhancing diffusion and penetration within the system. This, in turn, enhances extraction performance in separation methods like leaching and liquid–liquid extraction [56]. Conversely, high viscosity can hinder diffusion rates and complicate solid–liquid separation, leading to increased energy demands, additional processing steps during recovery, and operational difficulties in handling, pumping, stirring, and filtering [15,69]. Most DESs exhibit high viscosity at room temperature, often exceeding 100 mPa·s (for comparison, the viscosity of water at 293 K is 1 mPa·s) [72], making viscosity a vital factor in selecting suitable solvents for these applications [69]. Compared to water, DESs typically show higher viscosities due to the larger size and volume of their ionic components [63,76]. However, formulating DESs with low viscosity is challenging, as reducing hydrogen bonding is necessary to achieve this, which can also narrow the solvent's liquid range [59]. The elevated viscosity of many DESs is largely due to strong hydrogen bonding and other intermolecular forces, such as van der Waals interactions [61]. These interactions limit ionic mobility and reduce the number of free-moving species within the already compact liquid structure [68]. Several parameters influence the viscosity of DES, including the nature of the constituent components, temperature, molecular weight, molar ratio, and water content [59,68]. Specifically, the structure of the HBD, as well as its size and molecular weight, strongly affect overall fluidity [66]. In general, higher molar ratios of the HBA lead to greater viscosity due to denser molecular packing and reduced free volume, while increasing the proportion of HBD tends to lower viscosity [66]. Adding water to DESs is an effective method of reducing viscosity. For instance, the viscosity of a ChCl: oxalic acid system is notably higher than that of its hydrated counterpart, ChCl: oxalic acid dihydrate [15,61]. Temperature also plays a crucial role: increasing it decreases internal resistance, enabling smoother fluid flow [67]. Raising the temperature from 298 K to 353 K has been shown to increase leaching rates by up to 15 times, primarily due to the drop in viscosity. This temperature-dependent behavior can be explained by the Hole Theory, which describes molecular mobility within liquids [62]. The theory accounts for both steric effects and molecular interactions in determining viscosity. According to Hole Theory, when ionic materials melt, voids or “holes” form as liquid density changes with rising temperature. These holes vary in size and location and remain in constant motion. At lower temperatures, the holes are generally smaller than the DES components, restricting molecular movement, which significantly reduces mobility and results in high viscosity values, typically ranging from 100 to 1000 mPa·s. With increasing temperature, the average size of the cavities becomes more comparable to that of the DES components,

thereby enhancing molecular mobility. At any given temperature, a DES component can move only if a sufficiently large cavity is available [66,106]. As illustrated in Fig. 7(a), the viscosity of TBAB: octanoic acid (OC) DESs decreases exponentially with temperature and can be accurately described by the VFT model. The reduction in viscosity at elevated temperatures results from enhanced molecular mobility and weakened hydrogen-bond interactions. Furthermore, increasing the molar fraction of OC leads to a pronounced decrease in viscosity. This occurs because additional OC molecules, with their flexible long alkyl chains, disrupt the strength and continuity of the hydrogen-bond network between TBAB and the HBD, thereby lowering resistance to flow. As a result, DESs with higher OC content display reduced viscosities and better handling properties, which are beneficial for metal leaching and separation applications [107]. The typically high viscosity of DESs is largely attributed to the relatively large molecular size of their components compared to the radii of voids within the liquid structure [63]. While viscosity usually increases with longer hydrocarbon chains, this trend can be counterbalanced by the presence of carbonyl or hydroxyl groups, which weaken strong intermolecular interactions. For example, despite having one of the shortest alkyl chains (two carbons), the ChCl: oxalic acid DES is among the most viscous due to its extensive hydrogen-bonding network [62]. Therefore, careful molecular design is essential to optimize viscosity, ensuring a balance between efficient mass transfer and desirable solvation characteristics [64]. To address the issue of high viscosity in hydrophilic DESs, low-viscosity HDESs have been developed, some exhibiting viscosities below 20 mPa·s [60]. Fatty acid- and DL-menthol-based DESs, in particular, offer promising alternatives. For instance, an HDES composed of DL-menthol and octanoic acid (in a 1: 3 ratio) exhibits a viscosity of only 7.61 mPa·s at 298 K. In contrast, a hydrophilic DES made from ChCl: sorbitol (1: 1 ratio) exhibits an extremely high viscosity of 12,730 mPa·s at 303 K [58,108]. The most applied models to analyze the temperature dependence of viscosity in DESs are the Arrhenius model and the Vogel–Fulcher–Tamman (VFT) model. Studies have shown that the Eyring-based model provides better accuracy for low-viscosity DESs. In contrast, the VFT model offers a closer fit to experimental data for more viscous systems [66]. Generally, the viscosity of DESs follows an Arrhenius-like trend, where increasing temperature results in a decrease in viscosity [76]. Various instruments have been employed to measure DES viscosity, including Brookfield and Anton Paar viscometers, Ubbelohde viscometers (SI Analytics), TA Instruments AR-G2 rheometers, automatic viscometers such as the NDJ-8S and NDJ-79 models (Hengping Co., Shanghai, China), PM Tamson Instruments viscometers, as well as the Pinkevitch method, Bohlin rheometers, and Ostwald viscometers [68].

3.2.4. Water content

One of the most critical parameters to manage in DESs is their water content, as even small amounts can significantly alter their properties

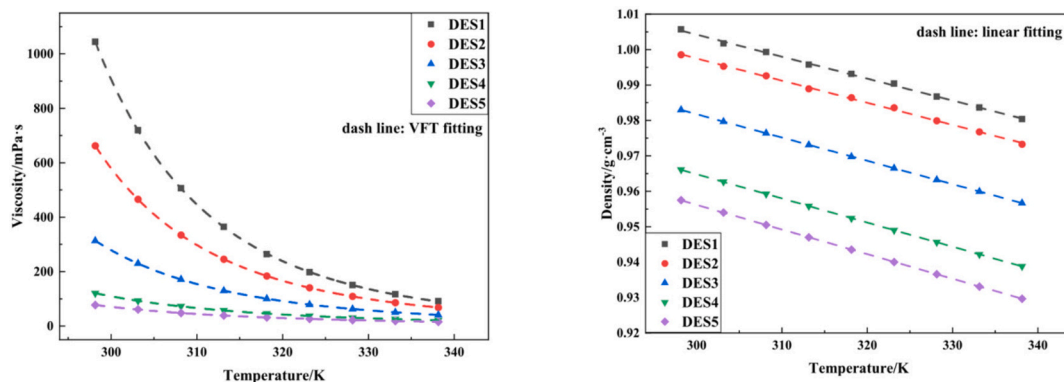


Fig. 7. Temperature-dependent (a) viscosity and (b) density of TBAB: OC DESs at different molar ratios: DES1 (1: 1.25), DES2 (1: 1.5), DES3 (1: 2), DES4 (1: 3), and DES5 (1: 4) [107].

and affect metal recovery processes [77]. Due to the hygroscopic nature of many DESs and their components, water uptake is inevitable, making complete dehydration a challenging task. While traces of water are often considered impurities, several studies have deliberately introduced water to improve DES performance for specific applications [20]. For example, adding water can reduce viscosity while preserving other key characteristics.

However, due to its high polarity and strong interactions with hygroscopic constituents, water can disrupt the molecular structure and integrity of DESs, resulting in discrepancies in the literature that arise from differences in preparation methods. Understanding the effects of water on eutectic systems is therefore essential, as water strongly influences their physicochemical properties and should always be carefully reported [20,58]. Water can act as both an HBA and an HBD, disrupting the hydrogen-bond network in DESs and altering the system's entropy by interfering with ion coordination [56]. During preparation, DESs readily absorb atmospheric moisture, which lowers density and viscosity [76]. Although DESs can maintain their nanostructure with water contents up to 50 wt.%, even small amounts can trigger major changes.

For instance, ethaline (ChCl: EG) tolerates up to 30 wt% water while preserving most of its eutectic features, though its properties shift significantly at higher water content [62]. Generally, increasing water content reduces the melting point, density, and viscosity, while enhancing ionic mobility, polarity, and solubility [58,68]. For example, the addition of only 10 wt% water can lower viscosity by 80 % and triple conductivity compared to dry DESs [20]. Experimental evidence further illustrates this effect: after three weeks of air exposure, a ChCl: urea DES absorbed 20 wt% water, and even 6 wt% water was enough to markedly reduce viscosity and increase conductivity. Similarly, the melting point dropped from 303 K (dry) to 288 K at 5 wt% water, while conductivity initially increased with hydration, peaking at 60 wt% due to enhanced ionic dissociation, before declining again as the electrolyte became overly diluted [58]. Overall, water uptake enhances polarity and ionic mobility but disrupts hydrogen bonding, with profound impacts on DES stability [72]. Water also affects biodegradability and cytotoxicity, as dissolution of DESs in aqueous environments can increase toxicity [68]. These interactions have direct implications for both environmental safety and economic feasibility. For instance, one study showed that DESs exposed to open air for a month absorbed 14–20 wt% water, significantly altering their density [66]. Understanding these effects is especially important for assessing DES reuse in hydrometallurgical processes, where water is naturally introduced during the dissolution of metal oxides [20,37]. Accordingly, Karl Fischer titration remains one of the most effective techniques for accurately measuring water content, especially in binary DES systems with low hydration levels [67].

3.2.5. Density

Density is a key physicochemical property of DESs that plays a critical role in solvent selection and performance evaluation, particularly in biphasic extraction systems [20]. It is a crucial parameter that affects process design, separation efficiency, and large-scale applications, as it influences diffusivity and the solvent's ability to mix with other liquids [69,108]. The higher density of hydrophilic DESs affects operational processes, resulting in increased energy consumption for agitation and pumping during extraction [69]. The efficiency of extraction with HDESs is closely tied to their immiscibility with water, which is influenced by differences in density [60]. Typically, hydrophilic DESs have a higher density than water, causing the DES-rich phase to settle at the bottom [64]. However, in liquid-liquid metal extraction, HDESs often remain in the upper phase because their density is lower, generally ranging from 0.90 to 0.95 g/cm³ [60,64]. HDESs are commonly used in sample pre-treatment due to their densities, which are similar to or lower than water, typically differing by more than 0.05 g/cm³ [57,60]. This facilitates easier processing, reduces emulsification, and enhances phase separation [60]. Experimental data on the density of DESs,

including their temperature dependence, are vital for studying liquid-liquid equilibrium, designing processes and equipment, mass transfer, and developing predictive models. These data also help to determine other thermodynamic parameters, such as viscosity, thermal expansion, isothermal compressibility, and free volume [62]. Higher density typically increases thermal conductivity and the solubility of polar and ionic compounds due to more compact molecular arrangements. However, the solubilization capacity of DESs is also influenced by factors like polarity, hydrogen bonding, molecular size and shape, and the nature of the solute [56,61]. The molecular interactions and packing of components in DESs significantly affect their density [57]. The density of DESs can be adjusted by altering the components and their ratios [64]. For example, using more hydroxyl groups in HBAs increases density, while adding aromatic groups or lengthening alkyl chains tends to reduce it. The higher the number of hydrogen bonds, the fewer the free spaces, resulting in an increase in density [72,106,108]. For instance, oxalic acid-based DESs are denser than those based on malonic acid [106]. The anionic type also significantly influences the density of DESs [64]. Additional carboxylic groups and ether bonds in HBAs can also increase the density of DESs [72]. Moreover, higher HBD ratios strengthen hydrogen bonding, thereby reducing available free space and altering the molecular structure, which in turn impacts density [56,66]. The temperature and molar ratio of HBA and HBD during preparation have a significant impact on the density of DESs [66,108]. For example, as shown in Fig. 7(b), the density of TBAB: OC-based DESs shows a typical linear decrease with increasing temperature, consistent with the behavior of most molecular solvents. This is attributed to thermal expansion, which increases the average intermolecular distance at higher temperatures, thereby reducing the overall packing efficiency. In addition, the density systematically decreases with the increase of the molar fraction of the HBD (OC). This effect arises because the long alkyl chains of OC introduce greater free volume and structural irregularity into the DES network, reducing molecular packing compared to shorter-chain HBAs. Consequently, DESs with higher OC content exhibit lower densities than those with lower HBD fractions [107]. Longer alkyl chains increase molar volume, which in turn lowers DES density, as steric hindrance reduces hydrogen bond interactions [69]. Hydrated DESs tend to have lower density than their pure counterparts due to the reduction in hydrogen bonding between the HBA and HBD upon the addition of water [66]. Density can be measured using a specific gravity meter [64]. Several instruments have been used for density measurements, including Anton Paar vibrating-tube density meters (Models: 60/602, DMA4100, DMA 4500, DMA4500M, DMA5000, DSA 5000, and SVM3000), the Anton Paar Lovis 2000 M/ME rolling-ball viscometer, the Mettler-Toledo density meter, and the KEM-DA100 vibrating tube density/specific gravity meter from Kyoto Electronics Manufacturing Co. Additionally, standard pycnometers are commonly employed to measure densities at room temperature [62].

3.2.6. Hydrophilicity/hydrophobicity

A key factor influencing the use of DESs in metal separation and extraction is their hydrophilicity or hydrophobicity [62]. Due to their strong hydrogen bonding ability, DESs generally exhibit a high degree of hydrophilicity and are often miscible with water [64]. DESs can be classified as either hydrophilic or hydrophobic depending on their affinity for water, which is influenced by the nature and interactions of the HBAs and HBDs [61,64]. Hydrophilic DESs are particularly effective for extracting metals in solid and non-aqueous solutions. For example, ChCl-based hydrophilic DESs that incorporate urea, EG, and carboxylic acids as HBDs have demonstrated excellent capabilities in dissolving and extracting polar metal oxides.

In contrast, HDESs are ideal for metal extraction and recovery from water-soluble environments, as they are immiscible with water and offer high extraction efficiency for non-polar analytes [39,64]. HDESs are typically non-ionic and are often formed by combining DL-menthol with carboxylic acids such as acetic acid, pyruvic acid, lactic acid, or lauric

acid [64]. The presence of bulky alkyl groups in HDESs lowers their viscosity, facilitating mass transfer between phases. Additionally, their hydrophobic nature eliminates the risk of water contamination [59]. Furthermore, increasing the hydrophobicity of the HBD reagents can enhance the overall hydrophobicity of the DESs [62]. Compared to hydrophilic DESs, HDESs typically have lower melting points, leading to reduced viscosity. Most HDESs have melting points below 298 K, making them suitable for use at room temperature as solvents or reaction media [60]. The hydrophobicity of DESs is affected by the chemical nature of both HBAs and HBDs, as well as the molar ratio of the eutectic mixture. Long alkyl chains in HBAs contribute to the overall hydrophobicity by creating steric hindrance, which reduces the interaction between water and the core salt [68]. HDESs tend to have low densities and moderate viscosities at room temperature. The physical properties of these HDESs are also influenced by the size of the anion; larger anions result in higher viscosity, and this effect is further influenced by longer alkyl chains in the HBAs [68]. The hydrophobicity of HDESs was evaluated by Ma et al. (2024) [109] through a simple water-separation method. Equal volumes of ultrapure water and the HDES phase were mixed, and the resulting liquid phases were visually distinguished using a dye-based technique. Specifically, 1-(2-Pyridylazo)-2-naphthol was used to stain the HDES (oil) phase, while chromium trichloride was added to colour the aqueous phase. The mixture was stirred at 363.15 K for 30 min with a magnetic stirrer to ensure thorough contact between the HDES and water. After stirring, the mixture was allowed to stand for 48 h. The hydrophobicity or hydrophilicity of the HDES was assessed based on the clear separation of the two stained phases, resulting in two stable layers with no noticeable volume loss [62]. The contact angle measurement method on glass can be used to determine the hydrophilicity or hydrophobicity of DESs in comparison to each other or to reference substances [110].

In conclusion, the performance of DESs in metal recovery is governed by a set of critical physicochemical properties that collectively determine dissolution efficiency, selectivity, and process sustainability. Among these, viscosity is a major determinant of mass transfer: hydrophilic ChCl-based DESs typically range from 10 to 30,000 mPa·s at 25 °C, with lower viscosities (<500 mPa·s) favoring faster dissolution and leaching kinetics, while high viscosities hinder mixing, pumping, and solid-liquid separation. Acidity and proton activity directly control

oxide dissolution, with Hammett acidity functions (H_0) spanning a range of -2 to $+2$. Strongly acidic DESs (ChCl: PTSA) accelerate leaching but may compromise stability, whereas weaker systems often result in incomplete metal dissolution. The coordination ability of the HBD: HBA is equally critical: systems with urea, ethylene glycol, or organic acids enhance selectivity for Cu, Co, or REEs, with coordination strength dictating the stability of the complex and its downstream separation.

Hydrophilicity/hydrophobicity dictates phase behavior and separation strategy: hydrophilic DESs are highly effective for oxide dissolution but are challenging to separate from aqueous phases, while HDESs (menthol: decanoic acid) enable biphasic extraction, minimizing wastewater and simplifying purification. Conductivity, typically ranging from 0.1 to 20 mS/cm, facilitates electrochemical recovery and is often inversely correlated with viscosity. Density, typically within the range of 1.0–1.8 g/cm³, plays a minor role in kinetics but significantly contributes to handling, mixing, and phase separation. Water content exerts a dual influence: moderate additions (10–30 wt%) lower viscosity and enhance conductivity, whereas excess water (>50 wt%) disrupts hydrogen-bond networks and destabilizes the DES structure.

As illustrated in Fig. 8, these properties exert decisive and interdependent effects on recovery outcomes. High acidity accelerates oxide dissolution but may compromise selectivity, while low acidity risks incomplete leaching. Viscosity governs mass transfer, while water content modulates both viscosity and stability. Coordination strength controls selectivity, and hydrophilicity/hydrophobicity dictates whether DESs are best suited for leaching or biphasic extraction. Conductivity underpins electrochemical applications, and density influences phase behavior. Taken together, these relationships demonstrate that DESs represent a tunable solvent platform, where targeted adjustments of key properties enable the optimization of leaching efficiency, selectivity, and industrial feasibility.

3.3. Kinetics

Kinetics, as a fundamental requirement of nearly all industrial activities and engineering research, plays a great role in leaching processes [111]. In hydrometallurgical and solvometallurgical processes, the solid-liquid system is commonly modeled using the unreacted shrinking

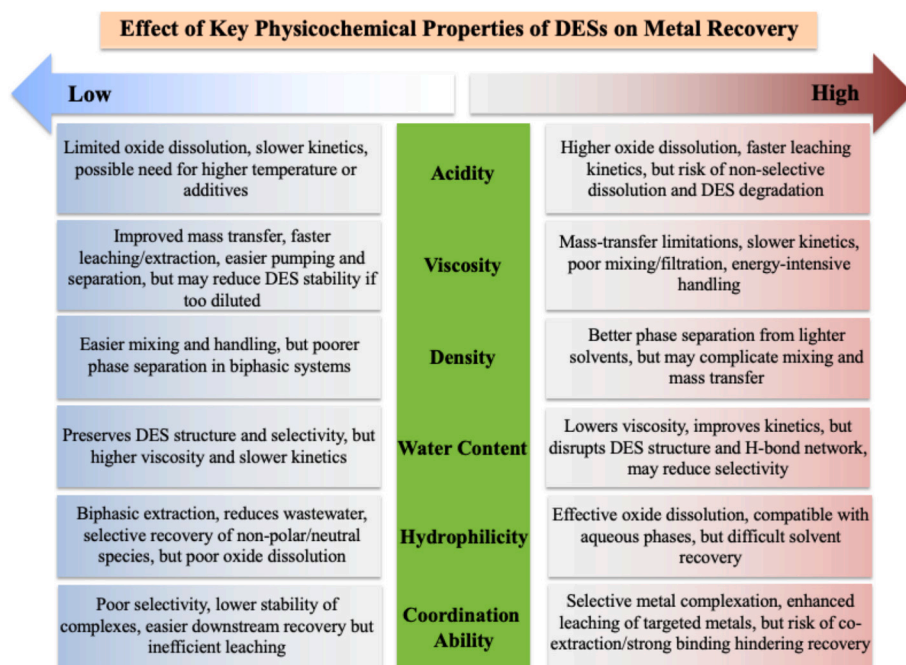


Fig. 8. Key physicochemical properties of DESs and their influence on metal recovery performance.

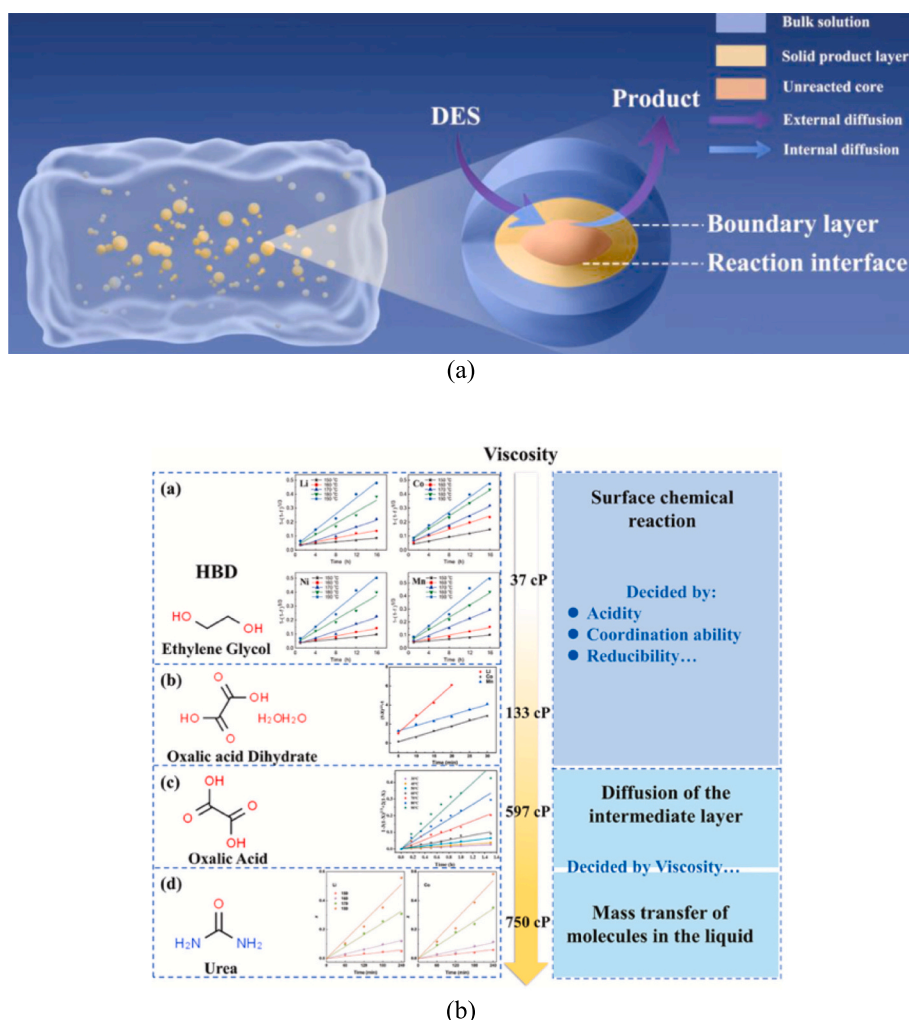


Fig. 9. (a) Illustration of the Shrinking Core Model applied to leaching with DESs [31]. (b) Comparison of leaching kinetic models of LIBs in various DESs [15].

core model, which is widely recognized for analyzing leaching kinetics. Throughout the metal extraction process, the solid particle gradually diminishes in size until it is fully dissolved [112]. Metal extraction starts on the particle's exterior, so the active reaction layer gradually shrinks inward over time, leaving the dissolved metals in solution. The shrinking-core model relies on two key simplifications: (1) the solid particle is treated as a perfect sphere, making the math easier by assuming identical reaction rates across its surface, and (2) that surface reaction rate is considered uniform everywhere, even though fundamental particles may have irregular shapes [113]. As shown in Fig. 9(a), the leaching process in DESs takes place at the solid-liquid interface and can be broadly divided into two sequential steps: diffusion of the active components and their subsequent interfacial chemical reactions with the solid phase. During leaching with DESs, molecules or ions first diffuse from the solution layer through the boundary layer to the outer region of the solid product layer. Then, these molecules or ions move from the solid product layer to the reaction interface, where internal diffusion takes place. At the interface of the compounds being leached, a chemical reaction occurs involving the molecules or ions. The resulting products then diffuse from the solid product layer into the inner part of the boundary layer. Finally, the products move from the boundary layer to the outer region of the solution layer. This sequence demonstrates the movement and transformation of molecules and ions during the leaching process in DESs [31]. According to the kinetic mechanism of the shrinking core model, leaching initiates at the outer surface of the particles and gradually progresses inward toward the core [114].

Heterogeneous reactions typically proceed through multiple steps, with the slowest step controlling the overall reaction rate. The shrinking core model's rate control steps can be classified into three types: diffusion control (Eq. (1), where k is related to the effective diffusion coefficient), chemical reaction control (Eq. (2)), and mixed control models (surface reaction control and diffusion through the product layer) (Eq. (3)). The activation energy (E_a) for different metals can be determined using the Arrhenius equation (Eq. (4)).

$$1 - 2/3X - (1 - X)^{2/3} = kt \quad (1)$$

$$1 - (1 - X)^{1/3} = kt \quad (2)$$

$$(1 - X)^{-1/3} - 1 + 1/3 \ln(1 - X) = kt \quad (3)$$

$$\ln(k) = \ln(A) - E_a/RT \quad (4)$$

where X is leaching efficiency for different metals (%), k is the rate constant of chemical reactions (min^{-1}) and t is leaching time (min), A is the frequency factor, E_a is the activation energy (kJ/mol), R is the universal gas constant ($8.314 \text{ J}/(\text{mol} \cdot \text{K})$), T is leaching temperature (K). The activation energy provides valuable insights into the kinetics and thermodynamics of chemical reactions. The step controlling the reaction kinetics can also be evaluated using activation energies. A higher activation energy indicates that temperature has a more significant effect on the reaction rate. Mass transport plays a critical role in the overall rate of many reactions. For example, diffusion, a mass transport process,

Table 3

Summary of research on leaching kinetics in metal recovery using DESs.

DESs	Molar ratio	Diffusion Control (DC) $1-2/3X-(1-X)^{2/3} = kt$	Mixed Control (MC) $(1-x)^{-1/3}-1 + 1/3\ln(1-x) = kt$	Chemical Reaction Control (CRC) $1-(1-X)^{1/3} = kt$	Ea (kJ/mol) of metals					μ (mPa·s) @ 298 K	Refs.
ChCl: Sulfamide	1: 2	×	×	✓	Au 44.52	—*	—	—	—		[112]
EG: Tartaric Acid	3: 1	×	×	✓	Li 40.85	Ni 42.98	Co 43.44	Mn 41.31	—		[117]
PEG-400: ρ -Toluene Sulfonic Acid (PSTA)	1: 1	×	×	✓	Ni 19.61	—	—	—	97		
ChCl: PTSA: EG	1: 2: 1	×	✓	✓	Ni 32.25 (CRC) 56.78 (MC)	—	—	—	114		[115]
BeCl: Formic Acid	1: 9	×	×	✓	Li 38.9	Mn 66.1	Co 70.3	Ni 94.1	4.3		[118]
ChCl: Citric Acid	1: 1	×	×	✓	Li 73.53	Co 63.34	—	—	69.1		[119]
ChCl: Urea: EG	1: 2: 1	×	✓	×	Li 21.15	—	—	—	—		[120]
ChCl: EG	1: 1	×	✓	×	Li 19.8	Mn 41.6	Co 24.4	Ni 36.5	—		[121]
EG: Sulfosalicylic Acid	12: 1	×	×	✓	Li 79.54	Co 77.38	—	—	44.3		[97]
MA: GUC: EG	1: 1: 2	×	×	✓	Li 50.77	Co 72.57	—	—	66.6		[122]
ChCl: Oxalic Acid	1: 1	×	×	✓	Li 86.22	Co 88.03	Mn 101.47	—	133		[123]
ChCl: H_3PO_4	1: 1	✓	×	×	Co 54.54	—	—	—	—		[124]
ChCl: Pyruvic Acid	1: 1	×	×	✓	Li 50.95	Co 60.68	—	—	—		[125]
ChCl: EG	2: 1	×	×	✓	Li 17.90	Co 28.20	Ni 29.45	Mn 28.53	—		[17]
ChCl: Oxalic Acid	1: 1	✓	×	×	Co 47.61	—	—	—	579		[126]
ChCl: Succinic Acid: EG	1: 1: 1	×	×	✓	Co 11.77	—	—	—	—		[127]
EG: Citric Acid	2.5: 1	×	×	✓	Li 47.84	Co 49.32	Ni 49.11	Mn 48.38	—		[128]
UCl ₃ : Formic Acid	—	×	×	✓	Li 50.05	—	—	—	—		[129]
Dimethyl-beta-Propiothetin: EG	—	×	×	✓	Li 117.59	Co 173.09	Ni 167.64	Mn 153.81	—		[130]
ChCl: Benzenesulfonic Acid (BSA): Ethanol	1: 1: 2	×	×	✓	Li 40.74	Co 44.01	—	—	—		[131]

*— Not Investigated or Not Available.

typically has a low activation energy, generally less than 25.08 kJ/mol. In contrast, reactions governed by surface chemical reactions usually have a much higher activation energy, typically greater than 30 kJ/mol. As a result, temperature has less impact on diffusion-controlled processes compared to chemical-controlled processes. Mixed control processes exhibit activation energies ranging from 20 to 33 kJ/mol. Activation energy is commonly used as a metric to determine whether mass transport or reaction control dominates a process [112,113,115]. The high viscosity of many DESs has raised concerns about their potential impact on the kinetics and mechanism of metal dissolution. As it is well-established, increased viscosity impedes mass transfer, which can reduce the efficiency of metal oxide dissolution [116]. As illustrated in Fig. 9(b), high viscosity can impede mass transfer and diffusion, resulting in slower processes during these stages. Ultimately, mass transfer and diffusion are the key controlling steps in the leaching reaction. In contrast, for DESs with lower viscosity, the chemical reactions themselves become the dominant controlling factor. Thus, as the viscosity of DESs decreases, the controlling steps shift from mass transfer and diffusion through the liquid and intermediate layers to surface chemical reactions. In these surface chemical reactions, the acidity, coordination, and reducibility of the DES are believed to govern the

reaction speed [15]. However, for highly viscous DESs, increasing temperature can lower viscosity, enhancing mass transfer and diffusion within the solution. This change can shift the controlling step of the leaching process from diffusion to interfacial chemical reactions [31]. It is essential to note that some studies yield results that do not conform to the typical trends of viscosity and activation energy. Table 3 summarizes several studies that explore the leaching kinetics of metal recovery using various DESs.

3.4. Mechanisms of metal recovery using DESs

Research on the application of DESs has primarily centred on maximizing metal dissolution yields, but the underlying mechanisms of metal dissolution remain inadequately explored. It is essential not only to improve performance but also to understand the dissolution processes themselves. Such understanding provides insights into the consumption of DESs during use, as DESs often function more as reagents than as solvents in these contexts [34]. The leaching mechanisms are still poorly understood, though common patterns have been observed in studies on acid-based DESs for metal leaching. These include: (i) acid-based DESs typically exhibit shorter reaction times than their non-acidic

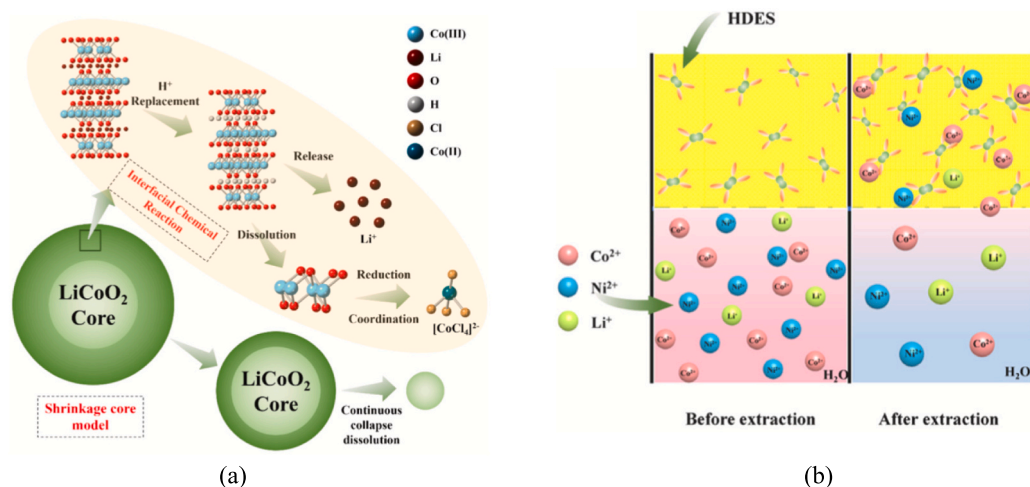


Fig. 10. (a) Schematic diagram of the LCO dissolution process by MA: GUC: EG DES [122]. (b) Schematic representation of cobalt extraction by HDES through a cation-exchange mechanism: before extraction (left) and after extraction (right). Co²⁺ ions selectively migrate into the HDES phase, while Ni²⁺ and Li⁺ largely remain in the aqueous phase [133].

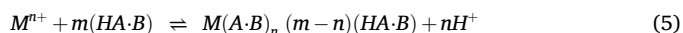
counterparts; (ii) metal oxide dissolution often involves the reduction of transition metals from higher to lower valence states; and (iii) the selective separation of valuable metals is closely related to the coordination forms of the resulting metal ions.

When cathode materials are treated with acidic aqueous solutions, two critical reactions generally occur: proton attack on the oxide moiety of the cathodes and reduction of transition metals to their divalent state, facilitating the formation of water-soluble complexes. Both reactions are essential for successful leaching and dissolution [29]. In the case of LCO leaching with DESs, the process primarily involves (i) acidic attack by H⁺ ions, (ii) reduction of Co(III) to Co(II), and (iii) complexation of Co(II) with Cl⁻ ions. The overall process, illustrated in Fig. 10(a), can be broken down into three steps: (1) H⁺ ions disrupt the lamellar LiCoO₂ structure by attacking Li—O and Co—O bonds, with Li⁺ ions being displaced due to their similar radii and activity; (2) protons adsorb onto and attack the oxygen of the transition metal oxide, forming H₂O and breaking metal—oxygen bonds; and (3) under acidic and high-temperature conditions, EG acts as a reducing agent, while delocalized Cl⁻ ions promote ligand substitution, stabilizing Co²⁺ as [CoCl₄]²⁻. This overall process indicates a redox reaction during the leaching process. Metal oxides with lower lattice energy and less negative Gibbs free energies of formation are particularly susceptible to proton attack from DESs [15,33,122]. Even after metal recovery (via precipitation or electrodeposition), the transformation is not fully reversible. Nevertheless, DESs can still be reused: partially oxidized compounds accept additional oxygen during electrodeposition, enabling reused DESs to leach LIB cathodes with efficiencies nearly identical to the original solvent [33].

To achieve optimal leaching efficiency, acidic DESs should be designed with low viscosity, strong coordination ability, and reducing capacity [25]. In contrast, neutral or basic DESs weaken the contribution of protons, making coordination and reducibility more critical for dissolution. For DESs composed of HBAs and HBDs without readily available protons (e.g., alcohols), the mechanisms are more complex and cannot be explained in the same manner as acid-based systems [15,116]. Ultimately, high-valence transition metal oxides are dissolved as coordination complexes of lower valence states, since low-valence oxides form more stable complexes with coordination ligands. Thus, reducibility plays a central role in enhancing leaching efficiency [15]. Interestingly, acidity does not always exhibit a direct quantitative relationship with leaching efficiency, especially when sufficient protons are already available or when the HBD provides strong complexation. Coordination ability enables the conversion of low-valence metal ions such as Co, Ni, or Mn into soluble forms. This ability is influenced by

both the HBA and HBD, as oxygen atoms in these components often serve as electron-donating ligands coordinating with metal cations [29]. For example, Tian et al. compared ChCl: lactic acid and GUC: lactic acid DESs in LCO leaching under identical conditions. GUC: lactic acid showed superior performance, attributed to stronger coordination of Cl⁻ in GUC, which stabilized delocalized π -bonds and increased the likelihood of coordination with Co(III). These differing coordination properties create opportunities for selective leaching [132]. However, precise details of how target metals are selectively extracted by HBAs or HBDs remain unclear. The dissociation of hydrogen-bond donors and acceptors clearly contributes to dissolution, and although fundamental mechanisms may be similar across systems, efficiency and selectivity vary depending on the DES composition [15,27]. Moreover, the solubility of metal oxides in DESs is linked to intrinsic characteristics such as stability, heats of fusion, melting points, metal charges, and ionic radii [39]. This suggests significant potential for improving efficiency through a deeper understanding of DES chemistry [33].

Beyond leaching, mechanistic studies have also addressed metal extraction using HDESs. For instance, Liu et al. (2024) demonstrated that cobalt extraction by HDESs proceeds via a cation-exchange mechanism, where HDES molecules act as ligands and directly coordinate with the metal ion. The overall reaction is (Eq. (5)):



where M^{n+} is the metal ion (Co²⁺), $HA \cdot B$ denotes the HDES molecule, n is the charge of the metal ion, and m is the number of HDES molecules participating in the coordination. As shown in Fig. 10(b), Co²⁺ ions migrate from the aqueous phase into the HDES phase, where they are stabilized by coordination with HDES molecules. During this process, protons (H⁺) are released, creating an acidic microenvironment. Competing ions such as Ni²⁺ and Li⁺ remain mainly in the aqueous phase, confirming the selectivity of the HDES system toward cobalt extraction [133].

3.5. Selectivity

One of the most compelling features of DESs in metallurgy is their ability to selectively dissolve metals from secondary sources such as e-waste. This selectivity can significantly simplify the production of high-purity metal salts by reducing the need for additional refining steps after leaching and extraction. Owing to their versatile combinations of HBAs and HBDs, DESs can demonstrate strong selectivity for specific metals of interest [29,34]. Transition metals in waste materials often share similar

physicochemical properties, which makes it difficult for conventional leaching agents, such as mineral acids, to selectively dissolve a target metal. This usually results in a liquid phase containing all metals and impurities together [34]. By contrast, the selective leaching and separation achieved by DESs using only their constituent components provide a streamlined approach that avoids additional operations or the use of precipitants [30]. Recent research has focused on optimizing the solubility of different metals in DESs to enhance recovery, often by precipitating the metals immediately after dissolution. This is typically achieved by combining an organic acid HBD with its conjugate base to form metal salts with low solubility. However, as noted earlier, this method has limitations when applied to real waste materials or when the aim is to produce precursors for the resynthesis of cathode materials [34]. Several studies have reported the selective recovery of cobalt, manganese, and lithium from spent LIBs using a ChCl: oxalic acid DES. In this system, cobalt and manganese are readily dissolved and can be subsequently precipitated as metal oxalates by adding water, depending on the conditions. For cobalt, $[\text{CoCl}_4]^{2-}$ complexes may form in chloride-rich environments or at elevated temperatures; however, precipitation as $\text{CoC}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$ still occurs when the system is left standing at room temperature or when an anti-solvent, such as water, is added. Lithium exhibits partial leaching, which improves with the addition of water, and can then be recovered through recrystallization [30,134,135]. Schiavi et al. proposed an alternative approach using a ChCl: EG DES, in which cobalt formed chloride complexes while nickel did not, thereby enabling their separation. This method not only facilitated efficient recovery but also offered potential for DES regeneration [136]. The inclusion of EG, with its lower viscosity and weaker coordination ability, prevented the dissolution of $\text{CoC}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$ while simultaneously reducing overall viscosity. In a related study, this approach also enabled the selective leaching of lithium and the efficient separation and recovery of transition metals [30]. The strong hydrogen-bonding properties of DESs further enhance their dissolution capabilities. Many compounds that are typically insoluble in water dissolve

readily in DESs, which is especially advantageous for dissolving metal oxides. This property supports selective metal extraction and recycling. For instance, aluminum and silicon oxides remain insoluble in DESs, providing an effective strategy for selectively extracting valuable metals from complex mixtures [59].

4. The process of separating metals using DESs

DESS are effective leaching agents and extractants due to their high solubility for metal oxides, versatile coordination abilities, and broad electrochemical windows. Beyond extraction, they also serve as efficient electrolytes in metal electrodeposition, with their components tunable for selectivity and efficiency in specific recovery applications [24,55]. Operating typically at lower temperatures than traditional leaching methods, DESS reduce energy consumption and operating costs [24]. In contrast, conventional organic solvents generally target either hydrophilic or lipophilic components, discarding the rest and contributing to resource waste and pollution. To overcome this, both hydrophilic and hydrophobic DESS have been developed to extract CRMs with varying polarities from secondary sources simultaneously. A notable innovation is the DES–DES two-phase system, combining hydrophilic and hydrophobic DESS to simultaneously extract metals of varying polarities from secondary sources. To date, only two studies have explored this concept: one for bioactive compounds and another for metals [137].

In this concept, Cao et al. (2018) first demonstrated a biphasic DES system using hydrophilic and hydrophobic DESs to extract bioactive compounds of different polarities from *Ginkgo biloba* leaves. Although not applied to metals, this study proved the feasibility of DES–DES biphasic extraction and phase separation in complex matrices [137]. Also, one example of hydrophilic DES–hydrophobic DES systems for metal recovery from secondary sources is shown in Fig. 11(a).

In 2025, Gholami et al. proposed a comprehensive flowsheet for the solvometallurgical separation and recovery of PGMs and REEs using DESs, building on previously established mechanisms and

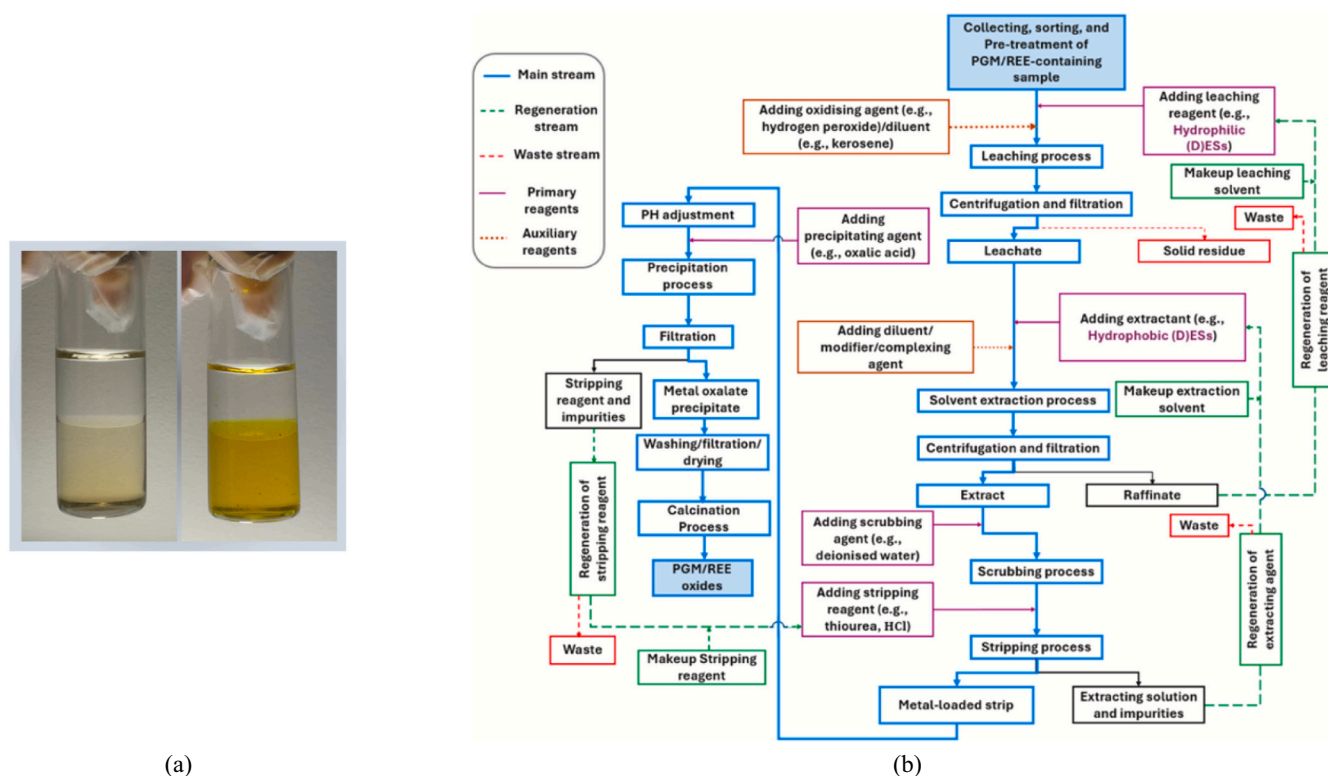


Fig. 11. (a) The two-phase DES system composed of a hydrophobic and a hydrophilic DES, shown before and after the extraction and centrifugation processes. (b) A conceptual flowsheet illustrating the overall solvometallurgical strategy for separating and recovering PGMs and REEs using DESs [37].

methodologies. As illustrated in Fig. 11(b), the proposed process involved an integrated leaching–extraction system that simultaneously utilized hydrophilic and hydrophobic DESs to recover valuable metals from secondary sources. Due to the complex composition of PGM- and REE-bearing EoL materials, the process began with pre-treatment steps designed to remove undesirable components and improve the leachability of target metals. This was followed by a three-step solvometallurgical approach. The first step involved leaching, where hydrophilic DESs acted as leaching agents, often enhanced by the addition of oxidizing or reducing agents, to selectively dissolve metal ions from the solid matrix. This step efficiently transferred target metals into the leachate. Beyond their role in leaching, DESs also exhibited strong performance in the purification and separation of metals from the leaching solution. In the second step, HDESs were used as extractants to selectively separate the target metal ions from the loaded leachate. This method, termed nonaqueous phase extraction, differed from conventional aqueous systems. To overcome the high viscosity of some HDESs and improve mass transfer, the use of environmentally benign diluents was often employed to enhance extraction efficiency. The third step was metal recovery, which involved a sequence of operations tailored to the specific composition of the extract phase and the desired purity of the final product. Initially, impurities were removed through scrubbing with agents like calcium chloride, water, or ethanol. This was followed by a stripping process using aqueous acidic or alkaline solutions, which served two purposes: (i) recovery of target metals from the DES phase and (ii) regeneration of the DES for reuse in subsequent cycles. While the aqueous solutions are partially mixed with hydrophilic DESs, careful process control was required to prevent degradation of the DES structure.

Depending on the stripping mechanism and DES type, replenishment of the DES components was sometimes necessary to maintain performance. For example, in anion exchange systems using quaternary ammonium chloride-based DESs, chloride ions may need to be replenished. Similarly, in lidocaine-based DESs, additional HBAs may be required to maintain system performance. The metal-rich strip solution is then subjected to precipitation, typically through the use of complexing agents such as oxalic acid, to form metal oxalates. These are subsequently calcined at high temperatures to produce high-purity PGM or REE oxides. Notably, recent findings suggest that the direct use of HDESs as lixiviants in nonaqueous leaching of PGMs and REEs can offer significant economic and environmental advantages, primarily by reducing the number of purification steps, particularly the need for solvent extraction [37]. Overall, integrated leaching–extraction systems combining hydrophilic and hydrophobic DESs represent a promising approach for efficient recovery of critical metals from diverse feedstocks, while minimizing impurity removal challenges and reducing environmental impacts.

4.1. Leaching

Leaching is a key process in extractive metallurgy that dissolves metals from ores or electronic waste using acidic or basic solutions, while leaving many impurities undissolved [3,4]. Hydrophilic DESs are particularly effective for dissolving metals from solid samples [39]. Once the pregnant leach solution (PLS) is obtained, techniques such as solvent extraction, ion exchange, or crystallization can be applied to purify and recover target metals [3]. Because the solubility of metal oxides varies significantly with DES composition, these solvents enable selective leaching. This selectivity enables the targeted extraction of specific metals, often circumventing the complex and costly separations required in fully aqueous systems [39].

Four main categories of DESs are commonly applied in leaching. Polyol-based DESs, typically composed of a polyhydric alcohol (e.g., EG) and a quaternary ammonium salt, exhibit low viscosity and good recyclability but require higher temperatures due to the weak reducing power of alcohols [31,32]. Organic acid-based DESs offer strong acidity

and reducing ability, enabling high efficiency under mild conditions; however, their high viscosity limits mass transfer and selective separation [31,32]. Polyol/acid-based DESs combine the low viscosity of polyols with the reducing capacity of organic acids; however, side reactions such as esterification can cause solvent loss, slow kinetics, and reduced throughput [31,32]. Amide-based DESs, utilizing urea or acetamide as HBDs, are inexpensive and moderately reducing, but suffer from poor thermal stability, the possible release of hazardous gases at elevated temperatures (413–443 K), and high hygroscopicity [31].

External assistance plays a crucial role in enhancing the leaching efficiency of all types of DESs. The incorporation of reducing agents or the application of energy-efficient technologies, particularly microwave and ultrasound irradiation, has reduced reaction times from several hundred hours to just minutes, while enabling operation under milder conditions. These methods not only conserve energy and lower operational costs but also significantly improve leaching kinetics and overall metal dissolution efficiency. Consequently, integrating such external enhancements into DES-based processes is increasingly regarded as a key strategy for developing sustainable and cost-effective metal recovery systems [32].

Extensive research has focused on achieving high leaching efficiency at minimal cost while maintaining environmental compatibility [32]. For DESs, this goal requires precise adjustment of several variables, including the HBA: HBD molar ratio, water content, temperature, contact time, and solid-to-liquid (S/L) ratio [29]. The HBA: HBD ratio is a critical factor in leaching efficiency. Ratios that are too low or too high can suppress metal dissolution by disrupting the balance between acidity, viscosity, and coordination ability. In the case of LCO leaching, the extraction efficiencies for Co and Li remained stable when the HBA: HBD (EG: sulfosalicylic acid dihydrate) ratio varied between 12:1 and 10:1. At these ratios, the DES still contained sufficient HBD content to provide the necessary protons and coordinating functional groups (carboxyl and sulfonic acid) that promote the decomposition and leaching of LiCoO_2 . However, when the HBA: HBD ratio was further increased to 16:1, the leaching efficiency decreased markedly due to the lower fraction of the acidic HBD component, which reduced the available protons and consequently hindered LCO decomposition. However, adjusting the ratio is only feasible within the range where the DES remains in a stable liquid phase, as extreme compositions may lead to crystallization or phase separation. In both hydrophilic and hydrophobic DESs, a small amount of water reduces viscosity and enhances mass transfer. However, excessive water competes with DES molecules for interaction sites, diminishing their affinity for metal ions and consequently lowering extraction yields [29]. Temperature and time also significantly influence the leaching process. Elevating the temperature promotes metal leaching by lowering the solvent's viscosity and accelerating reaction kinetics. Leaching proceeds rapidly at first, then approaches a plateau. Therefore, researchers typically select a contact time just before this limit to balance recovery and energy use. Beyond that point, longer or hotter runs consume more energy without proportionate gains and may even dissolve impurities or volatilize the solvent. The S/L ratio is also a critical parameter, with values typically ranging from 1/10 to 1/100 according to the literature. When too much solid is present, the slurry becomes viscous, hindering mixing and surface contact. However, an overabundance of DES wastes the reagent, complicating downstream separation. Leaching generally improves as the S/L ratio is reduced until an optimal threshold is reached. Particle size plays a crucial role, as smaller particles provide a greater surface area and improved solvent accessibility, thereby enhancing metal extraction. However, excessive grinding beyond the optimal range can lead to particle aggregation and processing challenges, ultimately reducing recovery efficiency [29,30,36]. For instance, increasing the particle size from 0.25 to 0.5 mm to over 0.5 mm led to a decrease in the Cu leaching rate, dropping from 73.5 % to 63.5 %. In contrast, the highest Cu yield of 94.3 % was achieved with particle sizes in the range of 0.100–0.250 mm [36].

4.2. Extraction

In conventional LLE, the extractant typically acts as a ligand that selectively binds to the target metal ions, enabling their transfer from the aqueous to the organic phase. However, in HDES-based extraction, the use of additional ligands may not be necessary. This is because HDESs can be designed to possess functional groups with a natural affinity for metal ions, allowing them to serve both as the extraction medium and the active extractant. This bifunctionality simplifies the process and reduces the need for toxic or complex ligands, offering a significant advantage over conventional LLE systems [3,47]. The first demonstration of this concept showed that indium can be efficiently extracted from hydrochloric or oxalic-acid media into a hydrophobic DES composed of a quaternary-ammonium salt and DL-menthol [3]. Some LLE studies have used real leachates or PLS, whereas others have relied on model or synthetic solutions [3]. Not every hydrophobic DES is well-suited to solvent extraction. Lengthening the alkyl chains on either the quaternary ammonium/phosphonium salt (HBA) or the HBD raises the solvent's melting points and lowers its density. On an industrial scale, this implies the need for additional heating to maintain HDES fluidity and potential challenges in phase separation when its density closely approaches that of water, which can prolong settling time and increase operating costs. If the HDES density is significantly lower than water, it may form the upper phase, which also requires careful design of the separation process. In comparative studies, HDESs built from alkanolic acids consistently outperformed those based on thymol: the acids are more polar and stronger HBAs, so they interact more effectively with chlorophenols or metal complexes and deliver higher extraction yield [47]. Compared with conventional LLE, liquid-liquid microextraction (LLME) uses only microliters of solvent, offering higher enrichment factors with reduced sample and solvent consumption. Hydrophobic DESs have increasingly replaced traditional organic solvents in LLME protocols. Dispersive LLME (DLLME) disperses the extractant into fine

droplets with a disperser solvent, rapidly reaching equilibrium. Variants include DLLME-SF (solidified floating droplets), EADLLME (CO₂-assisted dispersion), TADLLME (heat-enhanced dispersion), and UADLLME (ultrasound-assisted, no disperser needed). Vortex-assisted (VALLME) and shaker-assisted (SALLME) methods use physical agitation to disperse solvents, while air-assisted (AALLME) employs air bubbles for efficient droplet formation. Emulsification via LLME (ELLME) utilizes emulsifiers for dispersion, although toxicity is a limitation. Reactive LLE (RLLE) introduces interfacial reactions between solutes and extractants, enabling efficient recovery of carboxylic acids [47–49,53].

HDESs, such as DL-menthol and lauric acid, have demonstrated strong potential as green solvents for metal extraction. In hydrochloric acid media, that solvent showed effective extraction of Tl(III), moderate extraction of Tl(I), and weaker performance for In(III), with selectivity following the order Tl(III) > Tl(I) > In(III). Incorporating bis(2-ethylhexyl)phosphoric acid (HDEHP) into HDESs significantly enhanced the extraction yields of In(III) and Tl(I), particularly under low acidity, while Tl(III) extraction remained largely unaffected. Comparative studies further revealed that kerosene-based systems markedly improved In(III) extraction under low acidity, whereas at higher acid concentrations, eutectic-based systems retained superior efficiency. These results highlight the tunability of eutectic solvents and their potential advantages over conventional solvents for selective metal recovery [138].

The efficiency of metal extraction from solutions is significantly influenced by the nature and structure of the HBA and HBD, with factors such as competing anions and pH levels playing important roles. Despite their effectiveness in LLE, DESs present challenges such as high viscosity, and cation exchange can lead to contamination of the aqueous phase. The pH level affects the formation of metal complexes and precipitation, which in turn impacts the removal efficiency of target contaminants. For instance, the extraction of Cu(II) decreases as the alkyl chain length of carboxylic acids increases. The ratio of HBA to HBD plays

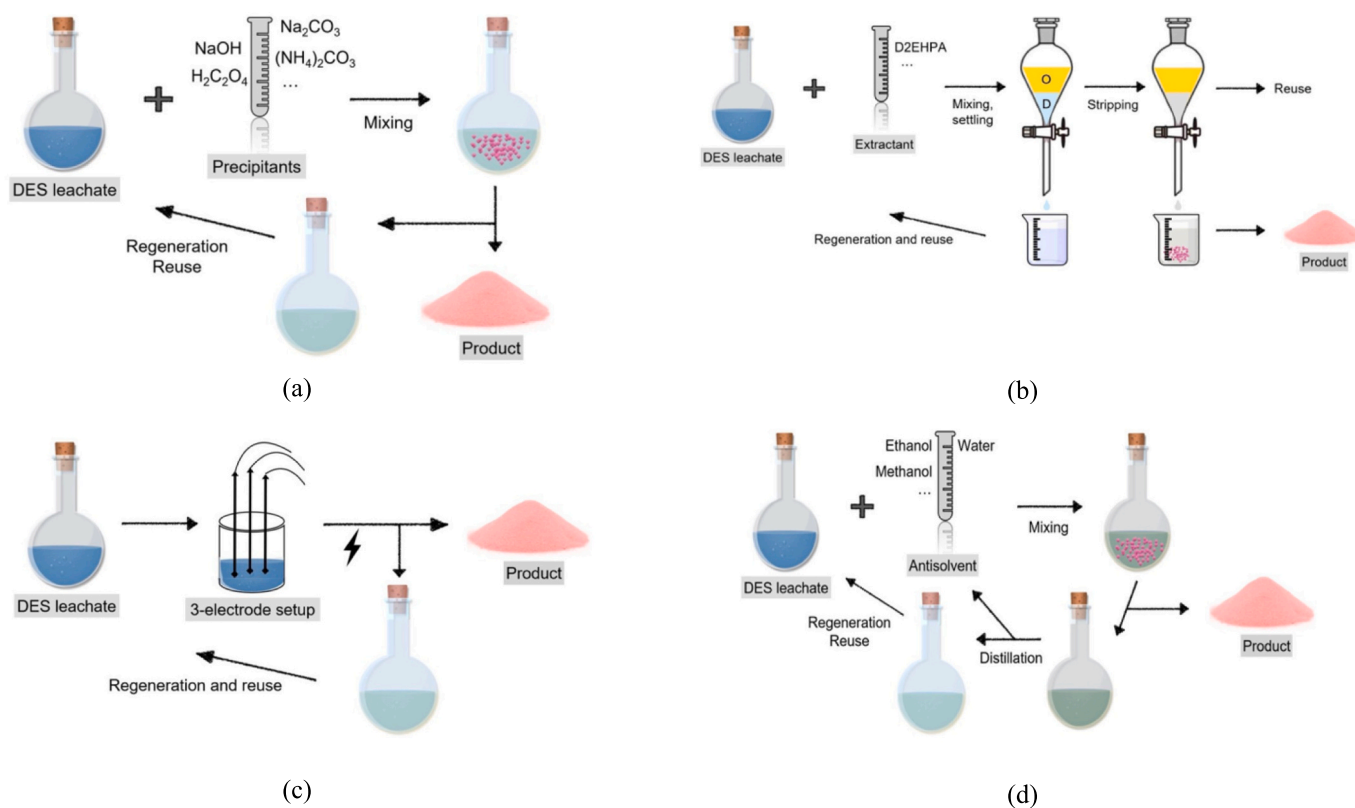


Fig. 12. Schematic of lab scale, (a) Chemical precipitation process, (b) Solvent extraction, (c) Electrodeposition, (d) Antisolvent crystallization-based metal recovery processes [34].

a key role in metal extraction. Optimizing factors such as solvent composition, sample volume, and extraction time is essential for maximizing analyte recovery and enhancing solvent efficiency [44,49].

4.3. Metals recovery processes

After the formation of leachate or the extraction of metal ions, an essential step is the recovery of the elements. An integrated process involving DES extraction followed by precipitation or leaching and subsequent precipitation has been widely used for metal recovery, including metals such as Co, In, and Sn [5,24]. In addition to efficiency and the final metal product, the choice of separation technique is crucial for the potential to regenerate the DES, allowing it to retain its solvent properties and be reused in multiple leaching cycles [34]. To enable the recycling of DESs while separating valuable metals, researchers have proposed methods such as electrodeposition, adding non-destructive components, utilizing the coordination of DES components, precipitation, and adsorption [30]. However, chemical precipitation, antisolvent crystallization, and solvent extraction are commonly used for metal recovery from leachates in solvometallurgy processes [34]. Dissolved metals can be selectively precipitated and separated, with further purification techniques, such as recrystallization or calcination, applied afterward [24]. A notable gap remains in studies focusing on the recovery and extraction of valuable metals from leachates after dissolution [30]. Common precipitation agents include NaOH, Na_2CO_3 , and $\text{H}_2\text{C}_2\text{O}_4$, with $\text{H}_2\text{C}_2\text{O}_4$ proving more effective due to its high selectivity for complexation and precipitation, better crystallinity of the final product, absence of impurities, and easier regeneration of the DES [29,33]. In general, metal ion precipitation occurs by adding precipitants that introduce ions to form insoluble compounds with metals [33]. When selecting a precipitator, the target metal must be taken into consideration. In chemical precipitation (shown schematically in Fig. 12(a)), supersaturation necessary for solid formation is achieved through a chemical reaction, either by the separate addition of a reagent or by in situ precipitation during leaching, as observed in a few DES leachate systems [34]. It is essential to note that chemical precipitation frequently results in the loss of HBDs and HBAs in DESs [24]. Because transition metal oxalates, such as cobalt oxalate, exhibit low solubility, oxalic acid, in either solid form or as a saturated solution, is commonly employed as a precipitant for metal separation from DES leachates [34]. When oxalic acid is used as the HBD in DES, it facilitates precipitation without the need for additional agents by cooling during the separation process. This is because metal oxalates have low solubility at lower temperatures, allowing for the selective separation of transition metals from lithium without the need for Na_2CO_3 or pH adjustments [29,126]. However, Li tends to accumulate after reuse, requiring its separate removal by adding an H_3PO_4 solution to form a phosphate phase [139]. The addition of sodium-based precipitants, such as NaOH, can lead to the buildup of sodium in the DES, which, unless removed in a separate step, reduces the dissolving power of the DES and contaminates the metal product in subsequent cycles. Although NaOH is an effective precipitation agent, removing sodium ions from the products is challenging [34]. Solvent extraction, or LLE, is a separation technique used to recover metals from DESs. This process, illustrated in Fig. 12(b), involves isolating components of a mixture based on their solubility in two immiscible liquids. For example, Cu(I/II) can be selectively extracted using LIX 984, and Co(II) can be extracted using Aliquat 336 [34].

Electrodeposition, also known as electrowinning (Fig. 12(c)), involves either electro-oxidation recovery at the anode, where metal anions lose electrons, or electro-reduction recovery at the cathode, where metal ions accept electrons and deposit as metallic elements. Electrodeposition within a DES system enables selective metal separation by exploiting its wide potential windows and the abundance of chloride ions, which can form complexes with metal ions [34]. Although the technique is relatively energy-intensive, it uniquely allows the DES to be regenerated and reused in successive extraction cycles. Compared with

aqueous electrolytes, DESs offer broader electrochemical windows and essentially zero vapor pressure [34]. Electrochemical deposition is more efficient than precipitation methods, providing higher yields and faster recovery rates. Furthermore, electrochemical deposition from DES-based electrolytes has a lower environmental impact than precipitation methods, as it can be conducted under milder conditions and with fewer chemical additives, utilizing eco-friendly systems. However, detailed characterization of DES before and after regeneration, as well as the electrochemical formation of oxidation and reduction products related to DES components, has not been thoroughly reported. These degradation products can hinder the reusability of DES and negatively affect both leaching and extraction processes [24,34].

Antisolvent crystallization or salting-out (Fig. 12(d)) offers a new way to recover metals from DES leachates. A second, fully miscible liquid in which the metal salt has poor solubility is introduced; the sudden drop in solubility makes the salt crystallize out. Only a handful of papers, all quite recent, have explored this approach in DES systems. Water is the most common antisolvent with oxalic acid DESs because it both thins the viscous medium and triggers precipitation. Ethanol and other short-chain alcohols have also been tested. Since these antisolvents are far more volatile than the DES, they can later be removed by gentle distillation, ideally at low temperatures to avoid degrading the solvent [34,64].

4.4. Reuse of DESs

The recovery and reuse of DESs in metal recycling processes offer significant environmental and economic benefits. Environmentally, recycling DESs reduces hazardous waste and lowers emissions from new solvent production. It also conserves natural resources and supports process efficiency by ensuring consistent solvent availability. Economically, recycling cuts costs by reducing the need to purchase new solvents and improves profit margins. Although DESs can be recovered through methods such as electrodeposition, the complexity of some recovery steps may pose challenges to their industrial-scale adoption [5,24]. Most DESs cannot be cycled back into service after metal recovery by precipitation. Combining extraction with precipitation requires dilution, pH adjustment, redox steps, and repeated additions of precipitants; these operations can dilute, co-precipitate, or chemically alter the DES components, potentially breaking down the eutectic structure [5]. Any process that introduces extra reagents inevitably leaves residues in the solvent, thereby shortening its useful life. In chloride-rich DESs, for example, chlorine can form trichloride anions, and electrochemical work has revealed volatile by-products such as chloromethane, dichloromethane, and chloroform compounds that raise serious environmental, health, and safety concerns. Further side reactions at both electrodes may generate alcohols, ethers, cyclic ethers, and amines, all of which compromise the solvent's integrity and recyclability [34].

Decomposition products not only erode the "green" character of a DES but can also hamper metal dissolution. Avoiding degradation is therefore critical to enable reuse of the DES across multiple leaching cycles. The applied temperature during DES synthesis can induce thermal degradation through undesired reactions, resulting in the formation of side products. For instance, microwave-assisted leaching can allow for low reaction times. Since regeneration steps involving extra reagents introduce new impurities, the best strategy is to minimize conditions that break down the eutectic. Maintaining a stable proton activity (pH) is especially important, as organic acids consumed during leaching can be replenished, but the accompanying formation of water is more troublesome. Once the water content exceeds approximately 40 wt%, the hydrogen-bond network collapses, and the mixture behaves like an ordinary aqueous solution rather than a DES. Calculations show that, if proton depletion is ignored, ten successive leaching cycles could exhaust all available protons while raising the water content by roughly 10 wt%, already eroding the solvent's unique properties [34]. To keep a DES in

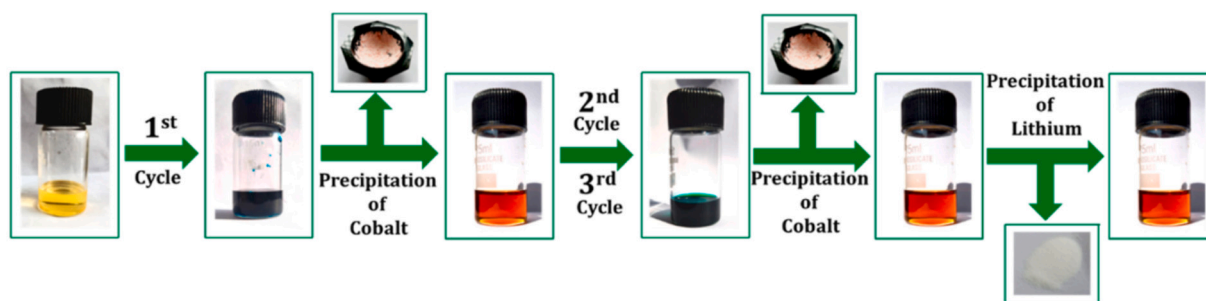


Fig. 13. Scheme of multistep extraction experiments (three cycles) to assess the ChCl: pyruvic acid DES reusability [125].

service for many leaching cycles, two things are essential: (i) the proton activity must be restored, and (ii) the water generated during leaching must be removed. Protons can be replenished by adding fresh HBD, and water can be driven off by evaporation. Both remedies, however, are costly. Organic acids, which are commonly used as HBDs in DESs, are more expensive than mineral acids. Moreover, adding a mineral acid would undermine the key benefit of using a DES as a greener alternative to conventional acids. Likewise, evaporating water demands substantial energy, and the required heating risks degrading the solvent, consuming its components and generating troublesome by-products. Additional losses occur when the HBD serves as a precipitant [34]. Despite these challenges, the ability to recover and reuse DESs greatly simplifies downstream leachate treatment by eliminating steps such as neutralizing acidic waste or removing ammonia, while also reducing reagent consumption and overall process costs. Although many DES formulations cost more upfront than standard mineral or organic acids, their longevity, smaller environmental footprint, and process efficiency gains can more than repay the initial outlay. A comprehensive life-cycle economic assessment could therefore support DESs as a greener and ultimately more cost-effective option for metal recycling. That said, systematic studies on DES recyclability are still sparse, and further work is needed to optimize regeneration strategies and quantify long-term performance [15,24].

Fig. 13 outlines the DES-based extraction sequence. In the first pass,

cobalt oxalate (pink CoC_2O_4) was precipitated in 99.1 % yield, while lithium remained in the liquor. After the water was removed under vacuum at 50 °C, the lithium-rich DES was reused. The second leach still yielded excellent recoveries, approximately 97 % for Co and 96 % for Li, indicating that the solvent can be recycled at least once without noticeable deterioration. In the third run, however, efficiencies fell sharply (68 % Co and 62 % Li). Analysis indicated that solvent breakdown, primarily oxidation and evaporation of propionic acid, as well as side reactions such as ester formation, occurred. Even so, lithium could still be isolated as LiHC_2O_4 from ethanol in 98.6 % yield, and both final products were of high purity (≈ 99.6 % Co and 99.1 % Li). In short, the DES retains its performance for two cycles, but beyond that, its components must be replenished or stabilized to maintain high extraction levels [125].

Despite the challenges associated with DES degradation during leaching, ongoing research and technological advances are steadily improving their prospects for use in circular economy approaches to metal recovery. With the development of effective strategies for stabilizing and regenerating DESs, their practical application in sustainable metal recycling is becoming increasingly feasible.

5. Current applications of DES for metal recovery

Each metal is utilized in varying concentrations across different

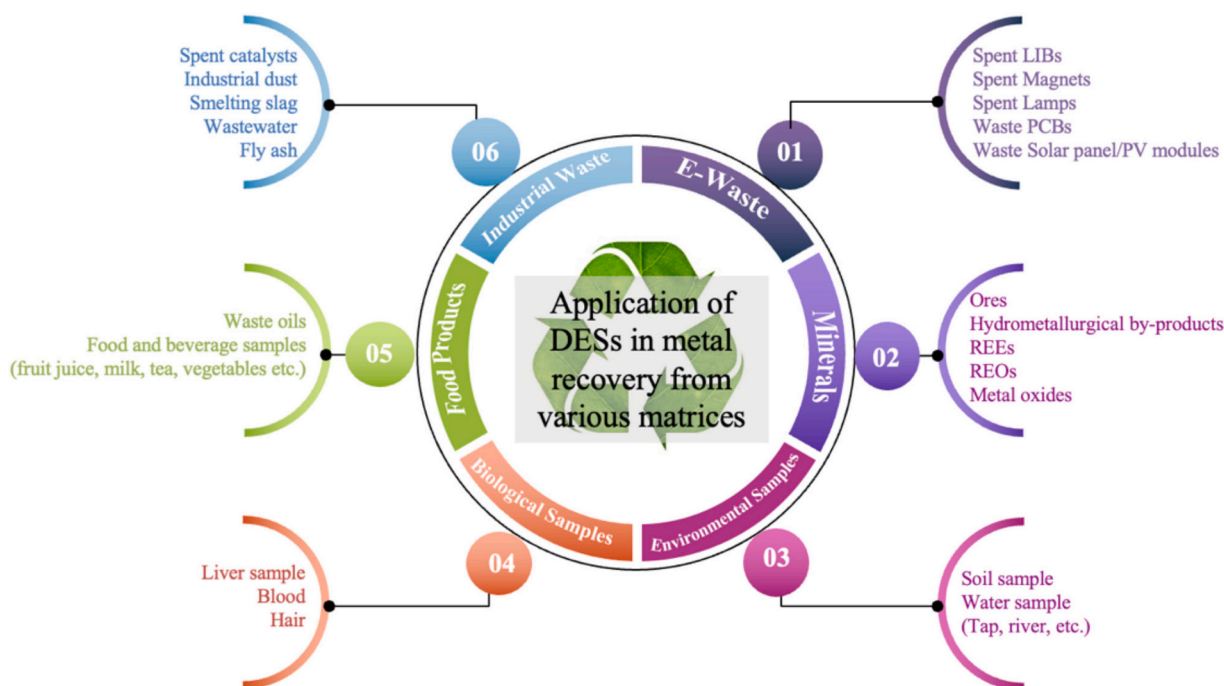


Fig. 14. Classification of various sources used in the metal recovery applications of DESs.

applications, and these concentrations are anticipated to change over time. The increasing release of metal-related waste not only raises concerns about human health and the environment but also accelerates the depletion of natural metal reserves [2]. To date, DESs have been explored in numerous metal extraction applications. This review highlights their potential as sustainable alternatives for metal separation from various sources. It also provides an overview of recent literature from 2024 to 2025, focusing on the use of DESs in metal separation, detection, and recovery processes. Fig. 14 presents a classification of the reported sources for DES-based metal recovery, which will be discussed in detail in the following sections.

5.1. Electric and electronic waste

E-waste refers to discarded or non-functional electronic devices that are no longer in use. It includes the entire equipment or components of electrical or electronic devices that users discard, as well as the by-products from the manufacturing, repair, and refurbishment processes. A report indicated that 42 % of e-waste comes from household appliances, 34 % from communication devices, 14 % from electronic gadgets, and 10 % from accessories [36]. Metals present in e-waste can be categorized into several key groups based on their chemical properties and economic significance: precious metals (e.g., Ag, Au, Pt, Pd), which constitute approximately 0.1–1 % by weight; base metals (e.g., Cu, Fe, Al), making up around 30 %; REEs (e.g., La, Dy, Gd, Pr, Nd, Sc, Y), typically present in smaller quantities; and critical and strategic metals (e.g., Co, Li, In, Sb), which, although often found in lower concentrations, are essential for modern technologies and possess high strategic value. This complex and valuable composition underscores the importance of e-waste as a secondary resource for sustainable metal recovery [39]. In 2022, global electronic waste generation reached 7.8 kg per capita and is expected to continue increasing due to shorter product lifecycles and higher consumption rates of electronic devices. Recovering metals from e-waste is crucial for both waste management and the retrieval of precious metals. Metal recovery from e-waste can significantly benefit national economies and meet the growing demand for metals used in the production of electronic and electrical equipment. For instance, from one ton of e-waste, 128 kg of Cu, 6 kg of Pb, 0.347 kg of Au, 15 kg of Ni, 0.15 kg of Pd, 3.63 kg of Ag, 1 kg of Sb, and 10 kg of Sn can be extracted [39]. E-waste contains metals that are 40–50 times more concentrated than natural ores, making it a more efficient source compared to traditional mining [25]. Additionally, extracting one ton of gold from natural mines releases approximately 17,000 tons of CO₂, while the extraction of palladium and platinum emits between 10,000 and 14,000 tons of CO₂. Therefore, recovering metals from e-waste can significantly reduce CO₂ emissions and mitigate environmental hazards [2,39].

The process of metal recovery from e-waste can be divided into three main stages: pre-treatment of raw materials, leaching and separation of metal ions, and refining the products into desired forms, such as pure metals or pure metal compounds. This refining stage can be achieved through methods like precipitation, solvent extraction, and electro-deposition [5]. Using DESs for metal recovery from e-waste has emerged as a promising approach, drawing significant attention [41]. This section focuses on the role of DESs in recovering valuable resources from various types of e-waste, including waste LIBs, PCBs, and FLs, as well as EoL permanent magnets and PV cells.

5.1.1. LIBs

The market for LIBs has rapidly expanded due to the widespread adoption of electric vehicles, portable electronics, renewable energy storage systems, and various other mobile devices [55]. This growth in LIB use has raised significant concerns about their proper disposal and the environmental impact they cause [2]. Notably, various funding sources have supported research initiatives focused on battery recycling, with a substantial portion directed toward projects, startups, and

industrial efforts aimed at improving LIB recycling [24]. While the scale of LIB recycling has grown, it still falls behind the rate at which batteries are being consumed. This growing demand for energy resources puts substantial pressure on raw materials, particularly metals like lithium, cobalt, and nickel, whose prices have surged. Without efficient recycling processes, the large-scale production of LIBs will deplete resources and generate vast amounts of waste. It is estimated that by 2030, over 11 million tons of spent LIBs will be retired, with only about 5 % of them being recycled. Therefore, developing efficient and environmentally friendly recycling technologies is crucial [33].

The lithium content in spent LIBs is approximately 10 times higher than that in primary lithium ore (lithium pyroxene) and 30 times higher than the lithium carbonate salts found in brine pools [25]. Among the various components of LIBs, cathode materials account for approximately 30–50 % of the total battery cost, making them the most valuable part of the battery. Currently, LIB cathodes are primarily composed of layered oxides containing lithium and one or more transition metals. The seven most common cathode materials are lithium iron phosphate (LFP), LCO, lithium manganese oxide (LMO), lithium nickel oxide (LNO), lithium nickel cobalt aluminate (NCA), lithium titanate oxide (LTO), and lithium nickel manganese cobalt oxide (NMC). Each of these materials has its unique strengths and weaknesses, and their selection depends on the specific application at hand. Given the non-renewable nature of these precious metals and the environmental pollution caused by heavy metals, there is an urgent need for greener methods to turn LIB waste into valuable resources [33]. Recent advancements in recycling processes, particularly solvometallurgical methods using eco-friendly DESs, show promise for the efficient recovery of materials from spent LIBs [140]. Table 4 summarizes recent studies that examine the use of DESs in the recycling of LIB cathodes, including details on the compositions of DESs and experimental conditions. It also covers the reuse of DESs in subsequent dissolution experiments after metal recovery and regeneration. This review highlights the latest advancements in applying DESs to recover CRMs from spent LIBs.

In 2025, Wu et al. proposed a water-DES solvent composed of ChCl and lactic acid for extracting Co and Li from spent LCO cathodes. The aqueous leaching system, using a 1: 3 molar ratio of ChCl to lactic acid and diluted with 90 vol% water, achieved high leaching efficiencies of 97.15 % for Li and 95.20 % for Co under optimal conditions. The high water content helps mitigate the viscosity challenges of traditional DESs, thereby improving mass transfer and ion diffusion. After leaching, cobalt and lithium were recovered as CoC₂O₄·2H₂O and Li₂CO₃, which were used to regenerate LCO. The regenerated LCO demonstrated excellent electrochemical performance, with an initial discharge capacity of 136.0 mAh/g at 0.5C and 87.43 % capacity retention after 100 cycles, matching the performance of commercial LCO, proving the effectiveness of this closed-loop recycling method [141]. In 2025, Liu developed a green and cost-effective method for recovering Li and Co from waste LIBs using a weakly acidic and reductive DES made from lactic acid and glycolic acid. Under optimal conditions (6 h, 100 °C, L/S = 10, molar ratio = 1: 1), over 99.5 % of Li was efficiently leached, driven primarily by interfacial chemical reaction rates. Cobalt spontaneously precipitated without the need for additional agents due to the weak reducibility and coordination effects of the DES. The leaching mechanism of DESs on LiCoO₂ was investigated using UV-vis, FT-IR, ¹H NMR, CV, XPS, and molecular orbital theory. Results showed that the weak acidity of DES promoted Li⁺ release, while its weak reducibility converted Co(III) to Co (II), which then formed precipitates with coordinating molecules in DES. Thermodynamic analysis revealed that the low solubility of Co complexes in DES enabled selective leaching. The DES achieved a maximum Li loading of 5.18 mg/L after cyclic leaching, demonstrating a green and efficient method for recovering valuable metals from spent LiCoO₂ [142]. Biniarz et al. (2025) reported a green route for recovering critical metals from spent LIBs with a ChCl: maleic acid DES. After nitrogen roasting of the black mass at 800 °C (which used residual carbon to reduce the metals), the cathode powder was leached and then simply

Table 4

List of prepared DESs along with their efficiency for recovering metals from LIBs.

No.	DESs	Molar ratio	Cathode type	Metals	Efficiency %	Optimum condition	Metal recovery process from leachate	Cyclic process	Refs.
Hydrophilic DESs (Leaching)									
1	(ChCl: Citric Acid) + 35 % water	2: 1	LCO	Co Al Cu Li	98 38 94 93	$t = 60$ min $T = 40$ °C S/L ratio: 1/ 20 g/mL $T = 180$ °C	Solvent Extraction: LIX 984, Aliquat 336 Stripping Agent: Oxalic Acid	–	[149]
2	ChCl: EG	1: 2	LCO NMC	Co Li	99.3 99.3	$t = 24$ h S/L ratio: 20/ 5 mg/g $T = 180$ °C	Precipitation: Na_2CO_3 Electrodeposition	2 cycles	[150]
3	ChCl: Urea	1: 2	LCO	Co Li	98 95	$t = 12$ h S/L ratio: 1/ 50 g/mL $T = 160$ °C	$\text{H}_2\text{C}_2\text{O}_4$ and NaOH Dilution, Precipitation, Calcination Process	–	[151]
4	PEG-200: Thiourea	2: 1	LCO	Co	60.2	$t = 24$ h S/L ratio: 1/ 50 g/mL $T = 90$ °C	Recovery of Co by Electrodepositing or Extraction	–	[152]
5	PTSA: $2\text{H}_2\text{O}$: ChCl	1: 2: 1	LCO	Co Li	100 100	$t = 15$ min S/L ratio: 60 g/L	Precipitation: Na_2CO_3 (NH_4) $_2\text{CO}_3$ Calcination	–	[153]
6	ChCl: EG	1: 2	Mixed Cathode of Spent LIBs	Ni Co	10 90	$T = 180$ °C $t = 20$ h	Precipitation: Oxalic Acid Solvent Extraction: Diluting D_2EHPA in kerosene	DES can be reused after a partial make-up	[136]
7	PEG-200: Phytic Acid	1: 1	LCO	Co Li	98 96	$T = 80$ °C $t = 24$ h S/L ratio: 1/ 50 g/mL $T = 80$ °C	–	–	[154]
8	PEG-200: Ascorbic Acid (AA)	6: 1	LCO	Co	84. 2	$t = 72$ h S/L ratio: 1/ 100 g/mL NMC $T = 70$ °C	–	PEG: AA cannot be recycled due to the oxidized AA in DESs	[155]
9	ChCl: Tartaric Acid	1: 1	LCO and NMC	NMC Co Ni Mn Li LCO Co Li	97 98 96 96 100 100	$t = 12$ h S/L ratio: 1/ 50 g/mL LCO $T = 80$ °C $t = 24$ h S/L ratio: 1/ 50 g/mL	Antisolvent Crystallization using Ethanol (Co,Ni,Mn)	2 cycles	[116]
10	ChCl: Lactic Acid	2: 1	LMO, LNMO and LNCO	Mn Co Ni Li Co	> 85	$T = 105$ °C $t = 5$ h	Precipitation: Oxalic Acid (for Co,Ni) and H_3PO_4 (for Li)	4 cycles	[139]
11	Betaine Hydrochloride: EG	1: 5	LNCM	Ni Mn Li Li	99 91	$T = 140$ °C $t = 10$ min S/L ratio: 1/ 40 g/mL $T = 180$ °C	–	–	[156]
12	ChCl: EG	1: 2	LNCM	Co Ni Mn	92 94 95	$t = 24$ h S/L ratio: 1/ 80 g/mL $T = 90$ °C	Solvent Extraction: P507 Stripping Agent: H_2SO_4	3 cycles	[17]
13	ChCl: Formic Acid	2: 1	LCO and LMO	Li Co Mn	>99	$t = 12$ h S/L ratio: 1/ 50 g/mL $T = 90$ °C	Precipitation: Saturated Na_2CO_3	6 cycles	[157]
14	ChCl: Oxalic Acid	1: 1	LCO	Li Co	96 96	$t = 2$ h S/L ratio: 1/ 50 g/mL $T = 200$ °C	Cooling Down (low solubility of metal oxalates at low temperatures)	5 cycles (by adding $\text{H}_2\text{C}_2\text{O}_4$)	[126]
15	ChCl: Glycerol	1: 1.5	LCO	Co	96	$t = 10$ h S/L ratio: 1/ 50 g/mL $T = 100$ °C	Precipitation: Na_2CO_3	–	[158]
16	PEG: PTSA	4: 1 1: 1	LCO	Li Co	>99	$t = 24$ h S/L ratio: 1/ 50 g/mL $T = 140$ °C	–	–	[159]
17	ChCl: Succinic Acid: EG	1: 1: 1	LCO	Co	99.62	$t = 16$ h	Precipitation: Sodium Oxalate	–	[127]

(continued on next page)

Table 4 (continued)

No.	DESS	Molar ratio	Cathode type	Metals	Efficiency %	Optimum condition	Metal recovery process from leachate	Cyclic process	Refs.
18	EG: Citric Acid	2.5: 1	LNCM	Li Co Ni Mn	99 96 97 98	S/L ratio: 1/ 300 g/mL T = 95 °C t = 10 h S/L ratio: 15 g/L	–	–	[128]
19	EG: Sulfosalicylic Acid Dihydrate	12: 1	Pure LCO and Spent LNCM	LNCM Co Ni Mn Li LCO Co Li	94 99 100 100 93 98	T = 110 °C t = 6 h S/L ratio: 1/ 40 g/mL	–	–	[97]
20	ChCl: Oxalic Acid Dihydrate	1: 1	NCM	Ni Co Mn	99 95 94	T = 120 °C t = 10 h S/L ratio: 1/ 50 g/mL	Precipitation: DMSO, Water	–	[114]
21	ChCl: L-Ascorbic Acid: H ₂ O	2: 1: 6	NCM	Li Mn Co Ni Mn	88 89 88 85 99.2	T = 30 °C t = 1 h S/L ratio: 1/ 25 g/mL T = 80 °C	Precipitation: Oxalic Acid	–	[160]
22	Betaine HydroChloride (BeCl): Citric Acid	1: 3	LNCM	Ni Li Co	99.1 99.8 98.8	t = 30 min S/L ratio: 1/ 50 g/mL T = 100 °C	–	–	[161]
23	PTSA: ChCl Malonic Acid: ChCl PTSA: Malonic Acid: ChCl	–	LCO	Co Li	98.61 98.78	t = 24 h S/L ratio: 1/ 50 g/mL T = 90 °C	Precipitation: Na ₂ CO ₃	–	[111]
24	EG: Oxalic Acid Dihydrate	5: 1	LCO and LNCM	Li (Co Mn Ni) LCO Li Co	94 < 1.2 > 99.1 98	T = 90 °C t = 12 h S/L ratio: 16 g/L T = 120 °C t = 150 min S/L ratio: 20 g/L	–	3 cycles	[92]
25	Urea: Caprolactam: Formic acid	1: 3: 2.4	LCO and NMC	NMC Li Ni Mn Co	93.4 87.8 89.6 90.1	T = 120 °C t = 150 min S/L ratio: 20 g/L	Precipitation: Oxalic Acid, EtOH	> 20 cycles	[129]
26	ChCl: EG: Tartaric Acid	2: 4: 0.2	NMC	Li Ni Co	99.9 95.6 98.1	T = 120 °C t = 10–600 min S/L ratio: 1/ 200 g/mL T = 150 °C	Electrodeposition	3 cycles	[162]
27	Maleic Acid: EG	1: 4	LCO	Co Li	98.3 98.4	t = 8 h S/L ratio: 60 g/L	Precipitation: Oxalic Acid	4 cycles	[163]
28	DMPT: EG	1: 3	NMC	Li Ni Co Mn	99.59 99.28 99.04 99.45	T = 100 °C t = 60 min	–	–	[130]
29	ChCl: BSA: Ethanol	1: 1: 2	LCO	Li Co	99 98	T = 90 °C t = 2 h S/L ratio: 20 g/L T = 90 °C t = 2 h	Precipitation: H ₂ C ₂ O ₄ Calcination: NaOH of Na ₂ CO ₃	–	[131]
30	ChCl: Oxalic Acid	1: 1	LCO	Li Co	nearly 100	S/L ratio: 0.02/3 g/mL (Ultrafast closed loop- T = 180 °C t = 10 s) T = 90 °C	Precipitation: Adding/ Evaporating Deionized Water	Can be reused	[164]
31	EG: Malonic Acid	3: 1	NMC	Li Ni Co Mn	95.63 84.63 92.74 91.31	t = 6 h S/L ratio: 30 g/L T = 100 °C	Precipitation: Oxalic Acid Sodium Phosphate	–	[165]
32	TetraButylammonium Chloride: Chloroacetic Acid	1: 3	LCO	Li Co	100	t = 7 h	Precipitation: Oxalic Acid Ultrapure water	2 cycles	[166]

(continued on next page)

Table 4 (continued)

No.	DESS	Molar ratio	Cathode type	Metals	Efficiency %	Optimum condition	Metal recovery process from leachate	Cyclic process	Refs.
33	ChCl: Lactic Acid: Citric Acid	7: 17: 2	LCO	Co	> 99	S/L ratio: 15 g/L T = 55 °C t = 3 h	–	–	[167]
34	(Dimethylthetin: Oxalic Acid) + 30 % water	1: 1	NMC	Li	99.98	T = 60 °C t = 15 min	Precipitation: Oxalic Acid, Water, Na ₃ PO ₄ /NaCO ₃	2 cycles	[168]
35	GUC: EG:Maleic acid	1: 2: 1	LCO	Li Co	>99.5	T = 100 °C t = 9 h S/L ratio:1/50 g/mL	Precipitation: Oxalic Acid	3 cycles	[122]
36	EG: Tartaric acid	1: 3	LNMC	Li Ni Co Mn	99.2 96.1 95.2 97.8	T = 118 °C t = 17 min S/L ratio: 42 g/L	Precipitation: Water, Na ₂ CO ₃	4 cycles	[117]
37	EG:Hydroxylamine Hydrochloride (NH ₂ OH·HCl)	4: 1	LCO	Li Co	99.7 88	T = 80 °C t = 8 h S/L ratio: 83.3 g/kg	Precipitation: Na ₂ CO ₃	Can be reused	[169]
38	ChCl: Malonic Acid	1: 2	LCO	Li Co	92.7 96.8	T = 90 °C t = 270 min S/L ratio: 1/100 g/mL	–	–	[170]
39	(ChCl: Citric Acid) + 25.7 wt % Water	1: 1	LCO	Li Co	100 97.6	T = 120 °C t = 4 h S/L ratio: 1/50 g/mL	Precipitation: Oxalic Acid	2 cycles	[119]
40	ChCl: Oxalic acid	1: 1	NMC	Li Ni Co Mn	99.95 99.88 98.1 90	T = 75 °C t = 3 h S/L ratio: 15 g/L	Precipitation	–	[123]
41	ChCl: H ₃ PO ₄	1: 1	LCO	Li Co	94.3 80.8	T = 100 °C t = 2 h S/L ratio: 1/30 g/mL	Precipitation: Ethanol Oxalic Acid NaOH	5 cycles	[124]
42	PTSA monohydrate: ChCl: EG	1: 0.5: 1	NMC	Li Ni Mn Co	100 98.2 97.8 96.1	T = 60 °C t = 6 h S/L ratio: 50 mg/g	–	–	[171]
43	TBAC: Monochloroacetic Acid (MCA)	1: 3	LNMC	Li Co Mn Ni	100 94 96 100	T = 100 °C t = 5 h S/L ratio: 15 g/L	Precipitation: Oxalic Acid	10 cycles	[172]
44	ChCl: Pyruvic Acid	1: 1	LCO	Li Co	99	T = 80 °C t = 5 h S/L ratio: 1/50 g/mL	Precipitation: Oxalic Acid	3 cycles	[125]
45	ChCl: Phenylphosphinic Acid	1: 2	NCM	Li Co Mn Ni	97 97 93 96	T = 100 °C t = 80 min S/L ratio: 90 mL/g	–	–	[173]
46	N-Methylurea: Acetamide	1: 1	LCO	Co	> 97	T = 180 °C t = 24 h	Precipitation	–	[174]
47	(ChCl: EG) + O ₃	1: 8	LiFePO ₄ Batteries	Li Fe	92.2 1.6	T = 40 °C t = 6 h S/L ratio: 20 g/L	Precipitation: Distillation, 150 °C, 0.5 h	10 cycles	[175]
48	GUC: lactic acid	1: 2	LCO	Solubility of LiCoO ₂	19.9 mg/g	T = 80 °C t = 624 h S/L ratio: 0.04/2 g/g	Precipitation: Oxalic Acid	2 cycles	[132]
49	ChCl: Formic Acid	1: 1	Cathode Material Powder (LiFePO ₄ ; Li(NiCoMn) _{1/3} O ₂)	Li Ni Co Mn P Fe	97.6 74.7 81.2 75.1 4.5 7.5	T = 80 °C t = 120 min S/L ratio: 1/200 g/mL	–	–	[176]
50	(GUC: lactic acid) + 1 wt% Ascorbic Acid	1: 2	LCO and NCM	Li Co	97.42 96.91	T = 50 °C t = 24 h S/L ratio: 1/50 g/mL	Precipitation: Oxalic Acid	3 cycles	[177]
51	ChCl: EG: PTSA	1: 2: 1	NCM	Li Ni Co Mn	<80 >99 >99 >90	T = 100 °C t = 72 h S/L ratio: 10 g/L	Precipitation: Oxalic Acid and Na ₂ CO ₃	–	[178]

(continued on next page)

Table 4 (continued)

No.	DESS	Molar ratio	Cathode type	Metals	Efficiency %	Optimum condition	Metal recovery process from leachate	Cyclic process	Refs.
52	EG: Tartaric Acid	5: 1	LCO	Li	98.34	$T = 120\text{ }^{\circ}\text{C}$ $t = 12\text{ h}$ S/L ratio: 20 g/L	Precipitation	5 cycles	[179]
53	Tetraethylammonium Chloride: Ascorbic Acid	1: 1	NCM	Li Ni Co Mn	99.1 99.6 99.4 99.3	$T = 80\text{ }^{\circ}\text{C}$ $t = 3\text{ min}$ S/L ratio: 40 mL/g	–	–	[180]
54	BeCl: Formic Acid	1: 9	NCM	Li Ni Co Mn	98.03 96.01 94.19 92.35	$T = 140\text{ }^{\circ}\text{C}$ $t = 360\text{ min}$ S/L ratio: 20 g/L	–	–	[118]
55	(Ascorbic Acid: Dimethyl-Beta-Propiothetin Chloride) + 0.75 Water	2: 1	LNCM	Li Ni Co Mn Ni Co	99.65 98.98 99.32 99.13	$T = 50\text{ }^{\circ}\text{C}$ $t = 14\text{ min}$ S/L ratio: 25 g/L	–	–	[181]
56	ChCl: EG: Maleic Acid	1: 1.5: 0.5		Mn Li Cu Fe Li	> 99	$T = 80\text{ }^{\circ}\text{C}$ $t = 2\text{ h}$ S/L ratio: 50 mL/g	Electrodeposition Precipitation	–	[144]
57	ChCl: Maleic Acid	1: 1	NMC	Li Ni Co Mn	99.18 94.97 65.36 67.88	$T = 80\text{ }^{\circ}\text{C}$ $t = 2\text{ h}$ S/L ratio: 20 g/L	–	–	[182]
58	ChCl: Oxalic Acid: Water	1: 1: 1	LCO	Li Co	91.2 94.6	$T = 130\text{ }^{\circ}\text{C}$ $t = 2\text{ h}$ S/L ratio: 20 g/L	–	5 cycles	[183]
59	GUC: EG: Maleic Acid	1: 2: 1	LCO	Li Co	>99.5	$T = 100\text{ }^{\circ}\text{C}$ $t = 6\text{ h}$ S/L ratio: 1/50 g/mL	Precipitation: Oxalic Acid	3 cycles	[122]
60	ChCl: Maleic Acid	1: 1	NMC	Co Ni Mn	>99	$T = 60\text{ }^{\circ}\text{C}$ $t = 24\text{ h}$ S/L ratio: 0.02 g/mL	Precipitation: Na_2CO_3	–	[143]
61	EG: Ethanol: Citric Acid	6: 6: 1	LCO and NMC	Li Ni Co Mn	94.4 84.6 85.5 47.4	$T = 150\text{ }^{\circ}\text{C}$ $t = 16\text{ h}$ S/L ratio: 8 g/L	Precipitation: Oxalic Acid	5 cycles	[184]
62	Lactic Acid: Glycolic Acid	1: 1	LCO	Li	99.5	$T = 100\text{ }^{\circ}\text{C}$ $t = 6\text{ h}$ S/L ratio: 1/10 g/mL	Precipitation	5 cycles	[142]
63	(ChCl: Citric Acid) + 35 % H_2O	2: 1	LCO	Li Co	96.51 99.83	$T = 80\text{ }^{\circ}\text{C}$ $t = 120\text{ min}$ $T = 90\text{ }^{\circ}\text{C}$	–	–	[185]
64	(ChCl: Lactic Acid) + 90 vol % H_2O	1: 3	LCO Samsung Phone	Li Co	97.15 95.20	$t = 24\text{ h}$ S/L ratio: 1/100 g/mL	Precipitation: Oxalic Acid and Na_2CO_3	–	[141]
Microwave and Ultrasound-Assisted Leaching									
65	ChCl: EG	3: 2	NCA	Li	100 (30 min)	$T = 220\text{ }^{\circ}\text{C}$ $t = 30\text{ s}$ S/L ratio: 10 mg/mL	Precipitation	6 cycles	[186]
66	ChCl: EG: Lactic Acid	–	LCO	Li Co	> 90	$t = 4\text{ min}$	–	–	[145]
67	ChCl: Oxalic Acid	1: 2	LMO	Mn Li	96 99	$T = 75\text{ }^{\circ}\text{C}$ $t = 15\text{ min}$ S/L ratio: 1/60 g/mL	–	–	[187]
68	PEG-200: Thiourea	2: 1	LCO	Li Co	74 71	$T = 160\text{ }^{\circ}\text{C}$ periodic ultrasound exposure S/L ratio: 0.02 g/g	–	–	[188]
69	(ChCl: Formic Acid) + 10 wt % Water	1: 2	LCO	Li Co	>99	$T = 60\text{ }^{\circ}\text{C}$ $t = 10\text{ min}$ S/L ratio: 20 g/L	Precipitation: Na_2CO_3	–	[146]

(continued on next page)

Table 4 (continued)

No.	DESS	Molar ratio	Cathode type	Metals	Efficiency %	Optimum condition	Metal recovery process from leachate	Cyclic process	Refs.
70	ChCl: Oxalic Acid: Water	1: 1: 8	LCO	Li Co	99.5 99.21	T = 100 °C t = 10 min	–	–	[134]
Hydrophobic DESs (Extraction)									
71	Decanoic: Lidocaine	2: 1	Synthetic Leach Liquor of Mixed Cathodes	Ni Li	> 90 < 20	O/A ratio: 2/1	–	Can be reused	[189]
72	Aliquat 336: DL-menthol D ₂ EHPA: DL-menthol (multistage cross-current extraction)	1: 1 1: 1	Leachate of Cylindrical Batteries and Charging Device in HCl	Ni Co Cu Fe Li Al Co Cu Mn Ni Li	> 98	O/A ratio: 1/1	–	–	[190]
73	TOPO: Decanoic Acid	1: 1	HCl-Loaded DES as a Leaching Medium	Co Mn Ni Li	48 102 82 89 98	T = 100 °C t = 240 min S/ L ratio: 44 g/L 1 mol/L HCl	Precipitated: KMnO ₄ , Oxalic Acid	5 cycles	[147]
74	Aliquat 336: DL-menthol	3: 7	Leachate of Cathodes of LIBs in HCl	Co Mn Fe	> 99	T = 25 °C t = 30 min O/A ratio: 1/5	Precipitation: NaH ₂ PO ₄ and H ₃ PO ₄	2 cycles	[99]
75	D ₂ EHPA: 1-butanol	2: 1	Simulated Solution Containing 1.2 g/L Co, 1.2 g/L Ni and 0.6 g/L Li	Co	95.64	pH = 5 t = 6 min O/A ratio: 2/1	Stripping: H ₂ SO ₄	5 cycles	[133]
76	TBAC: Oleic Acid	1: 3	From the Mother liquor of Li ₂ CO ₃	Li	76	O/A ratio: 1/1	Stripping: HCl Precipitation: Na ₂ CO ₃	5 cycles	[148]

cooled: cobalt, nickel, and manganese-maleate complexes crystallized directly from the liquor, resulting in > 99 % recovery from NMC cathodes and > 94 % from mixed black mass. Copper and aluminum remained primarily in solution, and calcining the crystals yielded pure oxides (Co₃O₄, NiO, Mn₃O₄, CoO). Lithium was later precipitated as lithium maleate via Na₂CO₃. Thus, ChCl served as a mild reductant, while maleic acid supplied the selective precipitating ligand [143]. Zou (2025) used a ternary DES of ChCl: EG: MA (1: 1.8: 0.2) to treat NMC battery scrap. At 80 °C for two hours with an L/S of 50 mL/g, > 99 % of Ni, Co, Mn, Li, Cu and Fe dissolved, while graphite remained intact. Spectroscopic and rheological studies showed that EG reduced the transition metals, thereby eliminating the need for extra reductants. Metals were then separated step-wise: Fe was first removed from the leachate by chemical precipitation. Then, Cu and Ni-Co alloys were sequentially recovered through electrodeposition, achieving deposition efficiencies of 99.36 % for Cu, 72.36 % for Ni, and 94.65 % for Co. Afterward, Mn and Li were separated by adjusting the pH. Lithium was recovered as high-purity Li₂CO₃. The remaining solid residue was identified as high-purity flake graphite. The entire flow sheet ran at a modest temperature, uses no toxic reagents, and yields high selectivity [144]. In recent work, binary and ternary ChCl-based DESs have dominated LIB recycling because they dissolve metals efficiently and, when diluted with water, become less viscous, allowing for faster mass transfer. Cobalt and nickel are typically precipitated with Na₂CO₃ or oxalic acid, and lithium with H₃PO₄. Many systems, e.g., ChCl with oxalic, formic, tartaric, or lactic acid, can be reused several times after topping up depleted components, whereas others, such as PEG-200: ascorbic acid, cannot, because ascorbic acid oxidizes during the leaching process.

Microwave and ultrasound can greatly intensify DES-based recycling of LIB materials. Zhu et al. showed that microwaving a urea/ lactic acid: ChCl: EG DES polarises its molecules, so they react more quickly with LCO [145]. Ma et al. reached the same or higher lithium and cobalt leaching yields in a ChCl: oxalic acid DES at much lower temperatures and in far less time than with conventional heating, while Liu et al. extracted virtually all the cobalt and lithium from LCO in just ten minutes using a ChCl: formic acid solvent [134,146]. Ultrasound provided

a complementary boost, as cavitation dispersed the organic extractants LIX 65 N and LIX 70 into fine droplets, increased the interfacial area, and resulted in a four- to seven-fold increase in nickel extraction efficiency compared to ordinary liquid-liquid extraction. Overall, microwave heating shortens leaching times from days to minutes and offers greater energy efficiency; however, identifying optimal reductants and fully elucidating the reaction pathways under microwave fields still requires further investigation [145,146].

Several hydrophobic DESs have proven effective for fast, reusable metal recovery. Formulations such as TOPO: decanoic acid [147], Aliquat 336: DL-menthol [99], D₂EHPA: 1-butanol [133], and TBAC: oleic acid [148] have been successfully used for metal recovery from simulated solutions, leachates, or Li₂CO₃ mother liquor in under two hours. These systems have also demonstrated good reusability, remaining effective for more than two cycles following metal precipitation and recovery.

5.1.2. Waste PCBs

PCBs dominate the e-waste stream: the UN Institute for Training and Research estimates they account for just over half of all discarded electronics. A typical mobile phone is 20–30 % PCB by weight, a computer about 20 %, and a television roughly 10 % [191]. PCBs contain a lucrative mix of recoverable materials, including polymers, ceramics, and various metals found in the board laminate, circuitry, components, solder joints, and connectors, with metals making up roughly 20–30 % of the total mass [192]. They carry a wide variety of components, such as capacitors, relays, resistors, and integrated circuits, whose composition can vary significantly, complicating the recycling process [36]. The rise of miniaturized electronic devices, including flexible printed circuit boards (FPCBs), presents both opportunities and challenges for resource recovery due to their complex design and manufacturing processes. Unfortunately, more than 5 million tons of waste PCBs are generated annually, mainly processed through incineration and landfilling, which pose environmental risks, including air and soil pollution from toxic chemicals and heavy metals [193]. Notably, spent PCBs are highly attractive resources due to their metal content and mineral value, making them a key focus for sustainable metal recovery. Waste PCBs are

Table 5

List of prepared DESs along with their efficiency for recovering metals from waste PCBs.

No.	DESs	Molar ratio	Metals	Efficiency %	Optimum condition	Metal recovery process from leachate	Cyclic process	Refs.
Hydrophilic DESs (Leaching)								
1	ChCl: Oxalic Acid ChCl: Glycolic Acid (Two-step leaching)	1: 2	ChCl: GA Cu Fe Zn Sn Pb ChCl: OA Zn Pb Cu Ag Al Fe Zn	85 75 83 22 38 > 90 > 90 8.8 26.4 91.5 19.7 2.3	T = 50 °C t = 2 h	Precipitation: Oxalic Acid, Water and Adding Fe for Sn-Oxalate Precipitation	—	[194]
2	ChCl: Malonic Acid	1: 1			T = 60 °C t = 2 h S/L ratio: 1/10 g/mL	Precipitation: NaCl	—	[198]
3	(ChCl: EG) + 5 wt% H ₂ O ₂	—	Cu	97.8	T = 20 °C t = 20 min S/L ratio: 1/100 g/mL	Electrodeposition	—	[195]
4	(Calcium Chloride Hexahydrate: EG) + Two oxidizing agents, FeCl ₃ and CuCl ₂	1: 1	Ni Au	—	—	—	—	[199]
5	(ChCl: EG) + CuCl ₂ ·2H ₂ O	1: 1	Au	95.3	T = 90 °C t = 2 h S/L ratio: 1/1000 g/mL	Precipitation	5 cycles	[193]
6	ChCl: Glycerol: Citric Acid	7: 10: 3	Ag	93.65	T = 120 °C t = 16 h S/L ratio: 1/20 g/mL	—	—	[191]
	ChCl: Formic Acid	1: 2	Cu Fe	90 89	T = 100 °C t = 16 h S/L ratio: 1/30 g/mL	—	—	
7	ChCl: Urea	1: 2	Zn	90	T = 100 °C t = 21 h S/L ratio: 1/20 g/mL	—	—	[113]
	ChCl: Oxalic Acid	1: 1	Sn	92.3	T = 80 °C t = 1 h S/L ratio: 1/20 g/mL	—	—	
8	(ChCl: EG) + I ₂	1: 2	Cu Ni Zn Sn	>75 >90 45 >90	T = 85 °C t = 72 h	—	—	[196]
9	(ChCl: Acetic Acid) + I ₂	1: 2	Cu Ni Au	99 92 90	T = 65 °C t = 96 h S/L ratio: 10 g/L	—	—	[197]
10	ChCl: Malonic Acid: H ₂ O ₂	1: 1	Cu Ag Al Fe Zn	15 17 >45 >45 <20	T = 45–65 °C t = 2–3 h	—	—	[200]

especially valuable, containing significant amounts of base metals such as Cu, Al, and Fe, along with precious metals like Ag, Au, and Pt. The concentrations of these metals in PCBs are often dozens of times higher than in natural ores or their corresponding concentrates. For example, the copper content in waste PCBs is approximately 13 to 26 times greater than that found in primary ores. Therefore, recovering metals, particularly gold, from PCBs offers substantial economic and environmental benefits [194]. The substantial value of silver lies in its essential function as a conductor and electrode, thanks to its exceptional electrical conductivity. Hence, the recovery of silver metal from electronic waste such as PCBs shows excellent potential [191,193,194]. Copper is pivotal to

clean-energy technologies. Electric vehicles (EVs), whose circuit boards, motor windings, rotors, wiring, and charging stations all rely on Cu, use about 83 kg each, roughly four times more than a conventional petrol car. To satisfy this surging demand, augmenting mining with copper recovery from waste streams offers a strategy that simultaneously eases waste-management pressures and supports long-term sustainability [195].

Table 5 reviews recent work on DES-mediated recovery of metals from waste PCBs. Sabzkoobi et al. (2021) leached metals from shredded boards with a ChCl: EG solvent. When iodine was added as an oxidant, the mixture extracted more than 75 % of the Cu, Ni, and Sn, and

approximately 45 % of the Zn, within 72 h. Without iodine, recoveries fell below 15 % for most elements, although Sn still dissolved almost completely [196]. Saffaj et al. (2024) compared five ChCl-based DESs (paired with EG, acetic, malonic, citric or levulinic acid) for solvo-leaching at 65 °C, 300 rpm and S/L = 10 g/L, again in the presence of I₂. The ChCl: acetic acid system yielded the best results, selectively dissolving 99 % of Cu, 92 % of Ni, and 90 % of Au from PCB powder in 96 h [197]. Adding oxidants such as FeCl₃, CuCl₂, or I₂ markedly accelerates metal dissolution in DESs. Water-rich media can form a passivating NiO/Ni(OH)₂ film on PCB surfaces; however, applying ultrasound breaks up this layer. Under sonication, for example, Cu leaching from PCBs in a CaCl₂·6H₂O: EG DES containing CuCl₂ proceeded 10,000 times faster than without ultrasound [41]. Wang et al. (2024) devised a green, high-efficiency route to reclaim gold from PCBs using a ChCl: EG DES doped with CuCl₂·2H₂O. Under optimized conditions, gold dissolved almost completely and was precipitated with a 95.3 % yield. Kinetic analysis revealed that the dissolution process follows a nucleation-controlled surface-reaction mechanism, with an apparent activation energy of 80.3 kJ/mol. The process was highly selective, with the two-step gold recovery exceeding 95 %, while copper and nickel remained below 2 %. Moreover, the leachate can be reused; after five cycles, the gold-dissolution efficiency remains above 97 % [193].

5.1.3. FLs

Discarded FLs are now considered secondary sources of REEs, such as Y, Eu, Ce, Tb, and La. Because they also contain toxic mercury, lamps are routinely collected and dismantled by specialist recyclers, who separate them into: (1) glass for new glass products, (2) metal parts from filaments and electrodes, (3) plastic components that are incinerated for energy, (4) phosphor powder, and (5) mercury. The phosphor fraction holds most of the REEs: Green phosphor LaPO₄: Ce(III), Tb(III) (LAP), Red phosphor Y₂O₃: Eu(III) (YOX), and Blue phosphor BaMgAl₁₀O₁₇: Eu(II) (BAM). It also contains halophosphate phosphor (Sr,Ca)₁₀(PO₄)(Cl, F)₂: Sb(III), Mn(II) (HALO), which lacks REEs. Given disposal rates at the time, stockpiled lamp-phosphor waste was estimated to contain about 25,000 tons of REEs by 2020, with europium, terbium, and yttrium being particularly abundant. Solvometallurgical studies have shown that the phosphors' acid solubilities decrease in the order HALO > YOX > LAP > BAM, guiding the choice of leaching strategies for REE recovery [201]. Thanks to their strong and versatile ability to complex metal ions, DESs are attractive non-aqueous media for the solvometallurgical recovery of REEs from spent FL phosphors. To date, however, most studies have focused on DESs using laboratory-prepared, impurity-free powders; when genuine lamp-phosphor waste is used, both leaching efficiency and element selectivity can differ significantly. For the technology to become commercially viable, future studies need to shift toward using real feedstocks and develop new, task-specific DES formulations or hybrid processes that can handle the full spectrum of contaminants present in industrial waste [41].

Pateli et al. 2020 studied the reclamation of REEs from discarded FLs using a DES (Table 6). They achieved about 70 % leaching of Y₂O₃: Eu(III) with ChCl: levulinic acid: (1:3) at 80 °C for 48 h. The study found that leaching efficiency improved with higher temperatures and longer durations. Using the DES, around 40 % of the actual lamp phosphor was

leached. The solubility of different phosphors was tested, and ChCl: levulinic acid showed high solubility for Y₂O₃: Eu(III) but low for (Sr, Ca)₁₀(PO₄)(Cl, F)₂: Sb(III)Mn(II). The researchers optimized leaching conditions and validated them using both synthetic and real lamp phosphor waste, finding higher co-dissolution of HALO in the real waste. Different methods for recovering metals were tested, including oxalic acid precipitation, which was unsuitable for the DES. However, solvent extraction using bis(2-ethylhexyl)phosphoric acid (D₂EHPA) has been shown to be effective. In comparison, pure levulinic acid proved to be more suitable than the ChCl-based DES for the selective recovery of YOX [201].

5.1.4. Magnets

Demand for NdFeB permanent magnets has soared with the expansion of wind-turbine and electric-vehicle markets, making these magnets the fastest-growing application for neodymium and other REEs. Production and end-of-life streams now generate large quantities of scrap: manufacturing alone discards about 30 wt% of the magnet alloy, which contains more than 20 % REEs (Nd, Pr, Dy, Tb). Given the finite nature of REE ores and the environmental burden of mining, recovering these metals from NdFeB waste has become essential [202]. NdFeB permanent magnets consist mainly of neodymium, iron, and boron, with iron making up about 60–70 % of the alloy. Small amounts of other elements, such as dysprosium, terbium, gadolinium, niobium, cobalt, copper, or gallium, are often introduced to tailor properties like thermal stability and magnetic performance for specific applications [203]. Samarium–cobalt magnets, prized for their strong magnetic performance and exceptional thermal and corrosion resistance, are seeing growing demand. Consequently, rising volumes of production scrap and EoL Sm-Co magnets are driving efforts to devise efficient recycling methods that can reclaim and refine samarium, cobalt, and other valuable constituents [204]. More broadly, recovering rare-earth permanent magnets has become a critical strategy for mitigating the looming global market shortage of rare-earth elements [205].

Researchers have begun testing DESs for recovering metals from spent rare-earth magnets (Table 7). Riaño et al. compared three ChCl-based DESs paired with lactic acid, urea, or EG for dissolving roasted NdFeB powder. Only the ChCl: lactic acid mixture proved effective, selectively solubilizing neodymium and iron. The lactic acid's protons form complexes with Nd₂O₃, producing an Nd–DES adduct and water; higher DES-to-solid ratios further raised dissolution yields. Metals were then stripped from the loaded solvent with ionic-liquid extractants. A 0.9 M Aliquat 336 thiocyanate solution in toluene efficiently removed boron, cobalt, and iron, while a 0.9 M Cyanex 923 in toluene extracted 99.9 % of dysprosium and 99.9 % of neodymium. The main drawbacks are the DES's high viscosity (≈ 0.156 Pa s) and the still-modest overall dissolution of Nd and Fe [205]. Shuping et al. (2025) formulated a tetra-ethyl-ammonium-chloride (TEAC): levulinic acid DES that selectively dissolves REEs from NdFeB scrap. At 132 °C, the solvent leached 97.6 % of neodymium while leaving iron contamination below 0.5 %, giving a separation factor > 9000 and yielding Nd₂O₃ of 99.65 % purity. The DES exhibited excellent performance, dissolving over 75 g/L of magnet powder. Its viscosity at the working temperature was only 12.8 cP, comparable to that of conventional organic solvents. It also displayed high thermal stability up to 132.4 °C and retained its selectivity through

Table 6

List of prepared DESs along with their efficiency for recovering metals from EoL FLs.

No.	DESs	Molar ratio	Metals	Efficiency %	Optimum condition	Metal recovery process from leachate	Cyclic process	Refs.
1	(ChCl: Levulinic acid) + 30 % water	1: 3	Y ₂ O ₃ : Eu (III)	>90	T = 80 °C t = 48 h S/L ratio: 1/10 g/mL	Precipitation: Oxalic acid Solvent Extraction: D ₂ EHPA	–	[194]

Table 7
List of prepared DESs along with their efficiency for recovering metals from waste magnets.

No.	DESs	Molar ration	Type of Magnet	Metals	Efficiency %	Optimum condition	Metal recovery process from leachate	Cyclic process	Refs.
Hydrophilic DESs (Leaching)									
1	TEAC: Levulinic Acid	1: 2	NdFeB Magnet	Nd Fe	93 <1	T = 90 °C t = 29 h S/L ratio: 2/100 g/ mL	Precipitation: Oxalic Acid	3 cycles	[202]
2	EG: Maleic Acid	4: 1	NdFeB Magnet	Nd Fe	97 0.8	T = 70 °C t = 5 h S/L ratio: 1/ 100 g/ mL	Precipitation: Oxalic Acid	3 cycles	[206]
3	ChCl: Salicylic Acid	1: 2	NdFeB Magnet	Nd	97	T = 60 °C t = 24 h S/L ratio: 1/ 100 g/ mL	–	–	[207]
4	EG: Maleic Acid	2: 1	NdFeB Magnet	Nd Fe	28.7 <1	T = 80 °C t = 5 h S/L ratio: 1/ 125 g/ mL	Solvent Extraction: P204-Sulfonated Kerosene Precipitation: Oxalic Acid	several cycles	[208]
5	GUC: Lactic Acid	1: 2	NdFeB Magnet	Nd Fe	95 < 1	T = 40 °C t = 6 h S/L ratio: 1/ 10 g/ mL	Stripping: Oxalic Acid	3 cycles	[103]
6	ChCl: Lactic Acid	1: 2	NdFeB Magnet	Nd Dy	99 99	T = 70 °C t = 12 h S/L ratio: 1/ 50 g/ mL	–	–	[205]
Hydrophobic DESs (Extraction)									
7	Decan-1-ol: Myristyl Alcohol: TOPO	5: 5: 4	NdFeB Magnet	Fe Nd Dy Pr Co Sm	<10 >50 >80 >40 <10 97.8	T = 25 °C t = 30 min A/O ratio: 1/1	Stripping: Na ₂ C ₂ O ₄	5 cycles	[203]
8	TOPO: HBTA	1: 2	SmCo Magnet	Co Cu Fe	99.1 95.6 99.5	T = 85 °C	Precipitation: Oxalic Acid	5 cycles	[91]
9	Thymol: TOPO	1: 1	NdFeB Magnet	Nd Dy	83 95	t = 10 min A/O ratio: 1/1	Stripping: H ₂ SO ₄	4 cycles	[209]
10	Dodecanol: TOPO Decanoic Acid: Lauric Acid	2: 1 2: 1	SmCo Magnet	Fe Sm Cu Co	99 99 96 99	t = 20 min A/O ratio: 1/1	Stripping: Oxalic Acid	5 cycles	[210]
11	TOPO:Phenol	1: 1	SmCo Magnet	Sm Co	98 <15	–	–	–	[204]
12	(HBTA: TBPO) + 2.5 % H ₂ O		NdFeB Magnet	Co Fe Nd Dy	<80 <10 100 80	T = 60 °C t = 3 h O/A ratio: 1/1	Stripping, and Scrubbing	2 cycles	[211]

at least three reuse cycles [202]. Liu et al. used a GUC: lactic acid DES to dissolve roasted NdFeB powder and found neodymium oxide far more soluble than iron oxide because of its higher ionic character. After six hours at 40 °C, 83 % of the Nd was recovered at 99.56 % purity. Spent solvent was regenerated by stripping with oxalic acid. Raising the S/L ratio suppressed iron dissolution but did not affect neodymium uptake. The DES's viscosity remained below 0.04 Pa·s at 50 °C, aiding mass transfer and extraction [103].

Yu et al. (2024) prepared and characterized 13 different HDES systems that combined TOPO with various alcohols. The optimization and characterization of these systems revealed several beneficial properties, including low viscosity, low water solubility, and a significant density difference from water, all of which enhanced their suitability as extractants. Among the systems tested, a ternary HDES, composed of decan-1-ol, myristyl alcohol, and TOPO in a 5: 5: 4 ratio, was identified as the most effective for extracting REEs and Fe. The extraction

Table 8
List of prepared DESs along with their efficiency for recovering metals from waste PV cells.

No.	DESs	Molar ratio	Metals	Efficiency%	Optimum condition	Metal recovery process from leachate	Cyclic process	Refs.
1	(ChCl: EG) + FeCl ₃ .6H ₂ O	1: 2	Ag	>99	T = 75 °C t = 6 h	Electrochemical	–	[212]
2	(ChCl: Urea) + CuCl ₂	1: 2	Ag	>99	T = 80 °C t = 60 min S/L ratio: 1/50 g/mL	Precipitation	5 cycles	[213]

conditions were optimized, reducing the concentration of REEs in the raffinate from 4390 mg/L to 34 mg/L, and achieving a recovery rate of 99 %. The HDES was stripped using 0.2 mol/L $\text{Na}_2\text{C}_2\text{O}_4$, which further separated a small amount of loaded Fe (133 mg/L) and REEs, resulting in a final product purity of over 99 %. The stripped HDES was recovered after washing with deionized water. Even after five cycles, the HDES maintained good stability. This innovative process did not require salting-out agents or toxic volatile solvents, offering new possibilities for efficient and clean separation [203].

5.1.5. PV Cells

As the world seeks to reduce massive CO_2 emissions from carbon-based power sources, solar PV has emerged as a leading renewable energy source. Installed PV capacity surpassed 1 TW in 2022 and is expected to reach about 9 TW by 2050. With an average lifespan of around 25 years, an enormous volume of EoL PV panels is anticipated by 2050. The International Renewable Energy Agency estimates that approximately 78 million tonnes of discarded modules will accumulate by then. Meanwhile, demand for silver, which was essential for the first generation of PV cells, continued to increase even though mine production remained almost unchanged. At the same time, newer generations of PV cells started using different metals instead of silver. Nowadays, most PV waste is landfilled due to cost considerations; however, this leads to environmental issues and a potential loss of thousands of tons of silver [212,213]. PV modules are made up of common materials like glass, copper, and aluminum, as well as essential elements such as silver, and energy-intensive, high-purity substances like silicon wafers. Silver is particularly valued in PV production for its excellent electrical and thermal conductivity, cost-effectiveness, and is typically found in amounts of about 630 g/t in PV panels, similar to the silver content in primary ore (around 700 g/t in 2015). The composition of silver ore is far more complex than that of PV panels, making it easier to recycle silver from the latter. Recycling waste PV panels can reduce energy

consumption, lower carbon emissions, and mitigate resource depletion over their lifecycle [213]. Dismantling first-generation solar panels for material recovery will become a crucial source of recycled silver for producing new panels, helping meet the rising demand for clean energy in the future [212,213].

As summarized in Table 8, Zhang et al. (2022) presented a more sustainable method for silver recovery. Under optimized conditions, silver leaching and precipitation rates reached 99 %. Kinetic analysis showed that silver leaching followed a classical shrinkage core model, with the chemical reaction as the rate-controlling step and an activation energy of 172.36 kJ/mol. In this process, Cu(II) acted as an oxidant, exhibiting a higher redox potential in the DES system compared to aqueous environments. Additionally, urea functioned as a non-solvating aprotic solvent and ligand, facilitating metal complex formation and enhancing silver leaching. This study provided a theoretical and practical framework for the environmentally friendly and cost-effective recycling of precious metals from waste solar panels [213]. Lemoine et al. (2024) explored a circular recycling method using an iron redox shuttle with various DES, particularly ethaline, and achieved a silver leaching efficiency of 99.9 % from crystallized silicon scraps at 75 °C. The study investigated the impact of Fe(III) chloride concentration on the kinetics of AgCl formation, emphasizing the importance of maintaining an oxygen-free atmosphere to achieve high-quality silver electrodeposition. Ultimately, the successful recovery of silver was demonstrated through leaching at 75 °C and electrodeposition at 50 °C using ethaline and $\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$ under argon [212].

The research underscores the importance of recycling PV panels, particularly for recovering silver, a valuable material used in first-generation PV cells due to its excellent conductivity and significant presence in panels. While silver recovery methods are well-developed and highly efficient, studies do not address the treatment or recovery of other key metals, such as copper, aluminum, and silicon. This represents a significant gap, as these materials also play vital roles in the

Table 9

List of prepared DESs along with their efficiency in recovering metals from spent catalysts.

No.	DESs	Molar ratio	Catalyst Type	Metals	Efficiency %	Optimum condition	Metal recovery process from leachate	Cyclic process	Refs.
Hydrophilic DESs (Leaching)									
1	(CHCl_3 : PTSA) + HNO_3	1: 1	Spent Automotive Catalysts	Pt Pd Rh	> 90 > 90 < 40	T = 80 °C t = 4 h S/L ratio: 1/5 g/L	–	4 cycles	[216]
2	PEG-400: PTSA ChCl: EG: PTSA	1: 1 1:2:1	Spent Hydroprocessing Catalysts	Ni	98 96	T = 100 °C t = 48 h S/L ratio: 20 g/L	Precipitation: Pure Al Powder	–	[115]
3	PEG-400: PTSA ChCl: PTSA	1: 1 2: 1	Spent Hydroprocessing Catalysts	Mo Co	87 93	T = 100 °C t = 48 h S/L ratio: 20 g/L	–	–	[214]
4	Combined IL (cy- D_2EH) and DES (Ph-Cl: PTSA: EG)	1: 2: 1	Spent Hydrodesulfurization Catalysts	Mo V Ni	91 82.5 81	T = 100 °C t = 24 h S/L ratio: 20 g/L	Precipitation and Cementation	–	[215]
Hydrophobic DESs (Extraction)									
5	[BTBA]Cl: Octanoic Acid	1: 2	Spent Catalysts	Mo	98.8	T = 50 °C t = 6 min S/L ratio: 1/10 g/mL	–	–	[218]
6	Trihexyl(tetradecyl) phosphonium Chloride: Decanoic Acid	–	Automotive Catalysts	Pt Pd Rh Al Mg La	90 100 89 <5 <5 <5	T = 80 °C t = 24 h	Solvent Extraction: Water Precipitation: NH_4NO_3 and NH_4Cl	2 cycles	[217]

function and sustainability of PV panels. Therefore, further research is needed to establish comprehensive recycling strategies that target all valuable components, supporting a circular economy and minimizing environmental impact.

5.2. Industrial waste

Industrial operations generate sizable amounts of metal-laden by-products, such as fly ash, electroplating sludges, spent catalysts, wastewater streams, etc., that are rich in Au, Ag, Pt, Pd, Cu, Zn, Pb, In, Sn, and other metals. DESs offer an environmentally friendly method for reclaiming these elements. For instance, a ChCl: EG DES can effectively dissolve Au, Ag, and Pd from such residues, after which the gold is readily recovered via electrodeposition [5].

5.2.1. Spent catalysts

Spent industrial catalysts contain valuable metals including Ni, Co, Mo, W, V, and several REEs. A catalyst's service life ends when its activity falls below the required threshold, typically due to prolonged operation leading to sintering, poisoning, fouling, or mechanical damage. Once deactivated, a catalyst can be regenerated or processed to recover its metal content; if neither route is chosen, it is simply discarded, adding substantially to industrial waste streams [214]. A significant amount of hazardous waste is generated from spent hydrosulfurization (HDS) catalysts during petroleum refining processes. It is estimated that approximately 120,000 tons of spent HDS catalysts are produced globally each year. This waste contains valuable metals, including Mo, V, Ni, Fe, and Al. The recovery of these valuable metals from spent HDS catalysts has become a vital area of research, with a focus on reducing waste and minimizing environmental impact [215]. Platinum, palladium, and rhodium, along with ruthenium, osmium, and iridium, form the group of PGMs. Among these, platinum, palladium, and rhodium are currently the most important from both financial and industrial perspectives [216]. Furthermore, there is a growing demand for efficient processes to recycle PGMs from secondary resources, such as spent automotive catalysts (SACs). PGMs are valuable due to their limited natural resources and uneven distribution of deposits. In particular, Pt, Pd, and Rh are extensively used in automotive catalysts [217]. The automotive industry utilizes the largest percentage of PGMs in the production of catalytic converters, where they are essential components [216]. The PGM content in natural ores is about 0.001 wt%, but it can be as high as 0.5 wt% in automotive catalysts. Therefore, SACs are considered extremely valuable resources. In summary, recycling SACs offers a crucial and potentially efficient method for recovering PGM resources on a global scale in the long term [217].

Based on Table 9, in a study by Abbas et al. (2024), a novel method combining ILs and DESs was explored for the recovery of metals from spent HDS catalysts. The process, spanning the leaching to precipitation stages, was conducted exclusively with ILs and DESs, without the use of conventional solvents. First, the spent catalyst was leached with an IL, which selectively extracted molybdenum and vanadium. The remaining material was then leached further using a DES to recover nickel. Various parameters influencing the leaching efficiency were examined and optimized for both IL and DES. Vanadium was recovered as NH_4VO_3 from the IL phase, precipitating selectively over molybdenum with NH_4Cl solution, while molybdenum was precipitated as BaMoO_4 by adding excess BaCl_2 to the filtrate. For nickel recovery, the IL-leached residue was treated with DES, and nickel was then recovered through cementation using aluminum [215]. The study by Kamisono et al. (2024) explored the use of an HDES composed of trihexyl(tetradecyl) phosphonium chloride and decanoic acid as an effective leaching medium for PGMs. This hydrophobic DES, when combined with an organic acid and an oxidant additive, demonstrated efficient leaching of PGMs from a SAC, while significantly reducing the leaching of unwanted metals, such as Al and Mg. Under optimal conditions: 24 h at 80 °C with stirring at 400 rpm, the leaching efficiencies for Pt, Pd, and Rh reached

90 %, 100 %, and 89 %, respectively, while the leaching of other metals was kept below 5 % for each one. The individual PGMs were isolated from the HDES through straightforward procedures involving contact with various aqueous solutions. The recovery rates of each metal into the aqueous phase were approximately 80 % for Rh and Pt, and 90 % for Pd. Additionally, the HDES could be regenerated and reused for further leaching of the SAC [217]. Lanaridi et al. (2022) presented a process for extracting PGMs from spent car catalysts. Their studies showed that several systems achieved high extraction efficiencies for Pt and Pd; however, complete extraction of Rh was not feasible under any conditions. This is attributed to both the inherent inertness of Rh and the presence of the poorly soluble Rh_2O_3 species in the car catalyst sample. Despite this challenge, the leaching system of (ChCl: PTSA) + HNO_3 was identified as the optimal choice due to its high extraction efficiency, relative simplicity of scalability, and reasonable cost. The study demonstrated that the addition of small amounts of an oxidizing agent enables DESs to act as effective leaching media for the quantitative extraction of platinum group metals. Although Rh is typically inert to acidic and oxidizing agents, it can still be extracted in significant amounts, and recovery can be further enhanced by physical pre-treatment of the spent car catalyst material [216]. Ebrahimi et al. (2023) screened eight binary and ternary DESs for stripping nickel from two spent hydro-processing catalysts. The best performers were p-toluenesulfonic-acid systems, specifically PEG-400: PTSA and a ChCl: EG: PTSA blend, each of which extracted more than 90 % of the Ni when the slurry was 20 g/L, heated to 100 °C, and stirred for 48 h [115]. In a follow-up study, the same PTSA-based DESs leached 93 % of the cobalt and 87 % of the molybdenum under identical conditions, confirming their broad effectiveness for metal recovery from spent catalysts [214].

5.2.2. Industrial dusts and sludge

Industrial dust and sludge from processes such as blast furnace, electric arc furnace, and cupola furnace operations contain numerous valuable and critical metals, including bismuth, zinc, lead, chromium, cadmium, indium, and tin. For instance, blast furnace dust in China contains approximately 20–30 thousand tons of bismuth annually, which is three to four times more than that obtained from bismuth ores. Nevertheless, only a small portion is currently recovered due to inefficient and costly extraction methods [219]. Blast furnace sludge, a byproduct of the iron extraction process, contains significant amounts of iron and carbon, along with trace metals like Mn, Zn, Pb, and Ag. This sludge can be recycled in furnaces after removing excess elements, with a focus on selectively removing Zn, which is problematic due to its high volatility. Zn evaporates rapidly and condenses on the furnace walls, disrupting the loading process and potentially damaging the furnace. The maximum allowable Zn is 120 g per ton of raw materials to prevent such issues. Efficient recovery of Fe, C, and removal of harmful elements are crucial for recycling sludge and mitigating environmental risks, as long-term disposal poses health hazards due to toxicity [220]. The electric arc furnace dust, formed during steelmaking from scrap metal, typically contains iron, zinc, lead, nickel, chromium, and cadmium [55]. Cupola furnaces, widely used in cast iron production, also generate considerable dust, with zinc oxide (ZnO) contents reaching up to 65 wt% when galvanized steel is used. These dusts contain various metal oxides, including Fe oxides, SiO_2 , CaO, and MnO, and form complex compounds such as Zn-Fe spinels. Additionally, zinc oxide dust from other sources (like zinc, lead, and copper smelting) is rich in valuable metals such as Zn, Pb, Cu, Ni, and In, though often contaminated with impurities like Cl and F, making extraction processes challenging [221,222]. Meanwhile, indium and tin, which are heavily used in electronics (e.g., in indium tin oxide for display technology), are primarily by-products of zinc refining. With indium listed as a critical raw material due to supply limitations, developing efficient recycling strategies for these metals from industrial dust is becoming increasingly vital for economic and environmental sustainability [223].

Table 10 highlights the effectiveness of ChCl and oxalic acid in

Table 10

List of prepared DESs along with their efficiency for recovering metals from industrial dust and sludge.

No.	DESs	Molar ratio	Type	Metals	Efficiency %	Optimum condition	Metal recovery process from leachate	Cyclic process	Refs.
Hydrophilic DESs (Leaching)									
1	ChCl: Oxalic Acid	1: 1	Blast Furnace Dust	Bi Zn Fe Pb	94.9 99.4 85.2 94.1	T = 70 °C t = 20 h L/S ratio: 5 mL/g	Hydrolysis using NaOH	–	[219]
2	ChCl: Malonic Acid	1: 1	Blast Furnace Sludge	Zn Fe	85 15	T = 60 °C t = 3 h S/L ratio: 1/4 g/mL	–	–	[220]
3	ChCl: Oxalic Acid	1: 1	The Dust of Ore-Thermal Melting of Ilmenite Concentrate	Ti V Fe Mn Pb Zn Si	0.14 66.2 2.2 9.0 1.47 10.1 32.5	T = 70 °C t = 10 h S/L ratio: 1/10 g/mL	–	–	[224]
4	ChCl: Urea	1: 2	Electric Arc Furnace Dust	Pb Zn	39 60	T = 60 °C t = 48 h	Electrowinning	–	[225]
5	ChCl: Urea: EG	1: 1.5: 0.5	Cupola Furnace Dust	Zn	38	T = 60 °C t = 48 h T = 79 °C	Electrowinning	–	[221]
6	ChCl: Urea	1: 2	Zinc Oxide Dust	Zn	87.9	t = 35.4 h S/L ratio: 1/12 g/mL	Electrodeposition	–	[222]
7	ChCl: Oxalic Acid	1: 1	Zinc Flue Dust	Fe Zn Pb Cu In Sn	83 88 93 76 92 88	T = 0 °C t = 48 h S/L ratio: 1/10 g/mL	Precipitation	–	[223]
8	ChCl: Urea: EG	2: 4: 1	Zinc Oxide Dust	Zn	85.2	T = 80 °C t = 48 h S/L ratio: 1/20 g/mL	Electrodeposition	3 cycles	[227]
9	ChCl: Fructose	1: 1	Ferromolybdenum Furnace Dusts	Mo Fe Zn	80 >30 50	T = 80 °C t = 6 h L/S ratio: 135 mL/g	–	–	[228]
10	ChCl: Urea (Ultrasonic Leaching)	1: 1	Zinc-Containing Metallurgical Dust Sludge	Zn	98	T = 60 °C t = 240 min	–	–	[226]

recovering metals from various industrial residues, including blast furnace dust [238], dust generated from the ore-thermal melting of ilmenite concentrate [224], and zinc flue dust [223]. Zürner et al. (2019) examined three ChCl-based solvents paired with EG, urea, or oxalic acid for leaching indium and tin from an oxide flue dust sample. ChCl: oxalic acid delivered the best results, dissolving 92 % of In and 88 % of Sn within 48 h. A two-step precipitation scheme then removed the major impurities (Fe, Zn, Pb, Cu) while keeping In and Sn in solution. When the leachate was diluted with water, a white solid containing Zn, Pb, and Cu formed rapidly; exposure to a photoinitiated reduction reaction produced a yellow Fe precipitate. Neither solid contained detectable indium or tin, so both target metals remained concentrated in the DES phase and could be recovered later by conventional solvent extraction. The high chloride level in the DES suppresses the co-precipitation of Fe with In, resulting in higher indium retention and better tin recovery compared to comparable aqueous processes [55,223].

In addition, ChCl: urea-based DESs have potential for recovering zinc from electric arc furnace dust [225], cupola furnace dust [221], and zinc oxide dust [222]. After leaching, the metal recovery process involved both electrowinning and electrodeposition. In the study by Niu et al. (2022), researchers focused on zinc-containing dust and sludge generated from iron and steel smelting. They developed ChCl: urea as a leaching agent. The study compared conventional leaching methods with ultrasonic leaching of zinc and analyzed how various factors,

including liquid-to-solid ratio, temperature, time, and ultrasonic power, influenced the zinc leaching rate. The results indicated that the ChCl: urea effectively solubilized ZnO, achieving a zinc leaching efficiency of over 98 % at a temperature of 60 °C, an ultrasonic power of 350 W, and a leaching time of 240 min. Kinetic analysis of the ChCl: urea leaching process demonstrated that it followed the nucleation contraction model under both conventional and ultrasonic conditions [226].

5.2.3. Wastewater

As previously discussed, traditional metallurgical industries have faced difficulties in meeting the growing demand for precious metals solely by extracting mineral resources. Consequently, waste catalysts, electronic components, and alloy materials have become crucial secondary sources of these valuable metals. The recovery process primarily involves hydrometallurgical methods, such as leaching with aqua regia, strong acids, or alkalis. These methods, however, generate large volumes of wastewater containing precious metal ions. In 2020, China alone produced around 980 million tons of such wastewater, which is highly saline and acidic, with precious metal ion concentrations ranging from 2 to 1000 mg/L. In addition to these precious metals, the wastewater also contains heavy metals, with concentrations ranging from 500 to 30,000 mg/L. Improper disposal of this wastewater not only results in the loss of precious metals but also contributes to significant environmental pollution [229]. Industrial sectors such as battery manufacturing, textiles, electroplating, and metal plating release

Table 11

List of prepared DESs along with their efficiency for recovering metals from wastewater.

No.	DESs	Molar ratio	Type	Metals	Efficiency %	Optimum condition	Metal recovery process from leachate	Cyclic process	Refs.
Hydrophobic DESs (Extraction)									
1	TAC (N263): Oleic Acid	1: 1	Acidic Leaching Solution of Solid Waste	Ir(IV) Ru(III) Pt(IV) Rh(III)	97.57 94.26 99.55 15.1	t = 5 min O/A ratio:1/ 1	Stripping	5 cycles	[229]
2	Lidocaine: Heptanoic Acid + Toluene (50 % v/v)	1: 1	Wastewater	Cd(III) Pb(II) Ni(II)	>90	t = 30 min O/A ratio:1/ 1	Stripping: Na ₂ C ₂ O ₄	5 cycles	[230]
3	DL-menthol: Acetic Acid DL-menthol: Acetic Acid : Pyruvic Acid	–	Sewage Sludge	Cd	>99	t = 15 min S/L ratio: 150 mg/ 4.0 mL HDES	–	–	[232]
4	Decanoic Acid: Lidocaine	2: 1	Non-Buffered Water	Co Cu Fe Mn Ni Zn	–	t = 5 s	Stripping: Na ₂ C ₂ O ₄	Can be reused	[233]
5	ChCl: Urea and Betaine: Urea (LLE)	1: 2 1: 2	Electroplating Wastewater	Cr	>85	T = 25 °C	–	–	[234]
6	Methyltriocetyl Ammonium Bromide: Palmitic Acid (LLME)	1: 1	Industrial Wastewater	Cr(VI)	97.4	t = 3 min	–	8 cycles	[235]
7	ChCl: Phenol Ultrasound assisted (ELPME)	1: 3	Chromium Plating Factory Wastewater	Cr(III) Cr(VI)	95 >99	–	–	–	[236]
8	(Betaine: Sorbitol) + 10 % water (v/v) (Ultrasound microextraction)	1: 3	Dental Wastewater Industrial Wastewaters	Hg	>99 >95	–	–	–	[237]
9	ChCl: Oxalic Acid (VAME)	1: 1	Wastewater	As(III/ V) Sb(III/ V)	>95 >99	–	–	–	[238]
10	Methyl Anthranilate: Lidocaine DL-menthol: Lidocaine DL-menthol: Proton Sponge (LLE)	9: 1 1: 1 7: 3	Hydrochloric Acid Solutions	In	99	–	–	–	[93]
11	Decanoic Acid: Lidocaine	2: 1	From Aqueous Media	Fe(III) Mn(II)	99.3 8.31	–	–	–	[239]
12	ZnCl ₂ : Acetamide (UA-DLLME)	1: 3	Wastewater	Cr(III)	>95	–	–	–	[240]
13	TOPO: 1-butanol (Fatty alcohol)	1: 1	Secondary Resources in HCl Medium	Pt(IV)	98.9	–	Stripped: NaOH Solution	4 cycles	[241]
14	TBAC: Decanoic Acid (LPME)	1: 4	Wastewater	Ni	97	–	–	–	[242]
15	TOPO: Decanoic Acid	1: 1	Synthetic Pregnant Leach Solution	Mo Re Cu Fe	>99 >99 <10 <10	T = 25 °C t = 2 h	Precipitation	–	[243]
16	DL-menthol: Lauric Acid (LLE)	1: 3	Lithium Extraction from Aqueous Medium	Li	>80	T = 25 °C t = 30 min O/A ratio: 0.5	Stripping: HCl	–	[244]
17	TOPO: Propanol TOPO: DL-menthol	1: 1 4: 1	From aqueous media	Ce(III)	>93 >93	T = 20 °C t = 15 min O/A ratio: 1/1	–	–	[231]
18	[N ₈₈₈₈][Br]: C ₆ OOH	1: 2	From Aqueous Media	Cr(VI) Pt Pd Ru Ir Rh	99.7 99.7 99.7 87.3 45.1 25.1	O/A ratio: 1/66	Precipitation	8 cycles	[245]
19	Tetraoctylammonium Bromide: Hexanoic Acid	1: 2	Model Hydrochloric and Chloride Media	Pd(III)	>90 >90	T = 25 °C t = 20 min	Hydrazine Hydrate Reduction	5 cycles	[246]
20	[N ₈₈₈₁][Cl]:1-hexanol [N ₈₈₈₁][Cl]: Hexanoic Acid	1: 1 1: 2	Hydrochloric Acid Solution	Pd(III)	>90 >90	T = 25 °C t = 20 min	Hydrazine Hydrate Reduction	5 cycles	[247]
21	HBTA: TRPO	1: 1	Mother Liquor of Lithium Carbonate	Li	>90	t = 5 min O/A ratio: 1/3	Stripping: HCl	5 cycles	[248]

various heavy metals, including Ni(II), Cu(II), Zn(II), Cd(II), Hg(II), and Pb(II), into water bodies, especially in developing nations. These metals can accumulate in rivers, lakes, and oceans, causing serious health risks, including damage to the lungs, liver, kidneys, and immune system [230]. Solvent extraction remains a preferred method for recovering these metals due to its high recovery efficiency, large processing capacity, and safety [229]. Table 11 demonstrates the use of HDESs in

extracting metals from various wastewater types, as well as from model systems in aqueous and hydrochloric acid solutions.

In this regard, Edgecomb et al. (2020) explored the use of hydrophobic binary liquid mixtures incorporating Proton Sponge® (one of the strongest organic bases) for the extraction of In(III) from hydrochloric acid solutions. These mixtures, which include components such as DL-menthol, lidocaine, ibuprofen, and methyl anthranilate, demonstrated

high extraction efficiencies of over 99 % in some cases. The extraction mechanisms were systematically studied, with an emphasis on factors such as charge, solubility, and indium complexation. Classical solubility principles have been shown to effectively predict liquid formation, aiding in solvent design. Additionally, glass transition temperatures were linked to hydrogen bonding networks. The research highlighted the strong extraction abilities of lidocaine and Proton Sponge® when combined with suitable hydrogen bond donors. A mathematical model was also developed to quantify the contributions of different indium species, revealing that neutral indium hydroxide and mixed hydroxy-chloride species predominated at low HCl concentrations, while methyl anthranilate enhanced extraction efficiency at higher acid levels (>0.5 M) [93]. Guidugli et al. (2024) addressed the urgent need for environmentally friendly and efficient methods to extract REEs, focusing on Ce (III) as a model element using the HDESs based on TOPO, combined with green HBDs such as propanol, DL-menthol, and octanoic acid. The extraction efficiencies of these HDESs were systematically evaluated under varying experimental conditions, including different solvent compositions, HBA: HBD molar ratios, and settling times. Among the tested systems, the TOPO: propanol (1: 1) mixture achieved the highest extraction efficiency (93.8 ± 1.54 %) for Ce(III) from a 30 mM nitrate solution. COSMO-RS modeling was employed to analyze sigma surfaces and potentials, providing molecular-level insights into solvent interactions. The findings highlighted that increasing the TOPO content improved extraction efficiency, whereas higher proportions of HBDs reduced it. Furthermore, viscosity and phase separation kinetics were examined, with TOPO: DL-menthol exhibiting the fastest equilibrium time and lowest viscosity, while TOPO: octanoic acid showed the

slowest kinetics and highest viscosity [231].

In 2024, Shaibuna et al. developed four non-ionic HDESs using lidocaine or ibuprofen combined with heptanoic acid in different molar ratios to efficiently remove metals such as Cd(II), Pb(II), and Ni(II) from aqueous solutions. Among the formulations tested, HDES1, which contained lidocaine and heptanoic acid in a 1: 1 molar ratio, exhibited the highest extraction performance, particularly when combined with 50 % v/v toluene. This mixture's low viscosity and density enhanced its efficiency in extracting both individual and mixed metal ions, even under high contamination levels that mimicked real industrial wastewater conditions. The extraction process was optimized by adjusting key parameters, including HDES concentration, extraction time, volume ratio, and pH. Importantly, HDES1 demonstrated consistent efficiency across a wide pH range and retained its performance after multiple reuse cycles, highlighting its robustness and sustainability. Additionally, metal ion stripping with 0.1 M sodium oxalate confirmed the potential for solvent recovery and reuse. This study highlighted the potential of HDES-based LLE as a cost-effective and environmentally friendly approach for large-scale wastewater treatment in industrial applications [230]. Also, Wang et al. (2023) developed an HDES composed of trioctylmethylammonium chloride (N263) and oleic acid to recover precious metals Ir(IV), Pt(IV), Ru(III), and Rh(III) from acidic leaching solutions of solid waste. The extraction performance of the HDES was evaluated under various conditions, including different N263: OA molar ratios, aqueous phase pH, phase ratios (O/A), and contact times. At an optimized 1: 1 molar ratio, pH 1, and a 5-min contact time, the HDES demonstrated high single-stage extraction efficiencies of 97.57% for Ir(IV), 94.26% for Ru(III), and 99.55% for Pt(IV), with a comparatively lower efficiency of 15.15 %

Table 12
List of prepared DESs along with their efficiency for recovering metals from waste Fly Ash and Smelting Slag.

No.	DESS	Molar ratio	Type	Metals	Efficiency %	Optimum condition	Metal recovery process from leachate	Cyclic process	Refs.
1	ChCl: Humic Acid	1: 1	Coal Ash	Lu	90	T = 90 °C t = 2 h	–	–	[249]
2	ChCl: PTSA	1: 1	Coal Fly Ash (Thermal power plant in southern India)	REE LREE HREE	94.8 96.8 89.6	T = 100 °C t = 5 h	Precipitation: Oxalic Acid and Sodium Fluoride	–	[250]
				Fe Cu Pb Li Al	49 89 63 16 24	T = 40 °C t = 30 min			
3	(ChCl: EG) + Nitrilotriacetic Acid (as a chelating agent)	1: 1	Blast Oxide Slag	Mn Ni Zn Pb Al Cr Cd	15 24 64 18 26 6 26	L/S ratio: 75 mL/g	–	–	[253]
						T = 48.5 °C t = 40.1 h L/S ratio: 62.3mL/g			
4	ChCl: Urea	1: 2	Waste Smelting Slag	Cu Zn	80 61	T = 40 °C t = 24 h S/L ratio: 1/10 g/mL	–	–	[254]
5	ChCl: Levulinic Acid	1: 3	Goethite Residue in the Zinc Industry	Zn	>50	T = 30 °C t = 24 h S/L ratio: 1/40 g/mL	–	–	[255]
6	ChCl: Urea	1: 2	Different Metal-Bearing Species (Sulfides, oxides, and sulfates)	Cu Zn Fe Pb	–	T = 90 °C t = 5 h	Electrowinning	–	[252]
7	ChCl:Glycerin	2: 1	Automotive Battery Recycling Slag	Pb	95	T = 90 °C t = 5 h			
8	TBAC: Citric Acid	1: 1	Waste Smelting Slag	Cu Zn Fe	77 49 1	T = 90 °C t = 2 h S/L ratio: 1/3 g/mL	–	–	[256]

for Rh(III). The HDES exhibited a high saturation loading capacity, achieving up to 18.44 g/L of precious metals. Mechanistic studies using UV-Vis, FTIR, and ESI-MS revealed that extraction occurred via ion exchange, where metal chloride complex ions displaced chloride ions electrostatically bound in the N263 structure. This process resulted in the formation of stable metal-organic complexes in the organic phase. The system also demonstrated excellent selectivity during scrubbing, efficiently removing Cu(II) and Fe(III) ions without stripping the precious metals. Subsequent stripping with various reagents resulted in an over 96 % recovery of the target metals [229].

5.2.4. Fly ash and smelting slag

As illustrated in Table 12, Mehmood et al. (2024) explored the use of coal ash as an alternative source for REEs, addressing the global supply-demand gap for these critical materials. Traditional extraction methods involving mineral acids and ILs are often hazardous, corrosive, and environmentally unfriendly. In this study, the researchers developed a DES composed of ChCl: humic acid, aiming to provide a sustainable alternative for REE recovery. Leaching experiments were conducted under controlled agitation (500 rpm) across varying temperatures (60–90 °C) and leaching durations (1–2 h). The ICP-MS analysis revealed that the coal ash sample contained a total REE concentration of 351.41 ppm, with light REEs (LREEs) at 231.91 ppm and heavy REEs (HREEs) at 119.5 ppm. Lutetium was the most abundant, at 96.56 ppm. The leaching efficiency of the DES increased with both time and temperature, reaching a maximum of 90 % at 90 °C after 2 h. SEM images of the raw and treated coal ash samples showed a more dispersed distribution of luminous and non-luminous mineral features after DES treatment, confirming enhanced leaching activity [249]. Karan (2022) studied the use of ChCl: lactic acid and PTSA for recovering REEs from coal fly ash, which had about 0.22 % REEs. Leaching experiments evaluated various factors influencing REE recovery. The ChCl: PTSA system demonstrated greater efficiency and faster kinetics compared to the ChCl: lactic acid system. Both DESs outperformed their individual components, achieving leachability of 85–95 %, which was 5–8 % higher than that of the individual HBDS and HBA. Additionally, they provided a 35 % increase in REE extraction efficiency compared to sulfuric acid. The REEs were selectively precipitated from the DES media using oxalic acid and sodium fluoride (NaF), leading to near-complete recovery. The study highlighted the leaching mechanism and promoted DESs as eco-friendly alternatives to conventional mineral acids for the recovery of REEs from industrial waste [250]. Topçu et al. (2024) conducted a study focused on optimizing the recovery of copper from copper smelting slag using a DES as a green leaching agent. Chemical and mineralogical characterization of the slag revealed it contained 0.9 % Cu, 3.3 % Zn, and 36.7 % Fe, primarily in the forms of Fe_2SiO_4 , Fe_3O_4 , and SiO_2 , with copper present as CuO and CuS. Under optimal experimental conditions (95 °C, 48 h, 1/25 g/mL solid/liquid ratio, and particle size of 33 μm), 80 % Cu and 61 % Zn recoveries were achieved, while iron and silicon dissolution remained minimal, demonstrating selective metal recovery. A predictive mathematical model was developed to optimize the copper leaching process. The model indicated that maximum copper recovery (up to 100 %) could be achieved at 48.5 °C, 40.1 h, and a liquid/solid ratio of 62.3 mL/g. This modeling approach showed flexibility in process parameters and allowed for predicting leaching efficiencies over a broad operational range [251]. Salgado et al. (2024) explored an environmentally benign approach to address the ecological concerns associated with automotive battery recycling, particularly the lead-rich foundry slag. Toxicity analysis revealed lead concentrations reaching up to 5.4 %, surpassing regulatory thresholds. Anglesite (PbSO_4) was identified as the predominant lead phase, as confirmed by XRF and XRD analyses. To mitigate this, the study employed a DES system composed of ChCl: glycerin as a green leaching agent. Through systematic optimization of five operational parameters, such as sample type, solvent composition, concentration, temperature, and time, the authors achieved up to 95 % lead dissolution at 90 °C

under specific conditions (5 % pulp, 470 rpm, 5 h). Additionally, 55 % of the dissolved lead was successfully recovered via electrowinning. This work highlights the potential of DES-based techniques for sustainable slag decontamination and metal recovery [252].

5.3. Mineral

Conventional metal extraction techniques, such as high-temperature smelting and leaching with strong acids, bases, or organic solvents, are associated with significant energy demands and the production of hazardous gaseous, liquid, and solid wastes, raising serious environmental concerns. REEs, primarily found in complex minerals like bastnäsite, monazite, and ion-adsorption clays, are particularly challenging to purify due to their similar physicochemical properties. Solvent extraction remains the predominant industrial method for REE production owing to its high throughput, effective separation capabilities, and rapid mass transfer. However, in response to environmental and economic challenges, the mining industry is increasingly prioritizing the development of greener and more cost-effective alternatives that ensure industrial feasibility. Among these, DESs have emerged as a promising class of green solvents. DESs offer several advantages, including low toxicity, high biodegradability, and cost-effectiveness compared to traditional extractants, making them a sustainable solution for recovering metals from complex mineral matrices [5,55].

In recent publications, based on Table 13, Drogobuzhskaya et al. (2024) conducted a comparative study on the efficiency of acid-based DESs and their corresponding saturated aqueous solutions in extracting transition metals and REEs from mining tailings, as well as various metals from biodiesel samples. The DESs were formulated using malonic, malic, tartaric, and citric acids with choline chloride as the HBA. While DESs demonstrated potential as green solvents, the study found that aqueous acid solutions generally exhibited higher metal extraction efficiencies than their DES counterparts. This was attributed to the lower viscosity and higher diffusivity of aqueous systems, as well as possible side reactions in DESs that may hinder metal complexation. The findings highlighted that DESs are not universally superior and emphasize the need for further investigation into the roles of extractant composition, matrix effects, and metal coordination behavior. The authors advocated for a nuanced understanding of when and where DESs can be most effectively applied in metal recovery processes [257]. Moreno et al. (2024) employed DESs as a greener alternative to conventional metallurgical methods for the recovery of zinc and lead from sphalerite and galena concentrates sourced from Ecuadorian mining operations. Three DES systems were evaluated: ChCl: urea, EG, glycerol. Leaching experiments were conducted at 30 °C for 24 h on both untreated and thermally pretreated (roasted at 600 °C) mineral samples. The influence of iodine as an oxidizing agent was also assessed. In untreated samples, DES leaching yielded low metal recoveries (2 % Zn and 14 % Pb), but the addition of iodine significantly enhanced the efficiency to 11 % for Zn and 99 % for Pb. Roasting the concentrate independently improved recoveries to 75 % for Zn and 90 % for Pb. When both roasting and iodine oxidation were combined, recoveries for both metals reached 99 %. These results demonstrate that DESs, particularly when combined with pretreatment and oxidizing agents, can offer a viable and environmentally friendly alternative to traditional acid leaching methods (e.g., H_2SO_4 and HNO_3) for metal extraction from primary ores [258]. Sudová et al. (2024) investigated the application of DESs for the environmentally sustainable leaching of antimony from mining residues, acknowledging antimony as a critical mineral frequently embedded in waste from antimonite ore flotation. Various ChCl-based DES with malonic acid, thiourea, and ethylene glycol in different molar ratios were tested. The study demonstrated that, under optimal conditions, specifically, a leaching temperature of 100 °C, a reaction time of 4 h, and the addition of up to 3 g of iodine as an oxidizer, antimony leaching efficiency reached 100 %. This study contributed to the development of greener practices in critical mineral recovery and

Table 13

List of prepared DESs along with their efficiency for recovering metals from mineral samples.

No.	DESs	Molar ratio	Type	Metals	Efficiency %	Optimum condition	Metal recovery process from leachate	Cyclic process	Refs.
1	ChCl: Urea: Malonic Acid	1: 1: 0.5	Multicomponent Mixtures of Y, La, Ce, Nd and Sm, and Bastnasite Mineral	Y La Ce Nd Sm	90 <20 <40 90 >90	$T = 50\text{ }^{\circ}\text{C}$ $t = 8\text{ days}$	Precipitation: Oxalic Acid	–	[263]
2	EG: Maleic Acid	4: 1	REOs	La Eu Na K Ca Mg Cu	>99	$T = 60\text{ }^{\circ}\text{C}$ $t = 120\text{ min}$	Precipitation: Oxalic Acid	3 cycles	[101]
3	ChCl: Malic Acid	1: 1	Mining Tailings Biodiesel Fuel	Fe Mn Pb Ba Ni Sr	>95	$t = 30\text{ min}$, S/L ratio: 1/ 1 g/L	–	–	[257]
4	ChCl: PTSA	2: 1	REEs from Caustic-Treated Monazite Concentrate	Ce La Nd	99 99 94	$T = 100\text{ }^{\circ}\text{C}$ $t = 72\text{ h}$ S/L ratio: 1/ 10 g/L	–	–	[260]
5	GUC: Lactic Acid	1: 2	REOs	La ₂ O ₃ Eu ₂ O ₃ Dy ₂ O ₃	> 90 < 10 < 10	$T = 50\text{ }^{\circ}\text{C}$ $t = 6\text{ h}$ S/L ratio: 1/ 50 g/L	Precipitation: Oxalic acid	3 cycles	[264]
6	TOMAC: Decanoic Acid	1: 2	From Laterite Ores	Mn Ni Co Mg Fe Si Cr Lu Yb Tm	> 90 > 90 > 90 < 20 > 90 < 10 < 40	$T = 60\text{ }^{\circ}\text{C}$ $t = 1\text{ h}$	Stripping and Scrubbing	4 cycles	[261]
7	1-Decyl alcohol: Oleic Acid: BEA	9: 1: 5	Separation of Yttrium from HREEs	Er Ho Dy Y	$90 > \text{Lu} \geq \text{Yb} > \text{Tm} > \text{Er} > \text{Ho} \geq \text{Dy} \gg \text{Y} > 50$	–	Stripping: Na ₂ C ₂ O ₄	5 cycles	[265]
8	ChCl: Oxalic Acid	1: 1	Secondary Mine Resources	W As	77 82	–	–	–	[266]
9	ChCl: EG: Malonic Acid	1: 1: 0.5	Pure Yttrium Carbonate Salts	Y	88.37	$T = 80\text{ }^{\circ}\text{C}$ $t = 60\text{ min}$	–	–	[267]
10	Decanoic acid: Aliquat 336	2: 1	Pressure Oxidative Leaching Solution of Molybdenite Concentrate	Re	>90	$T = 25\text{ }^{\circ}\text{C}$ $t = 10\text{ min}$ O/A ratio: 1/ 25	–	–	[268]
11	ChCl: PTSA: EG	1: 1: 1	Sphalerite Concentrate	Zn	99	$T = 100\text{ }^{\circ}\text{C}$ $t = 24\text{ h}$	–	–	[262]
12	FeCl ₃ : EG	–	Chalcopyrite and other ore minerals	Cu Fe	80 100	$T = 60\text{ }^{\circ}\text{C}$ $t = 24\text{ h}$	Electrochemistry	2 cycles	[269]
13	(ChCl: Urea) + I ₂	1: 2	Sphalerite–Galena Concentrate	Zn Pb	99	$T = 30\text{ }^{\circ}\text{C}$ $t = 24\text{ h}$	–	–	[258]
14	(ChCl: EG) + I ₂	1: 2	Mining Residues	Sb	100	$T = 100\text{ }^{\circ}\text{C}$ $t = 4\text{ h}$	–	–	[259]
15	ChCl: EG: Oxalic acid	1: 1: 2	Chalcopyrite	Cu	76	$T = 80\text{ }^{\circ}\text{C}$ $t = 48\text{ h}$	–	–	[270]
16	ChCl: Sulfamide	1: 2	Precious Metals from Ores	Au	>95	$T = 80\text{ }^{\circ}\text{C}$ $t = 25\text{ min}$	–	–	[112]
17	ChCl: Malonic Acid	1: 1	Indium Tin Oxide from the Mining Company	In Sn	99 >90	S/L ratio: 1/ 40 g/L	Precipitation	–	[271]
18	(ChCl: Citric Acid) + 30 % water	1: 1	Mining Tailings	Y Zr Nb La Ce Pr Nd Sm Eu Gd	95 61 25 55 48 44 53 57 64 49	S/L ratio: 1/ 10 g/L	–	–	[257]

(continued on next page)

Table 13 (continued)

No.	DESs	Molar ratio	Type	Metals	Efficiency %	Optimum condition	Metal recovery process from leachate	Cyclic process	Refs.
				Tb, Dy Ho Er Tm Yb Lu Hf Ta	58 95 83 93 90 89 82 61 20				
19	TOPO: 3,5-dinitrosalicylic acid (DNS)	1: 1	–	Yb Lu	>98 >98	T = 90 °C t = 4 h O/A ratio: 1/1	Stripping	5 cycles	[272]

supported more sustainable waste management strategies in the mining industry [259].

Shakiba et al. (2023) investigated DESs as an alternative to traditional mineral acids for extracting REEs from monazite. They synthesized eleven different DESs using various HBDs and HBAs. However, initial leaching tests revealed that none of the DESs were effective in directly extracting REEs from the monazite phosphate matrix. To address this challenge, the monazite concentrate was subjected to an alkaline roasting process with sodium hydroxide at 500 °C for 2 h, followed by water leaching. This method successfully converted the REE phosphates into rare earth hydroxides or oxides, enabling efficient extraction. Subsequent leaching using the synthesized DESs, particularly ChCl: PTSA, ChCl: PTSA: EG, and PEG-400: PTSA, yielded significant recoveries of REEs. Among them, ChCl: PTSA exhibited the highest extraction performance, achieving up to 99.9 % La, 99.8 % Ce, and 94 % Nd recovery under optimized conditions (100 °C, 72 h, 10 g/L solid loading). The study identified protonation as the likely extraction mechanism, particularly facilitated by the acidic character of PTSA DESs. The findings underscore the potential of DESs, especially strongly

acidic formulations, for high-efficiency, low-toxicity recovery of REEs from pretreated monazite [260]. Sakamoto et al. (2024) introduced a novel HDES system for the selective extraction and recovery of critical battery metals Ni, Co, and Mn from low-grade nickel laterite ores, specifically saprolite. The DES was synthesized using decanoic acid: TOMAC, with HCl as a co-leaching agent. This system enabled effective leaching of Ni, Co, and Mn while significantly suppressing the co-dissolution of common impurities such as Mg and silicon, offering a distinct advantage over traditional sulfuric acid leaching. The improved selectivity was attributed to the coordination behavior and metal affinity within the hydrophobic DES, which provided a unique solvent environment capable of stabilizing target metals. Additionally, the DES facilitated efficient downstream processing, including metal stripping and scrubbing, which allowed for the separation of desired metals from Fe and demonstrated excellent reusability across at least four leaching cycles [261].

Shahrezaei et al. (2024) explored the leaching behavior of sphalerite concentrate using a non-aqueous, biocompatible DES composed of ChCl, PTSA, and EG in a 1: 1: 1 molar ratio. They optimized several key

Table 14

List of prepared DESs along with their efficiency for recovering metals from water samples.

No.	DESs	Molar ratio	Type	Metals	Efficiency %	Optimum condition	Metal recovery process from leachate	Cyclic process	Refs.
1	Tri-Octylmethylammonium Chloride (TAC): Butyl 4-hydroxybenzoate	1: 3	River Water and Outlet Water of Sewage Treatment Plant	Cr(VI)	>90	T = 15 °C t = 1 day O/A ratio: 1/1	–	–	[273]
2	ZnCl ₂ : Acetamide (UA-DLLME)	1: 3	Ground Water, Seawater, Canal Water, Mineral Water, Tap Water Bottled Mineral	Cr(III)	>95	–	–	–	[240]
3	ChCl: Ethylene Glycol (UA-LLME)	1: 3	Water, Tap. Water, Seawater, Wastewater	Cr(III) Cr(VI)	>96	t = 5 min	–	–	[276]
4	TBAC: Decanoic Acid (LPME)	1: 4	Lake Water	Ni	93	–	–	–	[242]
5	DI-Menthol: Decanoic Acid (US-DLLME-SF)	1: 1	Tap, Mineral, Sea, and River Waters, Broccoli, Spinach	Ni Co	95.2–100.8	–	–	–	[277]
6	Trihexyl(tetradecyl)Phosphonium Chloride: Thiosalicylic Acid (UA-DLLME-SA)	1: 2	Lake Water River Water Well Water	Pb(II) Cd(II) Co(II) Ni(II)	91–101	–	–	–	[274]
7	ChCl: Oxalic Acid (VAME)	1: 1	Well, Tap, River, Bottled Water, Honey White and Brown Rice	As Sb	>94 >93	–	–	–	[275]
8	ChCl: Phenol (UA-LPME)	1: 3	Tap Water Mineral Water Iced Tea	Se(VI) Se(IV)	>97	–	–	–	[278]

operational parameters, including leaching time (20–1440 min), temperature (40–100 °C), and pre-leaching milling duration (0–24 h). The study found that higher temperatures and extended leaching times significantly improved zinc dissolution efficiency. Additionally, longer ball milling times increased recovery by reducing particle size and creating surface defects, which facilitated more effective leaching. Under optimal conditions (100 °C, 24 h of milling, 1440 min of leaching), a zinc recovery rate of 99.7 % was achieved. Infrared spectroscopy confirmed the chemical stability of the DES throughout the process. The leaching mechanism involved the formation of ZnCl_2 ion complexes and the release of H_2S gas. Residual analysis unexpectedly revealed the formation of lead sulfate, likely resulting from sulfide oxidation. Kinetic modeling using the Avrami model provided R^2 values between 0.87 and 0.97, indicating a diffusion-controlled dissolution process. Molecular dynamics simulations revealed strong electrostatic interactions between Zn, Fe, and Cl^- ions of ChCl, as well as complexation with O-S-O groups in PTSA. This research highlighted the potential of the ChCl: PTSA: EG DES for efficient and environmentally friendly zinc recovery from sulfide ores [262].

5.4. Environmental samples

Currently, most DESs are hydrophilic, which limits their ability to efficiently extract non-polar or weakly polar compounds from aqueous environments. To overcome this limitation and support the development of more sustainable extraction processes, recent advancements have introduced HDESSs as a significant breakthrough. HDESSs are more hydrophobic, enabling the effective extraction of a wide variety of micropollutants such as pharmaceuticals, pesticides, plastic additives, dyes, hormones, UV filters, polycyclic aromatic hydrocarbons, and explosives from water, and to a lesser extent, from soil samples. The hydrophobicity of HDESSs, which depends on factors like composition, hydrocarbon chain length, pH, and the solvent-to-water volumetric ratio, enhances separation performance and helps minimize secondary pollution. Recent studies have demonstrated that HDESSs exhibit high stability in aqueous environments, positioning them as a promising, eco-friendly alternative to traditional extraction solvents in environmental and analytical applications [46,51,54].

5.4.1. Water samples

The impact of trace elements on human health and the environment, both positive and negative, has gained increasing scientific attention. Based on Table 14, a notable example of the use of DESs for extracting metals from environmental samples is the work by Erbaş et al. (2019), who developed a method known as DES-LPME for preconcentrating and determining Ni(II) using MS-FAAS. The DES was composed of

tetrabutylammonium chloride and decanoic acid in a 1: 3 molar ratio. This solvent effectively extracted nickel when it was complexed with sodium diethyldithiocarbamate (NaDDTC). The method was optimized for various analytical parameters, such as pH, DES volume, and sample volume. It achieved a preconcentration factor of 60, with excellent sensitivity, demonstrated by a limit of detection of 0.13 $\mu\text{g/L}$ and a relative standard deviation of 3.2 %. The technique was successfully validated using certified reference materials and applied to the analysis of water and food samples, demonstrating superior performance compared to traditional methods [242]. Shi et al. (2020) created a new family of HDESSs by pairing parabens with the quaternary ammonium salt tri-octylmethyl-ammonium chloride. These solvents selectively pull Cr(VI) from water, reaching a loading of 66.7 mg/g and recovering more than 90 % of the chromium even at sub-ppm levels (0.51 mg/L). Their selectivity remains high in the presence of other metal ions or organics at concentrations up to 2000-fold greater than Cr(VI). DESs are prepared in a one-step, waste-free process that incorporates all starting atoms and remain stable and reusable over multiple cycles. Extraction is driven by the synergistic effects of electrostatic attraction, ion exchange, and hydrogen bonding, making these solvents an efficient and sustainable option for removing heavy metals from environmental waters [273]. Werner (2020) developed a green, high-efficiency protocol for concentrating and quantifying trace heavy metals in environmental samples. The procedure combined ultrasound-assisted DLLME-SFOD with subsequent HPLC detection. Central to the method was a low-density DES composed of trihexyl(tetradecyl)phosphonium chloride and thio-salicylic acid. Functioning simultaneously as both an extractant and a complexing ligand, this DES was ultrasonically dispersed throughout the sample, followed by phase separation through centrifugation and brief freezing, thereby eliminating the need for volatile organic solvents. The technique demonstrated excellent precision (RSD: 2.6–4.7 %), very low detection limits (0.05–0.13 $\mu\text{g/L}$ for Pb, Cd, Co, and Ni), and high enrichment factors (154–162). Recoveries on certified reference materials ranged from 91.4 % to 101.6 %, confirming the method's accuracy [274]. Altunay et al. (2020) developed an environmentally friendly VAME procedure that uses a DES to pre-concentrate arsenic and antimony in environmental waters, honey, and rice before measurement by hydride-generation AAS. At pH 10.5, As(III) and Sb(III) formed chelates with dithizone (H_2Dz), and these complexes were taken up by a ChCl: oxalic acid (1: 3). The method achieved detection limits of 7.5 ng/L for As and 15.6 ng/L for Sb, relative standard deviations of 2.1 % and 2.7 %, recoveries of 93.5 % (As) and 96.2 % (Sb), and enrichment factors of 104 and 85, respectively. Compared to conventional techniques, the DES-VAME approach is simpler, cheaper, and greener, yet it is fully suited for the trace-level speciation of inorganic arsenic and antimony [275].

Table 15

List of prepared DESs along with their efficiency for recovering metals from waste soil samples.

No.	DESs	Molar ratio	Type	Metals	Efficiency %	Optimum condition	Metal recovery process from leachate	Cyclic process	Refs.
1	(ChCl: Fructose: Water) + 2 % Saponin	5: 2: 5	Remove Lead from a Landfill Soil,	Pb	>72	–	–	–	[281]
2	Glycolic acid: L-proline	3: 1	Soil Heavy Metals (Simulated soil)	Pb	95.28	T = 40 °C t = 4 h	–	–	[90]
3	(ChCl: Glycerol) + 2 % Saponin	–	Contaminated Soil	Pb	>70	–	–	–	[282]
4	ChCl: Lactic Acid ChCl: Levulinic Acid	1: 2 1: 2 2: 1	Contaminated Soils	Pb	97 99	–	–	–	[279]
5	ChCl: Malonic Acid: Water	(DES: H_2O : 2: 1)	Contaminated Soils	Pb	92.12	T = 40 °C t = 30 min	–	–	[283]
6	ChCl: L-lactic Acid: Water	1: 2 (DES: H_2O : 2: 8)	Contaminated Soil	Cd Pb Cu Zn	43.42 94.73 90.72 96.44	T = 25 °C t = 1440 min S/L ratio: 1/20 g/L	–	–	[280]

5.4.2. Soil samples

DESs, particularly NADESs, are emerging as eco-friendly alternatives for soil washing in remediation efforts, especially for heavy metal contamination. While water remediation often employs hydrophobic DESs, soil remediation research focuses more on hydrophilic solvents. Studies, including those summarized in Table 15, demonstrate that NADESs derived from amines, diols, and polyols (such as urea-based ones) typically exhibit low efficiency in removing heavy metals and are unsuitable for soil washing. In contrast, NADESs using organic acids, especially lactic acid HBDs, demonstrate high removal efficiency (above 90 %) due to their strong acidity and minimal steric hindrance. Lactic acid's short hydrocarbon chain and strong ionization capability made it one of the most effective HBDs for producing NADESs in soil remediation [51]. For instance, Rashid et al. (2022) investigated the effectiveness of two DESs: ChCl combined with lactic acid and ChCl combined with levulinic acid. Both DESs demonstrated high efficiency in removing Pb(II), achieving removal rates of 97 % for ChCl: lactic acid and 99 % for ChCl: levulinic acid under optimized conditions. The performance of these solvents was influenced by several factors, including the concentration of the washing solution, the soil-to-liquid ratio, and the contact time [279]. Liu et al. (2024) investigated the use of water-enhanced DESs for the efficient and environmentally friendly removal of heavy metals from soil. By adding water to conventional DES systems, the study developed ternary solvents that exhibited improved extraction performance. Among the sixteen DESs tested, those based on ChCl combined with urea, L-lactic acid, and EG showed significant improvements in heavy metal removal when water was incorporated. Notably, the L-lactic acid-based system (with a 2: 8 ratio of DES to water) achieved the highest removal efficiencies for Cd, Pb, Cu, and Zn, reaching rates of up to 96.44 %. The mechanisms responsible for these improvements included hydrogen bonding, interactions between functional groups, ion exchange, changes in viscosity, and co-precipitation. Additionally, water-enhanced DESs preserved the integrity of essential soil minerals and nutrients, demonstrating effective reusability after multiple filtration stages [280]. Kang et al. (2023) investigated the effectiveness of NADESs composed of glycolic acid and L-proline in extracting lead from contaminated soil. Under controlled conditions, specifically, a pressure of 101.33 kPa, a concentration of 0.6 mol/L for the DES, a temperature of 313.15 K, and an extraction time of 4 h, the NADES achieved a lead removal efficiency of 95.28 % [90].

5.5. Biological and food samples

One of the major health concerns is the contamination of food, particularly plants and animals, by metals. While many metals are essential for life, acting as trace elements in enzymatic processes and immune functions, deficiencies in elements like zinc, chromium, and iron remain significant health issues, particularly in developing nations. However, some metals, such as mercury, lead, cadmium, and arsenic, have no biological function and can be toxic even in very small amounts. For instance, high lead levels can cause anemia, headaches, and cramps. Toxic metal pollution has a cumulative effect, meaning even small concentrations can be harmful over time with prolonged exposure. Therefore, it is crucial to monitor and regulate metal contamination throughout the food chain to protect human health [284]. This determination is crucial for assessing the environmental and health impacts, resulting in increased attention in recent decades. Inorganic arsenic (iAs) and antimony (iSb) are highly toxic even at ultra-trace levels. The toxicity of arsenic in its trivalent form, As(III), and antimony in its trivalent form, Sb(III), is greater than that of their pentavalent counterparts, As(V) and Sb(V). Because trace metals are present at only micro- to nanogram levels and real-world samples contain many interfering species, direct instrumental detection is often impractical. Before analysis, the target ions usually must be isolated and concentrated. DESs tailored to each metal's chemistry have proven effective for this task, selectively extracting and pre-concentrating ions such as Ag(I), Cr(III),

Co(II), Ni(II), Cu(II), Fe(III), and Zn(II) from complex matrices ranging from fish muscle, liver, hair, blood, leaves, and roots to foods like vegetables, coffee, and wine [5,273]. Due to their ability to extract both inorganic metallic and organic compounds, depending on the HBA: HBD molar ratio, DESs are also effective for food sample pretreatment and analysis. In these applications, water is typically added to reduce the viscosity of the DES, enhancing mass transfer between the target molecules and the solvent. An added benefit is that both hydrophilic and HDESs, being non-volatile, can be used in HPLC and UV-Vis spectrophotometry for rapid analysis [50].

One of the interesting properties of DESs is their ability to extract heavy metals from complex matrices, such as food [50]. Based on Table 16, Elik et al. (2024) developed a novel method called vortex-assisted HDES-DLPME for the selective and rapid extraction of both essential metal ions (Fe(II), Mn(II), Zn(II)) and toxic metal ions (Cd(II), Pb(II)) from coffee products. They synthesized and tested five different HDES, finding that the combination of phenylacetic acid and dimethylglycine exhibited the best performance. This method did not require heating or centrifugation, thereby reducing energy consumption and simplifying the extraction process. The researchers optimized key extraction parameters, including pH, solvent volume, and vortex time. The method demonstrated excellent sensitivity, with detection limits ranging from 0.01 to 0.27 µg/kg, depending on the metal ion. Validation using reference materials and real coffee samples confirmed high recovery rates of 94–98 % and low relative standard deviations of 1.3–1.9 % [285]. Zhang et al. (2025) explored the application of DESs for removing Cd from phytoremediation biomass, focusing on the mechanisms behind metal extraction and biomass deconstruction. The study compared acidic, neutral, and alkaline DESs, finding that acidic DESs, particularly ChCl: lactic acid, demonstrated the highest Cd removal efficiency. Under optimal conditions (90 °C for 1 h), ChCl: lactic acid left only 7.4 % of Cd in the biomass. Temperature played a critical role, with Cd residue decreasing from 60.5 % at 25 °C to 6.6 % at 120 °C. Multi-scale analysis revealed that cadmium accumulated primarily in the compound middle lamellae and secondary walls of poplar. During treatment, Cd migrated to the liquid phase through cell wall loosening and complex formation via coordination and proton exchange [286].

Shikov et al. (2022) investigated the potential of NADES for the extraction of trace elements from *Fucus vesiculosus*, a brown seaweed, and assessed the safety of the resulting extracts. Compared to conventional solvents such as water and 70 % acetone, NADES exhibited significantly lower extraction efficiencies for most trace metals. Notably, none of the tested NADES formulations were able to recover As or Co, and the recovery rates for elements including Ba, Ca, Fe, Mg, Mn, Sr, and Zn remained below 9 %. However, the incorporation of water (especially in the range of 60–80 %) into NADES formulations enhanced the extraction efficiency for some elements. While overall metal recovery was low, relatively higher extraction rates for Al and Cr, up to 59.9 %, were observed in selected NADES compositions. Importantly, toxicological risk assessments based on hazard quotients, hazard indices, and carcinogenic risk estimates demonstrated that all NADES extracts posed no health risks upon topical application. Additionally, compared to extracts produced with conventional solvents, NADES extracts were found to contain lower concentrations of trace metals, thereby offering a safer and more environmentally friendly alternative [287]. It should be noted that ChCl: phenol was tested as an extraction solvent during the determination of zinc in fish and eel items [288], the extraction of Pb in milk samples [289], and for the separation of Co from pharmaceutical supplements and tea samples [290].

Shishov et al. (2022) systematically screened ChCl-based DESs for pulling a wide spectrum of metal ions, Ag, Al, B, Ba, Ca, Co, Cr, Cu, Fe, K, Mg, Mn, Mo, Na, Ni, Pb, Sn, Ti, V, and Zn from oily matrices. Extraction efficiency was highly influenced by the HBD; DESs formulated with carboxylic acids or sugars exhibited superior performance compared to those prepared with polyols, urea, or urea derivatives. Introducing water as a third component further boosted performance. The best

Table 16

List of prepared DESs along with their efficiency for recovering metals from biological and food samples.

No.	DESs	Molar ratio	Type	Metals	Efficiency %	Optimum condition	Metal recovery process from leachate	Cyclic process	Refs.
1	ChCl: Phenol	1: 2	Fish and eel samples	Zn Ba Ca Cu Fe K Mg Mn Mo Na Ni Pb Sn V Zn	92.8–101.3	$t = 5 \text{ min}$	–	–	[288]
2	ChCl: Malic Acids	1: 1	Plant Tobacco Lettuce	Fe	80–90	$t = 30 \text{ min}$ $T = 70 \text{ }^{\circ}\text{C}$ S/L ratio: 1/5 g/L	–	–	[292]
3	ChCl: Lactic Acid (Ultrasound-assisted)	1: 1	Sheep, Bovine, and Chicken Liver Samples, Turkey	Fe	> 99	–	–	–	[293]
4	ChCl: Oxalic Acid (Microwave-assisted)	1: 2	Marine Biological Samples Fish and Algae Samples	Cu Fe Ni Zn	97.8 96.1 97.7 98	–	–	–	[294]
5	ChCl: Citric Acid Betaine: Citric Acid Betaine: Glucose Betaine: Glycerol B-alanine: Citric Acid 1-Octyl-3-Methylimidazolium Chloride: 1-Undecanol (VADLLME)	1: 1 1: 1 2: 1 2: 1	Powdered Barley Grass	Mn Cu Zn Mo	80–95	$t = 50 \text{ min}$ $T = 60 \text{ }^{\circ}\text{C}$	–	–	[284]
6	ChCl: Tartaric Acid: Water	1: 1: 2	Porphyra Haitanensis	Hg	90–109	–	–	–	[295]
7	ChCl: Urea	1: 2	The Human Hair Sample	Pb Cu Cd Cr As Pb(II) Cd(II)	87 79 100 48 78	S/L ratio: 1/20 g/L	–	–	[296]
8	Lactic Acid: Glucose: H ₂ O	5: 3: 1	<i>F. vesiculosus</i>	Ba Ca Fe Mg Mn Sr Zn	99–105	–	–	–	[297]
9	ChCl: Lactic Acid	1: 2	Metal-Enriched Plants	Mg Mn Sr Zn	60–80	–	–	–	[287]
10	ChCl: Lactic Acid	1: 2	Metal-Enriched Plants	Cd	60	$t = 1 \text{ h}$ $T = 90 \text{ }^{\circ}\text{C}$	–	–	[286]
11	Phenylacetic Acid: Dimethylglycine (VA-DLPME)	1: 1	Coffee Products	Fe Mn Zn Cd Pb Cd As	>95	$t = 3 \text{ min}$	–	–	[285]
12	TAC: DL-Lactic Acid (UA-DLLME)	1: 3	Wine Sample	Cd As	90.6–103.6	–	–	–	[298]
13	ChCl: Oxalic Acids	1: 1	Vegetable Sample: Basil Herb, Spinach, Dill, and Cucumber Barks	Mn	>97.0	$t = 2 \text{ h}$ $T = 95 \text{ }^{\circ}\text{C}$ S/L ratio: 1/25 g/L	–	–	[299]
14	TBAC: Decanoic Acid (LPME)	1: 4	Onion Parsley Cigarette	Ni	>90	–	–	–	[242]
15	ChCl: Phenol (VE-LPME)	1:2	Milk Samples	Pb	102.5–103.2	–	–	–	[289]
16	ChCl: phenol (UALPME)	1:4	Pharmaceutical Supplements and Tea Samples	Co(II)	101	–	–	–	[290]
17	ChCl: Oxalic Acid	1: 2	Fish Samples	Cu Zn Fe	96 99 95	–	–	–	[300]
18	DL-menthol: Sorbitol: Mandelic Acid (AALLME)	1: 2: 1	From Water and Fruit Juice Samples	Cd Zn	80 70	–	–	–	[301]
19	DL-menthol: Sorbitol: Mandelic Acid (LLME)	1: 2: 1	From Milk Samples	Cd(II) Cu(II) Pb(II)	> 85	–	–	–	[302]

(continued on next page)

Table 16 (continued)

No.	DESs	Molar ratio	Type	Metals	Efficiency %	Optimum condition	Metal recovery process from leachate	Cyclic process	Refs.
20	ChCl: Phenol (DILME)	1: 4	Water and Food Samples (Rice Cultivated Mushroom Chicken Meat)	Al	>97	–	–	–	[303]
				Na	> 80				
				K	100				
				Ca	100				
21	ChCl: Glycerol	1: 2	Waste Animal Fat; Waste Cooking Oil	Mg	> 70	–	–	–	[304]
				P	> 80				
				B	< 20				
				Fe	100				
22	ChCl: Urea	1: 2	Edible Oils	Pb	> 90	–	–	–	[305]
				Cd	> 90				
23	ChCl: Lactic Acids (80 % in water)	1: 1: 5.5	Vegetable Oil	Ag	35	–	–	–	[291]
				Al	88				
				B	100				
				Ba	99				
				Ca	99				
				Co	98				
				Cr	20				
				Cu	94				
				Fe	97				
				K	100				
				Mg	100				
				Mn	96				
				Mo	77				
				Na	100				
				Ni	98				
				Pb	97				
				Sn	93				
				Ti	96				
				V	94				
				Zn	100				

formulation, ChCl: lactic acid: water, stripped up to 99 % of the metals and allowed direct ICP-OES analysis of vegetable and motor oils without microwave digestion. Detection limits ranged from 0.02 to 17 mg/kg. This study demonstrated that hydrophilic DESs can significantly simplify sample preparation while enabling efficient and environmentally friendly extraction of trace metals from hydrophobic oils [291].

6. Challenges in advancing DESs for industrial applications and solutions

Despite over a decade of promising academic research, particularly regarding metal recovery from secondary sources such as spent LIBs, the application of DESs remains largely confined to laboratory-scale studies. While DESs offer notable advantages, including tunable properties, low volatility, and potential environmental benefits, several interrelated scientific, technical, economic, and regulatory challenges hinder their widespread adoption in industry. These limitations highlight the need for further research to improve their efficiency, stability, and scalability. Key areas for future investigation include optimizing the composition and properties of DES, understanding its long-term stability and recyclability, and developing cost-effective, scalable processes for industrial implementation [27,37,306].

• Challenges in DES Selection and Process Optimization for Metal Recovery

The performance of DESs in metal leaching is strongly influenced by the specific combination of HBDs and HBAs, which determine crucial properties such as viscosity, conductivity, acidity, and leaching temperature. However, the absence of standardized selection criteria or predictive guidelines poses a significant challenge when identifying suitable DES compositions for specific metal targets. This variability

hinders reproducibility, slows down process optimization, and limits industrial scalability [27,306]. Moreover, even after selecting an appropriate DES, a comprehensive understanding of how process variables, such as water content, residence time, heat treatment, and viscosity, affect leaching performance remains underdeveloped. The limited availability of integrated datasets on these parameters further restricts the ability to optimize extraction conditions and transition from lab-scale research to industrial application. To address these challenges, both experimental and data-driven strategies are needed. Systematic investigations can elucidate the relationships between DES composition and leaching behavior, while machine learning, AI-assisted modeling, and platforms such as DESignSolvents provide powerful tools for predicting physicochemical properties and guiding rational solvent design. In parallel, design of experiments approaches and advanced analytical techniques can help identify optimal process conditions and deepen mechanistic understanding. Additionally, expanding DES-based research beyond commonly studied metal oxides to include more refractory compounds, such as metal nitrides, phosphates, and spinels, may reveal novel dissolution mechanisms and broaden the applicability of DESs. Ultimately, fostering stronger collaboration between academia and industry is crucial to generate high-quality, application-specific datasets, which are essential for developing reliable, scalable, and cost-effective metal recovery systems.

• High Viscosity and Mass Transfer Limitations

One of the main drawbacks associated with DESs is their inherently high viscosity. This characteristic can significantly limit mass transfer and slow the kinetics of solid–liquid reactions, particularly in heterogeneous systems where leaching occurs at the solid–liquid interface. The sluggish diffusion of reactants and products within viscous media reduces the efficiency of metal extraction processes.

To address this issue, dilution of DESs with water or organic co-solvents is commonly employed to reduce viscosity. However, excessive dilution can disrupt the DES structure, weaken the hydrogen bonding network, and consequently compromise its leaching performance. Similarly, increasing the temperature is an effective strategy to reduce viscosity, as DESs tend to become more fluid at elevated temperatures. Nevertheless, maintaining high temperatures can be energy-intensive and may not be economically viable in large-scale applications unless integrated with waste heat recovery systems [24,306]. Despite these challenges, several practical solutions are available. One approach is the careful selection of DESs with naturally lower viscosity (among hydrophilic systems) or the use of acidic hydrophobic DESs, which generally exhibit better flow behavior while maintaining leaching activity. Furthermore, since many leaching reactions are already conducted at temperatures above 50–60 °C, the associated reduction in viscosity under these conditions can be leveraged to improve mass transfer and reaction rates. Overall, through informed solvent design, temperature optimization, and process integration, the limitations posed by DES viscosity can be effectively managed, enhancing the applicability of DESs in metal recovery processes.

• Handling, Separation, and Drying Difficulties

Despite the environmental advantages of DESs, their high viscosity presents several practical challenges in large-scale operations. The thick, sticky nature of DESs complicates fluid handling tasks such as mixing, pumping, and stirring. These issues become more pronounced in continuous systems or reactors with poor mixing zones. Furthermore, solid-liquid separation is hindered because viscous DESs adhere strongly to solid residues, resulting in solvent loss and inefficient recovery. Traditional filtration techniques are often inadequate for such systems, necessitating additional steps, such as pressurized filtration, centrifugation, or multiple washing cycles. Post-leaching, the drying of solid residues is another concern. Since DESs are non-volatile, they cannot be removed by simple evaporation, unlike conventional organic solvents. As a result, the solids must be washed with water or organic solvents to remove DES residues. This, in turn, introduces the need for extra separation and recycling steps to recover both the DES and the wash solvents, increasing process complexity, operational costs, and the potential for secondary waste streams [306].

To mitigate these issues, several approaches can be considered. One promising strategy involves the use of DESs composed entirely of low-toxicity, thermally decomposable organic compounds, such as certain organic acids or non-ionic molecules like thymol, as both the HBD and HBA. For example, systems based on lactic acid, such as glycolic acid [142], have shown potential. These components can degrade into non-toxic by-products under moderate heating, which may allow for easier removal from solid residues without the release of harmful gases. However, the feasibility of this approach depends on the thermal stability of the metals and matrix involved, as well as the avoidance of undesirable decomposition products. Further research is needed to validate such systems at a pilot scale.

Additionally, improving the rheological properties of DESs through structural modification or controlled dilution may facilitate better handling and separation. The development of hydrophilic DESs with self-emulsifying or phase-separating behavior may also simplify recovery processes. Exploring membrane-based separation or advanced solvent recovery techniques could further improve the sustainability and efficiency of DES-based leaching systems.

• Lack of Techno-Economic, Pilot-Scale Validation, and Instability of DESs

A critical obstacle to the industrial adoption of DESs is the lack of comprehensive techno-economic assessments (TEA) and pilot-scale demonstrations to validate their commercial feasibility. While some

low-cost DES formulations have shown promising results in small-scale experiments, such as 250 mL beaker tests, there is limited evaluation of their performance under real-world industrial conditions. Most research to date has focused primarily on the leaching step, often overlooking essential downstream processes, such as purification, recovery, and solvent regeneration. Additionally, DESs are still in the early stages of development for large-scale metal recovery, and significant knowledge gaps remain regarding their long-term behavior, particularly in terms of energy and material balances, solvent loss, and impurity accumulation. Some DESs, including those based on choline chloride and dicarboxylic acids, demonstrate strong leaching capabilities for transition metal oxides. However, their poor chemical stability remains a barrier to industrial use. For instance, the alcohol group in choline can react with carboxylic acids to form esters, leading to gradual degradation over time and compromising the reusability and reliability of these solvents in continuous operations [37,306]. To enable the industrial application of DESs in metal recovery, it is crucial to address their susceptibility to chemical and thermal degradation, particularly under the harsh conditions of hydrometallurgical processes. Many DESs, especially those based on choline chloride, are prone to decomposition at elevated temperatures (more than 180 °C), producing potentially hazardous byproducts such as trimethylamine and 2-chloroethanol.

Additionally, solvent components may leach into aqueous phases or degrade during operation, leading to reduced efficiency, the formation of precipitates, and difficulties in solvent recycling. These challenges are compounded by the inherently high viscosity of many DESs, which complicates mixing, separation, and solvent recovery, ultimately leading to increased energy and processing costs. To bridge these gaps, pilot-scale testing and integrated process simulations are urgently needed to assess the performance of DES-based leaching systems under realistic industrial conditions. Coupling leaching with downstream recovery and regeneration units will allow for full-cycle evaluation of material flows, energy consumption, and process economics. Comprehensive TEA and life cycle assessments will help identify cost drivers, environmental trade-offs, and energy bottlenecks in comparison to conventional systems. From a solvent design perspective, enhancing the chemical and thermal stability of DESs is essential. This could involve selecting more robust HBA: HBD combinations or incorporating stabilizers to mitigate degradation. For example, replacing choline chloride with alternative quaternary ammonium salts, or formulating non-esterifying DESs using sulfonic acids or amides, may significantly improve long-term reliability. Furthermore, implementing real-time monitoring of solvent composition and degradation products in closed-loop systems can support predictive maintenance and timely regeneration. By integrating solvent engineering, system design, and economic modeling, the transition of DESs from lab-scale success to industrial viability can be significantly accelerated.

• Redox Activity and Unintended Decomposition

A potential advantage of DESs in recycling LIB cathode materials, such as LCO and NMC, is the elimination of the need for an external reducing agent. However, the reduction of Co(III) to Co(II) within DESs implies that the DES itself acts as a reducing agent, leading to self-oxidation. This process causes the DES to oxidize and eventually decompose, contradicting its intended role as a stable and reusable solvent system [306].

To address this issue, future research should focus on understanding the redox mechanisms within DESs and identifying which solvent components are prone to oxidation under typical leaching conditions. By characterizing redox pathways through electrochemical analysis, researchers can distinguish between reversible redox systems and those that cause permanent degradation. One potential solution is the design of redox-stable DESs, incorporating components that are less susceptible to oxidation or capable of undergoing reversible redox cycling without breaking down. Alternatively, sacrificial additives or co-solvents may be

introduced to absorb the redox burden, thereby protecting the DES structure while still facilitating metal reduction. Another strategy is to carefully control reaction conditions, such as pH, temperature, and metal concentration, to minimize oxidative stress on the solvent.

• Precipitation Issues, Recycling Difficulties, and Reusability Challenges

Instability in metal-DES solutions presents another challenge, as metals such as cobalt and lead tend to form precipitates, disrupting the leaching process. These unwanted reactions complicate process control and hinder the reuse of DESs. Since DESs are mixtures and not pure compounds, their recycling is more complex than that of traditional solvents. The degradation products and metal residues further complicate the situation. Furthermore, the need to maintain the original molar ratio between the hydrogen bond donors and acceptors during DES recovery complicates the process, as it directly affects the solvent's physical and chemical properties. Moreover, the accumulation of organic degradation products and co-dissolved impurities makes it challenging to recover and recycle DESs effectively [24,306]. Opportunities for improvement include developing targeted strategies to prevent precipitation through careful control of process conditions and solvent composition. Advanced separation techniques such as membrane filtration or selective extraction could facilitate the removal of impurities while preserving DES integrity.

Gaining regulatory approval for processes involving DESs can be challenging due to their novelty, and regulatory bodies understandably require thorough safety and environmental data. However, the eco-friendly nature and tunability of DESs present promising advantages that continue to attract growing research and development efforts. At the same time, aligning these benefits with industrial goals, such as reducing chemical diversity and meeting strict regulations [306]. Economically, DESs are more affordable than ILs and hold potential for cost reduction and improved recyclability, especially for treating low-grade ores and residues. Although water remains the most widely used and cost-effective solvent in hydrometallurgy, DESs offer unique properties that could complement traditional systems in the future [306]. Ultimately, rather than immediately transforming industrial extractive metallurgy, the knowledge gained from DES research enriches our fundamental understanding of solvent–metal interactions. This insight can help optimize conventional aqueous processes and support a gradual transition toward more sustainable, low-energy, circular metal recovery methods [24,306]. In conclusion, while DESs show great potential for metal recycling, addressing these challenges through further research and development is essential for improving their practical use and scalability in industrial applications [325].

7. Conclusion and outlook

DESs have emerged as promising green alternatives for recovering critical and strategic metals from diverse secondary resources. A comparative analysis of recent studies reveals that performance is primarily governed by physicochemical properties, rather than solvent class labels alone. Acidic, low-viscosity DESs with strong coordination or reducing ability consistently achieve higher leaching efficiencies and selectivity, while viscous or weakly coordinating systems are limited by mass-transfer constraints and slower kinetics. Hydrophilic DESs are particularly effective for oxide dissolution through proton activity, whereas hydrophobic DESs enable biphasic extraction that minimizes wastewater and simplifies separation. Because these properties can be tuned through deliberate HBA/HBD selection and preparation methods, DESs represent a versatile and customizable solvent platform. This review, therefore, also serves as a data resource, linking solvent properties to recovery outcomes across a broad spectrum of metals and waste streams.

Nevertheless, significant challenges remain. The current literature

suffers from inconsistencies due to non-standardized solvent selection, variable reporting practices, and a limited understanding of the mechanisms involved. Stability issues, such as esterification in CHCl_3 -based systems or redox self-consumption during cobalt reduction, remain insufficiently explored. High viscosity continues to complicate pumping, mixing, and solid–liquid separation, while impurity accumulation hinders solvent regeneration. The lack of full-process flowsheet integration makes it difficult to benchmark DES systems against aqueous hydrometallurgy on critical performance indicators such as space–time yield, recyclability, and overall cost per kilogram of recovered metal. Moreover, there is a shortage of techno-economic assessments, life-cycle analyses, and pilot-scale demonstrations, all of which are essential for industrial deployment.

Future progress requires a transition from descriptive case studies to predictive, mechanism-driven design and engineering. Molecular simulation and machine learning approaches could play a transformative role by predicting property–performance relationships, guiding rational selection of HBA: HBD pairs, and accelerating the discovery of DES formulations with optimized viscosity, acidity, and selectivity. Coupling DES-based leaching with industrially relevant downstream technologies, such as electrochemical deposition for metal recovery or membrane separation for solvent reuse, provides practical routes to integrate DESs into scalable flow sheets. In parallel, systematic safety and toxicity assessments are urgently needed, not only for the DES components themselves but also for degradation products and by-products generated under operational conditions, to ensure regulatory compliance and environmental safety. Finally, the concept of synergistic DES–DES systems, where hydrophilic solvents drive oxide dissolution and hydrophobic solvents enable selective extraction and water minimization, remains an underexplored but promising strategy for achieving both higher efficiency and greater sustainability.

In conclusion, DESs offer credible advantages, tunability, low volatility, and the potential for wastewater minimization; however, their industrial adoption will depend on overcoming physicochemical and engineering barriers, establishing predictive design tools, validating performance in integrated pilot-scale flowsheets, and ensuring safety through robust toxicity assessments. By consolidating comparative insights and addressing these forward-looking perspectives, this review not only highlights current opportunities and limitations but also proposes a roadmap for transforming DESs from laboratory concepts into scalable technologies for the recovery of circular and sustainable metals.

CRediT authorship contribution statement

Fereshteh Moradi: Writing – original draft, Investigation, Data curation, Conceptualization. **Francis Bougie:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Funding

This work was supported by the Natural Sciences and Engineering Research Council of Canada.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Francis Bougie reports financial support was provided by Natural Sciences and Engineering Research Council of Canada. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

References

- [1] P. Pathak, P.R. Rout, *Urban Mining for Waste Management and Resource Recovery: Sustainable Approaches*, CRC Press, 2021.
- [2] P. Panda, S. Mishra, Deep eutectic solvents: Physico-chemical properties and their use for recovery of metal values from waste products, *J. Mol. Liq.* 390 (2023) 123070.
- [3] R.M. Dias, M.C. da Costa, Y.P. Jimenez, Perspectives of using DES-based systems for solid-liquid and liquid-liquid extraction of metals from E-waste, *Minerals* 12 (6) (2022) 710.
- [4] G. Zante, M. Boltoeva, Review on hydrometallurgical recovery of metals with deep eutectic solvents, *Sustainable Chemistry* 1 (3) (2020) 238–255.
- [5] Z. Yuan, H. Liu, W.F. Yong, Q. She, J. Esteban, Status and advances of deep eutectic solvents for metal separation and recovery, *Green Chem.* 24 (5) (2022) 1895–1929.
- [6] Z. Sun, Y. Xiao, H. Agterhuis, J. Sijtsma, Y. Yang, Recycling of metals from urban mines—a strategic evaluation, *J. Clean. Prod.* 112 (2016) 2977–2987.
- [7] N. Schaeffer, H. Passos, I. Billard, N. Papaiconomou, J.A. Coutinho, Recovery of metals from waste electrical and electronic equipment (WEEE) using unconventional solvents based on ionic liquids, *Crit. Rev. Environ. Sci. Technol.* 48 (13–15) (2018) 859–922.
- [8] L.H. Xavier, E.C. Giese, A.C. Ribeiro-Duthie, F.A.F. Lins, Sustainability and the circular economy: a theoretical approach focused on e-waste urban mining, *Resources Policy* 74 (2021) 101467.
- [9] S. Fang, et al., Selective recovery of gallium (indium) from metal organic chemical vapor deposition dust—a sustainable process, *ACS Sustainable Chemistry & Engineering* 7 (10) (2019) 9646–9654.
- [10] A. Priya, S. Hait, Comparative assessment of metallurgical recovery of metals from electronic waste with special emphasis on bioleaching, *Environ. Sci. Pollut. Res.* 24 (2017) 6989–7008.
- [11] M. Sethurajan, et al., Recent advances on hydrometallurgical recovery of critical and precious elements from end of life electronic wastes—a review, *Crit. Rev. Environ. Sci. Technol.* 49 (3) (2019) 212–275.
- [12] L. Zhan, F. Xia, Q. Ye, X. Xiang, B. Xie, Novel recycle technology for recovering rare metals (Ga, In) from waste light-emitting diodes, *J. Hazard. Mater.* 299 (2015) 388–394.
- [13] M. Baniasadi, F. Vakili, N. Bahaloo-Horeh, S.M. Mousavi, S. Farnaud, Advances in bioleaching as a sustainable method for metal recovery from e-waste: a review, *Journal of Industrial and Engineering Chemistry* 76 (2019) 75–90.
- [14] D. Prodius, K. Gandha, A.-V. Mudring, I.C. Nlebedim, Sustainable urban mining of critical elements from magnet and electronic wastes, *ACS Sustainable Chemistry & Engineering* 8 (3) (2019) 1455–1463.
- [15] A. Zhu, et al., The application of deep eutectic solvents in lithium-ion battery recycling: a comprehensive review, *Resources, Conservation and Recycling* 188 (2023) 106690.
- [16] K. Zheng, M.F. Benedetti, E.D. van Hullebusch, Recovery technologies for indium, gallium, and germanium from end-of-life products (electronic waste)—a review, *J. Environ. Manage.* 347 (2023) 119043.
- [17] K. Wang, T. Hu, P. Shi, Y. Min, J. Wu, Q. Xu, Efficient recovery of value metals from spent lithium-ion batteries by combining deep eutectic solvents and coextraction, *ACS Sustainable Chemistry & Engineering* 10 (3) (2021) 1149–1159.
- [18] A. Van den Bossche, W. Vereycken, T. Vander Hoogerstraete, W. Dehaen, K. Binnemans, Recovery of gallium, indium, and arsenic from semiconductors using tribromide ionic liquids, *ACS Sustain. Chem. Eng.* 7 (17) (2019) 14451–14459.
- [19] K. Binnemans, P.T. Jones, Solvometallurgy: an emerging branch of extractive metallurgy, *Journal of Sustainable Metallurgy* 3 (2017) 570–600.
- [20] M.I. Martín, I. García-Díaz, F. López, Properties and perspective of using deep eutectic solvents for hydrometallurgy metal recovery, *Miner. Eng.* 203 (2023) 108306.
- [21] J. Liu, X. Li, K.H. Row, Development of deep eutectic solvents for sustainable chemistry, *J. Mol. Liq.* 362 (2022) 119654.
- [22] T. Van Gerven, A. Stankiewicz, Structure, energy, synergy, time. The fundamentals of process intensification, *Ind. Eng. Chem. Res.* 48 (5) (2009) 2465–2474.
- [23] A. Stankiewicz, T. Van Gerven, G. Stefanidis, *The Fundamentals of Process Intensification*, John Wiley & sons, 2019.
- [24] A. Kityk, V. Pavlik, M. Hnatko, Reshaping the future of battery waste: deep eutectic solvents in Li-ion battery recycling, *J. Energy Storage* 97 (2024) 112990.
- [25] A. Kovačević, M. Tolazzi, M. Sanadar, A. Melchior, hydrometallurgical recovery of metals from spent Lithium-ion batteries with ionic liquids and deep eutectic solvents, *Journal of Environmental Chemical Engineering* 12 (4) (2024) 113248.
- [26] B. Zhou, H. Su, W. Liu, Z. Zhu, L. Wang, T. Qi, Solvent extraction of metal ions from the leaching solutions of waste lithium-ion battery materials: a review, *Sep. Purif. Technol.* 354 (2024) 129173.
- [27] C. Padwal, et al., Deep eutectic solvents: green approach for cathode recycling of Li-ion batteries, *Advanced Energy and Sustainability Research* 3 (1) (2022) 2100133.
- [28] H. Shi, Y. Luo, C. Yin, L. Ou, Review of the application of ionic liquid systems in achieving green and sustainable recycling of spent lithium-ion batteries, *Green Chem.* 26 (14) (2024) 8100–8122.
- [29] J. Wang, et al., Green recycling of spent Li-ion battery cathodes via deep-eutectic solvents, *Energ. Environ. Sci.* 17 (3) (2024) 867–884.
- [30] R. Su, S. Tang, M. Zhang, M. Guo, strategies for overcoming challenges in using deep eutectic solvents for the selective extraction of valuable metals from spent Lithium-ion batteries: a review, *Journal of environmental Chem. Eng.* 12 (4) (2024) 113200.
- [31] W.-H. Gao, C.-C. Nie, L. Li, S. Yan, W.-T. Zhou, X.-N. Zhu, Sustainable and efficient deep eutectic solvents in recycling of spent lithium-ion batteries: recent advances and perspectives, *J. Clean. Prod.* 464 (2024) 142735.
- [32] Y. Fan, et al., Development and challenges of deep eutectic solvents for cathode recycling of end-of-life lithium-ion batteries, *Chem. Eng. J.* 463 (2023) 142278.
- [33] Z. Wang, S. Li, T. Li, T. Hu, X. Ge, Deep eutectic solvents (DESs) for green recycling of wasted lithium-ion batteries (LIBs): progress on pushing the overall efficiency, *Mining, Metallurgy & Exploration* 39 (5) (2022) 2149–2165.
- [34] M. Svård, C. Ma, K. Forsberg, P.G. Schiavi, Addressing the reuse of deep eutectic solvents in Li-ion battery recycling: insights into dissolution mechanism, metal recovery, regeneration and decomposition, *ChemSusChem* 17 (20) (2024) e202400410.
- [35] B. Wang, Y. Zhang, C. Zhu, S. Ren, Y. Hou, W. Wu, Advances of deep eutectic solvents in lithium battery recycling field, *Sep. Purif. Technol.* 362 (2025) 131836.
- [36] E.A. Oke, H. Potgieter, Recent chemical methods for metals recovery from printed circuit boards: a review, *J. Mater. Cycles Waste Manage.* 26 (3) (2024) 1349–1368.
- [37] S. Gholami, M. Pérez-Page, C. D'Agostino, J. Esteban, (deep) eutectic solvents for the separation of platinum group metals and rare earth elements: characteristics, extraction mechanisms and state of the art, *Chem. Eng. J.* 505 (2025) 159497.
- [38] S. Cheshmekhezr, et al., Exploring ionic liquids and deep eutectic solvents for emerging contaminant removal from water and wastewater, *npj Materials Sustainability* 2 (1) (2024) 1–11.
- [39] S. Suffia, D. Dutta, Applications of deep eutectic solvents in metal recovery from E-wastes in a sustainable way, *J. Mol. Liq.* 394 (2023) 123738.
- [40] C. Wang, Z. Zhou, X. Zhang, H. Guo, G. Boczkaj, Metal extraction using deep eutectic solvents for metal recovery and environmental remediation—a review, *Sep. Purif. Technol.* 364 (2025) 132533.
- [41] M. Guo, R. Deng, M. Gao, C. Xu, Q. Zhang, sustainable recovery of metals from E-waste using deep eutectic solvents: advances, challenges, and perspectives, *Current Opinion in Green and Sustainable Chemistry* 47 (2024) 100913.
- [42] F.J. Alguacil, J.I. Robla, Recent work on the recovery of rare earths using ionic liquids and deep eutectic solvents, *Minerals* 13 (10) (2023) 1288.
- [43] G. Tian, H. Liu, Review on the mineral processing in ionic liquids and deep eutectic solvents, *Miner. Process. Extr. Metall. Rev.* 45 (2) (2024) 130–153.
- [44] I. Wazeer, H.F. Hizaddin, M.A. Hashim, M.K. Hadj-Kali, An overview about the extraction of heavy metals and other critical pollutants from contaminated water via hydrophobic deep eutectic solvents, *J. Environ. Chem. Eng.* 10 (6) (2022) 108574.
- [45] M. Marchel, M.P. Rayaroth, C. Wang, L. Kong, J.A. Khan, G. Boczkaj, Hydrophobic (deep) eutectic solvents (HDESs) as extractants for removal of pollutants from water and wastewater—a review, *Chem. Eng. J.* 475 (2023) 144971.
- [46] R. Rodríguez-Ramos, Á. Santana-Mayor, B. Socas-Rodríguez, M.A. Rodríguez-Delgado, Recent applications of deep eutectic solvents in environmental analysis, *Applied Sciences* 11 (11) (2021) 4779.
- [47] C.M. Chabib, J.K. Ali, M. Abi Jaoude, E. Alhseinat, I.A. Adeyemi, I.M. Al Nashef, “application of deep eutectic solvents in water treatment processes: a review,” *Journal of water, Process. Eng.* 47 (2022) 102663.
- [48] M. Devi, R. Moral, S. Thakuria, A. Mitra, S. Paul, Hydrophobic deep eutectic solvents as greener substitutes for conventional extraction media: examples and techniques, *ACS omega* 8 (11) (2023) 9702–9728.
- [49] S.C. Cunha, J.O. Fernandes, Extraction techniques with deep eutectic solvents, *TrAC Trends Anal. Chem.* 105 (2018) 225–239.
- [50] R. Castro-Muñoz, et al., Deep eutectic solvents for the food industry: extraction, processing, analysis, and packaging applications—a review, *Crit. Rev. Food Sci. Nutr.* 64 (30) (2024) 10970–10986.
- [51] Z.Y. Lai, C.L. Yiin, S.S.M. Lock, B.L.F. Chin, N.S.A. Zauzi, S. Sar-Ee, A review on natural based deep eutectic solvents (NADESs): fundamentals and potential applications in removing heavy metals from soil, *Environ. Sci. Pollut. Res.* 30 (55) (2023) 116878–116905.
- [52] L. Mu, et al., Research progress on deep eutectic solvents and recent applications, *Processes* 11 (7) (2023) 1986.
- [53] G. Li, K.H. Row, Utilization of deep eutectic solvents in dispersive liquid-liquid micro-extraction, *TrAC Trends Anal. Chem.* 120 (2019) 115651.
- [54] M.S. Jagirani, M. Soyak, Deep eutectic solvents-based adsorbents in environmental analysis, *TrAC Trends Anal. Chem.* 157 (2022) 116762.
- [55] S. Chen, et al., Deep eutectic solvents for separation and purification applications in critical metal metallurgy: recent advances and perspectives, *Int. J. Miner. Metall. Mater.* 32 (1) (2025) 1–19.
- [56] J. Ruan, et al., Applications and microscopic mechanism analysis of deep eutectic solvents in liquid-liquid extraction, *Journal of Industrial and Engineering Chemistry* 144 (2024) 18–31.
- [57] S. Affat, A review of deep eutectic solvents (DESs), preparation, classification, physicochemical properties, advantages and disadvantages, *University of Thi-Qar Journal of Science* 11 (1) (2024) 166–174.

- [58] T. El Achkar, H. Greige-Gerges, S. Fourmentin, Basics and properties of deep eutectic solvents: a review, *Environ. Chem. Lett.* 19 (2021) 3397–3408.
- [59] M.B.S. Mr, V.S.K. Mr, M. Chaudhary, P. Singh, A mini review on synthesis, properties and applications of deep eutectic solvents, *J. Indian Chem. Soc.* 98 (11) (2021) 100210.
- [60] M.H. Zainal-Abidin, M. Hayyan, W.F. Wong, Hydrophobic deep eutectic solvents: current progress and future directions, *Journal of Industrial and Engineering Chemistry* 97 (2021) 142–162.
- [61] K.A. Pishro, M.H. Gonzalez, Use of deep eutectic solvents in environmentally-friendly dye-sensitized solar cells and their physicochemical properties: a brief review, *RSC Adv.* 14 (21) (2024) 14480–14504.
- [62] K.A. Omar, R. Sadeghi, Physicochemical properties of deep eutectic solvents: a review, *J. Mol. Liq.* 360 (2022) 119524.
- [63] K. Zhou, et al., Recent advances in deep eutectic solvents for next-generation lithium batteries: safer and greener, *Prog. Mater. Sci.* 146 (2024) 101338.
- [64] M. Zhang, et al., Insights into the relationships between physicochemical properties, solvent performance, and applications of deep eutectic solvents, *Environ. Sci. Pollut. Res.* 28 (27) (2021) 35537–35563.
- [65] E.L. Smith, A.P. Abbott, K.S. Ryder, Deep eutectic solvents (DESS) and their applications, *Chem. Rev.* 114 (21) (2014) 11060–11082.
- [66] S.P. Jjardar, V. Singh, R.L. Gardas, Revisiting the physicochemical properties and applications of deep eutectic solvents, *Molecules* 27 (4) (2022) 1368.
- [67] A.T. Yeow, et al., A comprehensive review on the physicochemical properties of deep eutectic solvents, *Results in Chemistry* 7 (2024) 101378.
- [68] K.A. Omar, R. Sadeghi, Database of deep eutectic solvents and their physical properties: a review, *J. Mol. Liq.* 384 (2023) 121899.
- [69] T. Negi, et al., Deep eutectic solvents: preparation, properties, and food applications, *Heliyon* 10 (7) (2024).
- [70] A. Al-Bodour, N. Alomari, A. Gutiérrez, S. Aparicio, M. Atilhan, Exploring the thermophysical properties of natural deep eutectic solvents for gas capture applications: a comprehensive review, *Green Chemical Engineering* 5 (3) (2024) 307–338.
- [71] I.M. Pateli, D. Thompson, S.S. Alabdullah, A.P. Abbott, G.R. Jenkin, J.M. Hartley, The effect of pH and hydrogen bond donor on the dissolution of metal oxides in deep eutectic solvents, *Green Chem.* 22 (16) (2020) 5476–5486.
- [72] I. Cichowska-Kopczyńska, B. Nowosielski, D. Warmińska, Deep eutectic solvents: properties and applications in CO₂ separation, *Molecules* 28 (14) (2023) 5293.
- [73] Y. Cui, C. Li, J. Yin, S. Li, Y. Jia, M. Bao, Design, synthesis and properties of acidic deep eutectic solvents based on choline chloride, *J. Mol. Liq.* 236 (2017) 338–343.
- [74] K. Shi, C.M. Pedersen, H. Chang, J. Shi, Y. Wang, Y. Qiao, Characterization of the acidity and basicity of green solvents by NMR techniques, *Magnetic Resonance Letters* 1 (1) (2021) 81–88.
- [75] R. Qin, et al., Quantification of alkalinity of deep eutectic solvents based on (H–) and NMR, *Phys. Chem. Chem. Phys.* 26 (8) (2024) 7042–7048.
- [76] K. Chandran, C.F. Kait, C.D. Wilfred, H.F.M. Zaid, A review on deep eutectic solvents: physicochemical properties and its application as an absorbent for sulfur dioxide, *J. Mol. Liq.* 338 (2021) 117021.
- [77] A. Shishov, P. Makos-Cheistowska, A. Bulatov, V. Andruch, Deep eutectic solvents or eutectic mixtures? Characterization of tetrabutylammonium bromide and nonanoic acid mixtures, *J. Phys. Chem. B* 126 (21) (2022) 3889–3896.
- [78] M.M. Abdallah, et al., Physicochemical characterization and simulation of the solid–liquid equilibrium phase diagram of terpene-based eutectic solvent systems, *Molecules* 26 (6) (2021) 1801.
- [79] T. Brouwer, B.C. Diels, J.M. Bock, B. Schuur, Hydrophobic deep eutectic solvents for the recovery of bio-based chemicals: solid–liquid equilibria and liquid–liquid extraction, *Processes* 9 (5) (2021) 796.
- [80] T. Hanada, M. Goto, Synergistic deep eutectic solvents for lithium extraction, *ACS Sustainable Chemistry & Engineering* 9 (5) (2021) 2152–2160.
- [81] D.O. Abranches, J.A. Coutinho, Type V deep eutectic solvents: design and applications, *Current Opinion in Green and Sustainable Chemistry* 35 (2022) 100612.
- [82] A.P. Crema, et al., New family of type V eutectic solvents based on 1, 10-phenanthroline and their application in metal extraction, *Hydrometallurgy* 215 (2023) 105971.
- [83] A. Kianpoor, R. Sadeghi, Novel acidic deep eutectic solvents: synthesis, phase diagram, thermal behavior, physicochemical properties and application, *J. Iran. Chem. Soc.* 21 (9) (2024) 2385–2399.
- [84] X. Yao, et al., Efficient removal of trace olefins in aromatics by novel acetamide-based acidic deep eutectic solvents, *Fuel* 316 (2022) 123103.
- [85] M.B. Haider, D. Jha, R. Kumar, B.M. Sivagnanam, Ternary hydrophobic deep eutectic solvents for carbon dioxide absorption, *International Journal of Greenhouse Gas Control* 92 (2020) 102839.
- [86] E. Picciolini, G. Pastore, T. Del Giacco, G. Ciancaleoni, M. Tiecco, R. Germani, Aqueo-DESS: water-based binary natural deep eutectic solvents, *J. Mol. Liq.* 383 (2023) 122057.
- [87] M. Tiecco, A. Grillo, E. Mosconi, W. Kaiser, T. Del Giacco, R. Germani, Advances in the development of novel green liquids: thymol/water, thymol/urea and thymol/phenylacetic acid as innovative hydrophobic natural deep eutectic solvents, *J. Mol. Liq.* 364 (2022) 120043.
- [88] L.P. Silva, M.A. Martins, D.O. Abranches, S.P. Pinho, J.A. Coutinho, Solid-liquid phase behavior of eutectic solvents containing sugar alcohols, *J. Mol. Liq.* 337 (2021) 116392.
- [89] D.P. Arcon, F.C. Franco Jr., All-fatty acid hydrophobic deep eutectic solvents towards a simple and efficient microextraction method of toxic industrial dyes, *J. Mol. Liq.* 318 (2020) 114220.
- [90] K. Kang, X. Jia, K. Zheng, X. Wang, B. Liu, Y. Hou, Physical properties of natural deep eutectic solvent and its application in remediation of heavy metal lead in soil, *J. Contam. Hydrol.* 258 (2023) 104222.
- [91] R. Liu, X. Liu, J. Li, X. Yin, Y. Yang, Bifunctional hydrophobic deep eutectic solvents for selective recovery of Sm and Co from waste SmCo permanent magnets, *J. Ind. Eng. Chem.* 130 (2024) 191–202.
- [92] S. Tang, J. Feng, R. Su, M. Zhang, M. Guo, New bifunctional deep-eutectic solvent for in situ selective extraction of valuable metals from spent lithium batteries, *ACS Sustainable Chemistry & Engineering* 10 (26) (2022) 8423–8432.
- [93] J.M. Edgecomb, E.E. Tereshatov, G. Zante, M. Boltoeva, C.M. Folden III, Hydrophobic amine-based binary mixtures of active pharmaceutical and food grade ingredients: characterization and application in indium extraction from aqueous hydrochloric acid media, *Green Chem.* 22 (20) (2020) 7047–7058.
- [94] M. Gilmore, E.N. McCourt, F. Connolly, P. Nockemann, M. Swadźba-Kwaśny, J. D. Holbrey, Hydrophobic deep eutectic solvents incorporating trioctylphosphine oxide: advanced liquid extractants, *ACS Sustainable Chemistry & Engineering* 6 (12) (2018) 17323–17332.
- [95] N. Schaeffer, et al., Non-ionic hydrophobic eutectics—versatile solvents for tailored metal separation and valorisation, *Green Chem.* 22 (9) (2020) 2810–2820.
- [96] M.A. Martins, et al., Tunable hydrophobic eutectic solvents based on terpenes and monocarboxylic acids, *ACS Sustain. Chem. Eng.* 6 (7) (2018) 8836–8846.
- [97] S. Tang, M. Zhang, M. Guo, A novel deep-eutectic solvent with strong coordination ability and low viscosity for efficient extraction of valuable metals from spent lithium-ion batteries, *ACS Sustainable Chemistry & Engineering* 10 (2) (2022) 975–985.
- [98] A.P. Abbott, D. Boothby, G. Capper, D.L. Davies, R.K. Rasheed, Deep eutectic solvents formed between choline chloride and carboxylic acids: versatile alternatives to ionic liquids, *J. Am. Chem. Soc.* 126 (29) (2004) 9142–9147.
- [99] N. Milevskii, I. Zinov'eva, Y.A. Zakhodyaeva, A. Voshkin, Separation of Li (I), Co (II), Ni (II), Mn (II), and Fe (III) from hydrochloric acid solution using a menthol-based hydrophobic deep eutectic solvent, *Hydrometallurgy* 207 (2022) 105777.
- [100] E.L. Byrne, R. O'Donnell, M. Gilmore, N. Artioli, J.D. Holbrey, M. Swadźba-Kwaśny, Hydrophobic functional liquids based on trioctylphosphine oxide (TOPO) and carboxylic acids, *Phys. Chem. Chem. Phys.* 22 (42) (2020) 24744–24763.
- [101] W. Chen, J. Jiang, X. Lan, X. Zhao, H. Mou, T. Mu, A strategy for the dissolution and separation of rare earth oxides by novel Brønsted acidic deep eutectic solvents, *Green Chem.* 21 (17) (2019) 4748–4756.
- [102] L.P. Silva, et al., Design and characterization of sugar-based deep eutectic solvents using conductor-like screening model for real solvents, *ACS Sustainable Chemistry & Engineering* 6 (8) (2018) 10724–10734.
- [103] C. Liu, Q. Yan, X. Zhang, L. Lei, C. Xiao, Efficient recovery of end-of-life NdFeB permanent magnets by selective leaching with deep eutectic solvents, *Environ. Sci. Technol.* 54 (16) (2020) 10370–10379.
- [104] L.J. Kollau, M. Vis, A. Van Den Bruinhorst, A.C.C. Esteves, R. Tuinier, Quantification of the liquid window of deep eutectic solvents, *Chem. Commun.* 54 (95) (2018) 13351–13354.
- [105] L.J. Kollau, M. Vis, A. van den Bruinhorst, R. Tuinier, G. de With, Entropy models for the description of the solid–liquid regime of deep eutectic solutions, *J. Mol. Liq.* 302 (2020) 112155.
- [106] G. García, S. Aparicio, R. Ullah, M. Atilhan, Deep eutectic solvents: physicochemical properties and gas separation applications, *Energy Fuel* 29 (4) (2015) 2616–2644.
- [107] Q. Zheng, F. Yang, Q. Sun, H. Tan, X. Wang, The eutectic and physicochemical properties of the carboxylic-acid-based deep eutectic solvents, *J. Mol. Liq.* 410 (2024) 125639.
- [108] R. Biswas, Physicochemical properties and applications of deep eutectic solvents for CO₂ capture, *Chem. Eng. Technol.* 47 (1) (2024) 20–35.
- [109] W. Ma, S. Xu, Q. Wei, X. Ren, Density, acidity and theoretical calculation of hydrophobic deep eutectic solvents formed by amide and lactic acid, *J. Mol. Liq.* 409 (2024) 125547.
- [110] L. Cichocki, D. Warmińska, J. Łuczak, A. Przyjazny, G. Boczkaj, New simple and robust method for determination of polarity of deep eutectic solvents (DESS) by means of contact angle measurement, *Molecules* 27 (13) (2022) 4198.
- [111] B. Lu, et al., High-efficiency leaching of valuable metals from waste Li-ion batteries using deep eutectic solvents, *Environ. Res.* 212 (2022) 113286.
- [112] F. Feng, et al., Hydrogen bonding-based deep eutectic solvents for choline chloride/sulfamide and its application in the recycling of precious metals, *J. Environ. Chem. Eng.* 12 (5) (2024) 113611.
- [113] S. Mishra, T. Hunter, K. Pant, D. Harbottle, Green deep eutectic solvents (DESS) for sustainable metal recovery from thermally treated PCBs: a greener alternative to conventional methods, *ChemSusChem* 17 (8) (2024) e202301418.
- [114] X. Chang, et al., Selective extraction of transition metals from spent LiNi_{0.8}Co_{0.1}Mn_{0.1}—x—yO₂ cathode via regulation of coordination environment, *Angew. Chem. Int. Ed.* 61 (24) (2022) e202202558.
- [115] E. Ebrahimi, H. Abdollahi, S.Z. Shafaei, M. Ghanbarzad, E. Talebi, Green extraction of pure ferromagnetic nickel from spent hydroprocessing catalysts via deep eutectic solvents, *Sep. Purif. Technol.* 313 (2023) 123461.
- [116] C. Ma, M. Svárd, K. Forsberg, Recycling cathode material LiCo_{1/3}Ni_{1/3}Mn_{1/3}O₂ by leaching with a deep eutectic solvent and metal recovery with antisolvent crystallization, *Resour. Conserv. Recycl.* 186 (2022) 106579.
- [117] S. Gupta, K.K. Pant, G. Corder, A tandem approach for precipitant-free highly selective recovery of valuable metals from end-of-life lithium-ion batteries using a green deep eutectic solvent, *J. Clean. Prod.* 461 (2024) 142624.

- [118] C. Liu, et al., Ultra-low viscosity betaine hydrochloride-formic acid deep eutectic solvent for leaching critical metals from spent NCM lithium-ion batteries, *J. Environ. Chem. Eng.* 12 (3) (2024) 112586.
- [119] C. Wang, et al., Effective recycling of critical metals from LiCoO₂ batteries by hydrated deep eutectic solvents: performance, kinetic and mechanism, *J. Water Process Eng.* 59 (2024) 105088.
- [120] M. Jafari, S.Z. Shafaei, H. Abdollahi, A. Entezari-Zarandi, A green approach for selective ionometallurgical separation of lithium from spent Li-ion batteries by deep eutectic solvent (DES): process optimization and kinetics modeling, *Mineral Processing and Extractive Metallurgy Review* 44 (3) (2023) 218–230.
- [121] M. Jafari, S.Z. Shafaei, H. Abdollahi, A. Entezari-Zarandi, Green recycling of spent Li-ion batteries by deep eutectic solvents (DESS): leaching mechanism and effect of ternary DES, *J. Environ. Chem. Eng.* 10 (6) (2022) 109014.
- [122] R. Liu, J. Li, X. Liu, X. Yin, Y. Yang, Novel ternary deep eutectic solvents used for recycling lithium and cobalt from waste lithium-ion batteries, *Sep. Purif. Technol.* 354 (2025) 128934.
- [123] H. Yuan, et al., An effective tandem leaching method for recovering precious metals from depleted ternary lithium-ion batteries, *Sustain. Chem. Pharm.* 41 (2024) 101694.
- [124] Y. Zhang, et al., One-step selective separation and efficient recovery of valuable metals from spent lithium batteries by phosphoric acid-based deep eutectic solvent, *Green Chemical Engineering* 5 (3) (2024) 390–398.
- [125] B. Chenthamara, R.L. Gardas, Beyond the conventional leaching: exploring pyruvic acid-based deep eutectic solvents for sustainable recycling of spent lithium-ion battery cathode material, *ACS Sustainable Chemistry & Engineering* 12 (34) (2024) 12827–12836.
- [126] Q. Lu, et al., Sustainable and convenient recovery of valuable metals from spent Li-ion batteries by a one-pot extraction process, *ACS Sustainable Chemistry & Engineering* 9 (41) (2021) 13851–13861.
- [127] F. Huang, et al., Ternary deep eutectic solvent (des) with a regulated rate-determining step for efficient recycling of lithium cobalt oxide, *ACS, Omega* 7 (2022), 11452–11459.
- [128] G. Zeng, et al., Simultaneous recycling of critical metals and aluminum foil from waste LiNi_{1/3}Co_{1/3}Mn_{1/3}O₂ cathode via ethylene glycol–citric acid system, *ACS Sustainable Chemistry & Engineering* 9 (48) (2021) 16133–16142.
- [129] S. Cao, et al., Designing low-cost, green, and recyclable deep eutectic solvents for selective separation and recovery of valuable metals from spent Li-ion batteries, *ACS Sustainable Chemistry & Engineering* 11 (48) (2023) 16984–16994.
- [130] Y. Luo, Y. Deng, H. Shi, H. Yang, C. Yin, L. Ou, Green and efficient recycling method for spent Ni–co–Mn lithium batteries utilizing multifunctional deep eutectic solvents, *J. Environ. Manage.* 351 (2024) 119814.
- [131] Y. Liao, et al., A novel ternary deep eutectic solvent for efficient recovery of critical metals from spent lithium-ion batteries under mild conditions, *J. Environ. Chem. Eng.* 10 (6) (2022) 108627.
- [132] Y. Tian, et al., A weak acidic and strong coordinated deep eutectic solvent for recycling of cathode from spent lithium-ion batteries, *ChemSusChem* 15 (16) (2022) e202200524.
- [133] H. Liu, et al., Cobalt recovery from lithium battery leachate using hydrophobic deep eutectic solvents: performance and mechanism, *Process Safety and Environmental Protection* 190 (2024) 1–10.
- [134] W. Ma, M. Liu, X. Zhang, Q. Zhao, Z. Liang, An efficient and precipitant-free approach to selectively recover lithium cobalt oxide made for cathode materials using a microwave-assisted deep eutectic solvent, *Energy Fuel* 37 (1) (2022) 724–734.
- [135] D.L. Thompson, I.M. Pateli, C. Lei, A. Jarvis, A.P. Abbott, J.M. Hartley, Separation of nickel from cobalt and manganese in lithium ion batteries using deep eutectic solvents, *Green Chem.* 24 (12) (2022) 4877–4886.
- [136] P.G. Schiavi, et al., Selective recovery of cobalt from mixed lithium ion battery wastes using deep eutectic solvent, *Chem. Eng. J.* 417 (2021) 129249.
- [137] J. Cao, L. Chen, M. Li, F. Cao, L. Zhao, E. Su, Two-phase systems developed with hydrophilic and hydrophobic deep eutectic solvents for simultaneously extracting various bioactive compounds with different polarities, *Green Chem.* 20 (8) (2018) 1879–1886.
- [138] E. Tereshatov, M. Volia, C. Folden Iii, Menthol-based eutectic solvent for indium and thallium partition from hydrochloric acid media, *J. Mol. Liq.* 391 (2023) 123339.
- [139] R. Morina, D. Callegari, D. Merli, G. Alberti, P. Mustarelli, E. Quartarone, Cathode active material recycling from spent lithium batteries: a green (circular) approach based on deep eutectic solvents, *ChemSusChem* 15 (2) (2022) e202102080.
- [140] F. Zhou, et al., Machine learning models accelerate deep eutectic solvent discovery for the recycling of lithium-ion battery cathodes, *Green Chem.* 26 (13) (2024) 7857–7868.
- [141] X. Wu, et al., Green recycling of spent LiCoO₂ cathodes using a water-based deep eutectic solvent, *Sep. Purif. Technol.* 354 (2025) 128808.
- [142] R. Liu, J. Li, Y. Yue, X. Yin, Z. Gai, Y. Yang, A weakly acidic and reductive deep eutectic solvent used for one-step efficient separation of Li and co from spent lithium-ion batteries, *Waste Manag.* 197 (2025) 76–85.
- [143] P. Biniaz, R. Gol, S. Askari, Y. Hora, P.C. Banerjee, S. Bhattacharya, Selective recovery of critical metals from spent lithium-ion batteries using maleic acid-based deep eutectic solvent, *Resources, Conservation and Recycling* 217 (2025) 108177.
- [144] J. Zou, et al., High efficiency leaching of black powder from spent lithium-ion battery by ternary deep eutectic solvent and recovery of metals by precipitation and electrodeposition, *Sep. Purif. Technol.* 364 (2025) 132438.
- [145] A. Zhu, et al., Microwave-ultra-fast recovery of valuable metals from spent lithium-ion batteries by deep eutectic solvents, *Waste Manag.* 156 (2023) 139–147.
- [146] M. Liu, W. Ma, X. Zhang, Z. Liang, Q. Zhao, Recycling lithium and cobalt from LIBs using microwave-assisted deep eutectic solvent leaching technology at low-temperature, *Mater. Chem. Phys.* 289 (2022) 126466.
- [147] A.R. Carreira, A. Nogueira, A.P. Crema, H. Passos, N. Schaeffer, J.A. Coutinho, Super concentrated HCl in a deep eutectic solvent as media for the integrated leaching and separation of metals from end-of-life lithium-ion batteries, *Chem. Eng. J.* 475 (2023) 146374.
- [148] W. Chen, et al., Tailoring hydrophobic deep eutectic solvent for selective lithium recovery from the mother liquor of Li₂CO₃, *Chem. Eng. J.* 420 (2021) 127648.
- [149] N. Peeters, K. Binnemans, S. Riaño, Solvometallurgical recovery of cobalt from lithium-ion battery cathode materials using deep-eutectic solvents, *Green Chem.* 22 (13) (2020) 4210–4221.
- [150] M.K. Tran, M.-T.F. Rodrigues, K. Kato, G. Babu, P.M. Ajayan, Deep eutectic solvents for cathode recycling of Li-ion batteries, *Nat. Energy* 4 (4) (2019) 339–345.
- [151] S. Wang, Z. Zhang, Z. Lu, Z. Xu, A novel method for screening deep eutectic solvent to recycle the cathode of Li-ion batteries, *Green Chem.* 22 (14) (2020) 4473–4482.
- [152] Y. Chen, et al., Efficient dissolution of lithium-ion batteries cathode LiCoO₂ by polyethylene glycol-based deep eutectic solvents at mild temperature, *ACS Sustainable Chemistry & Engineering* 8 (31) (2020) 11713–11720.
- [153] M.J. Roldán-Ruiz, M.L. Ferrer, M.C. Gutiérrez, F.D. Monte, Highly efficient p-toluenesulfonic acid-based deep-eutectic solvents for cathode recycling of Li-ion batteries, *ACS Sustainable Chemistry & Engineering* 8 (14) (2020) 5437–5445.
- [154] Y. Chen, et al., Efficient recovery of valuable metals from lithium-ion battery cathodes using phytic acid-based deep eutectic solvents at a mild temperature, *Energy Fuel* 37 (7) (2023) 5361–5369.
- [155] Y. Chen, et al., Mild and efficient recovery of lithium-ion battery cathode material by deep eutectic solvents with natural and cheap components, *Green Chemical Engineering* 4 (3) (2023) 303–311.
- [156] Y. Luo, C. Yin, L. Ou, C. Zhang, Highly efficient dissolution of the cathode materials of spent Ni–co–Mn lithium batteries using deep eutectic solvents, *Green Chem.* 24 (17) (2022) 6562–6570.
- [157] L. Chen, et al., Engineering a tandem leaching system for the highly selective recycling of valuable metals from spent Li-ion batteries, *Green Chem.* 23 (5) (2021) 2177–2184.
- [158] H. Yu, S. Wang, Y. Li, Q. Qiao, K. Wang, X. Li, Recovery of cobalt from spent lithium-ion battery cathode materials by using choline chloride-based deep eutectic solvent, *Green Processes Synth.* 11 (1) (2022) 868–874.
- [159] Y. Chen, et al., Significant improvement in dissolving lithium-ion battery cathodes using novel deep eutectic solvents at low temperature, *ACS Sustainable Chemistry & Engineering* 9 (38) (2021) 12940–12948.
- [160] Y. Hua, et al., Ionization potential-based design of deep eutectic solvent for recycling of spent lithium ion batteries, *Chem. Eng. J.* 436 (2022) 133200.
- [161] Y. Luo, L. Ou, C. Yin, Extraction of precious metals from used lithium-ion batteries by a natural deep eutectic solvent with synergistic effects, *Waste Manag.* 164 (2023) 1–8.
- [162] J. Cheng, C. Zheng, K. Xu, Y. Zhu, Y. Song, C. Jing, Sequential separation of critical metals from lithium-ion batteries based on deep eutectic solvent and electrodeposition, *J. Hazard. Mater.* 465 (2024) 133157.
- [163] H. Li, et al., A reusable deep eutectic solvent for the regeneration of Li and co metals from spent lithium-ion batteries, *J. Alloys Compd.* 966 (2023) 171517.
- [164] T. Li, et al., Closed-loop cobalt recycling from spent lithium-ion batteries based on a deep eutectic solvent (DES) with easy solvent recovery, *J. Energy Chem.* 72 (2022) 532–538.
- [165] L. Dong, Y. Li, P. Shi, Z. Ren, Z. Zhou, Low-viscosity acidic deep eutectic solvent for extraction of valuable metals from spent NCM, *J. Power Sources* 582 (2023) 233564.
- [166] Y. Zhang, F. Wang, W. Zhang, S. Ren, Y. Hou, W. Wu, Recyclable deep eutectic solvents for recycling LiCoO₂ from spent lithium-ion batteries with high selectivity, *Sep. Purif. Technol.* 330 (2024) 125498.
- [167] L. Yurramendi, J. Hidalgo, A. Siriwardana, A sustainable process for the recovery of valuable metals from spent lithium ion batteries by deep eutectic solvents leaching, *Materials Proceedings* 5 (1) (2022) 100.
- [168] Y. Luo, C. Yin, L. Ou, Recycling of waste lithium-ion batteries via a one-step process using a novel deep eutectic solvent, *Sci. Total Environ.* 902 (2023) 166095.
- [169] Y. Tian, et al., Deep eutectic solvent with acidity, reducibility, and coordination capability for recycling of valuable metals from spent lithium-ion battery cathodes, *Sep. Purif. Technol.* 348 (2024) 127810.
- [170] Y. Wen, X. He, S. Di, K. Liu, D. Li, J. Du, Comparative of malonic acid aqueous solution and malonic acid-based deep eutectic solvent for LiCoO₂ cathode materials recovery: leaching efficiency and mechanism, *J. Environ. Chem. Eng.* 11 (5) (2023) 110979.
- [171] B. Xu, N. Díez, M. Sevilla, M.L. Ferrer, M.C. Gutiérrez, F. Del Monte, Recycling and reutilization of metals aided by deep eutectic solvents: from NMC cathodes of spent Li-ion batteries to electrolytes for supercapacitors, *ChemSusChem* 18 (2) (2025) e202401128.
- [172] Y. Zhang, F. Wang, W. Zhang, S. Ren, Y. Hou, W. Wu, High-selectivity recycling of valuable metals from spent Lithium-ion batteries using recyclable deep eutectic solvents, *ChemSusChem* 17 (9) (2024) e202301774.

- [173] X. He, Y. Wen, X. Wang, Y. Cui, L. Li, H. Ma, Leaching NCM cathode materials of spent lithium-ion batteries with phosphate acid-based deep eutectic solvent, *Waste Manag.* 157 (2023) 8–16.
- [174] S. Suriyanarayanan, M.P. Babu, R. Murugan, D. Muthuraj, K. Ramanujam, I. A. Nicholls, Highly efficient recovery and recycling of cobalt from spent lithium-ion batteries using an N-Methylurea-Acetamide nonionic deep eutectic solvent, *ACS Omega* 8 (7) (2023) 6959–6967.
- [175] S. Tang, Z. Yang, M. Zhang, M. Guo, A simple green method for in-situ selective extraction of Li from spent LiFePO₄ batteries by synergistic effect of deep-eutectic solvent and ozone, *Environ. Res.* 239 (2023) 117393.
- [176] M. Wang, et al., Selective extraction of critical metals from spent lithium-ion batteries, *Environ. Sci. Technol.* 57 (9) (2023) 3940–3950.
- [177] Q. Yan, A. Ding, M. Li, C. Liu, C. Xiao, Green leaching of lithium-ion battery cathodes by ascorbic acid modified guanidine-based deep eutectic solvents, *Energy Fuel* 37 (2) (2023) 1216–1224.
- [178] E. Ebrahimi, M. Kordloo, G. Khodadadmahmoudi, A. Rezaei, M. Ganjali, G. Azimi, Solvometallurgical recycling of spent LiNi_{0.8}Co_{0.1}Mn_{0.1}O₂ (NCM) cathode material using ternary choline chloride-ethylene glycol-p-toluenesulfonic acid deep eutectic solvent, *Hydrometallurgy* 222 (2023) 106184.
- [179] Z. Yang, S. Tang, X. Huo, M. Zhang, M. Guo, A novel green deep eutectic solvent for one-step selective separation of valuable metals from spent lithium batteries: bifunctional effect and mechanism, *Environ. Res.* 233 (2023) 116337.
- [180] T. Zhang, et al., Efficient leaching of valuable metals from NCM cathode materials by green deep eutectic solvent, *J. Clean. Prod.* 438 (2024) 140636.
- [181] B. Li, et al., High-efficiency leaching of valuable metals from waste lithium-ion ternary batteries under mild conditions using green deep eutectic solvents, *Green Chem.* 27 (1) (2025) 163–178.
- [182] W.A. Muryanta, A. Manaf, F. Bahfie, A.S. Handoko, E. Prasetyo, Polycarboxylic acid-based deep eutectic solvents for critical metal recovery from Lithium-ion batteries: kinetic and efficiency analysis, *Trends in Sciences* 22 (3) (2025) 9244.
- [183] M. Cheng, et al., One-step direct recovery of Lithium from spent Lithium-ion batteries via water-based deep eutectic solvent, *ACS Sustainable Chemistry & Engineering* 13 (4) (2025) 1719–1728.
- [184] R. Su, S. Tang, Y. Wang, M. Zhang, M. Guo, Ethanol-driven bifunctional deep eutectic solvents for efficient in situ selective recovery of valuable metals from spent lithium batteries, *Sep. Purif. Technol.* 364 (2025) 132577.
- [185] P. Panda, S. Mishra, Optimization of leaching of Lithium and cobalt from spent Lithium-ion batteries by the choline chloride-citric acid/malonic acid DES using response surface methodology, *Environ. Res.* 269 (2025) 120917.
- [186] S.H. Alhashim, S. Bhattacharyya, P.M. Ajayan, Microwave assisted ultrafast, selective lithium extraction in deep eutectic solvent for LIB cathode recycling, *Adv. Funct. Mater.* 34 (39) (2024) 2404570.
- [187] Z. Xu, H. Shao, Q. Zhao, Z. Liang, Use of microwave-assisted deep eutectic solvents to recycle lithium manganese oxide from Li-ion batteries, *Jom* 73 (2021) 2104–2110.
- [188] T. Ketegenov, K. Kamunur, L. Mussapyrova, A. Batkal, R. Nadirov, Enhanced recovery of Lithium and cobalt from spent Lithium-ion batteries using ultrasound-assisted deep eutectic solvent leaching, *Metals* 14 (9) (2024) 1052.
- [189] G. Zante, A. Braun, A. Masmoudi, R. Barillon, D. Trébouet, M. Boltoeva, Solvent extraction fractionation of manganese, cobalt, nickel and lithium using ionic liquids and deep eutectic solvents, *Miner. Eng.* 156 (2020) 106512.
- [190] A.V. Kozhevnikova, I.V. Zinov'eva, Y.A. Zakhodyaeva, V.B. Baranovskaya, A. A. Voshkin, Application of hydrophobic deep eutectic solvents in extraction of metals from real solutions obtained by leaching cathodes from end-of-life Li-ion batteries, *Processes* 10 (12) (2022) 2671.
- [191] A.A. Aziz, et al., Silver recovery from E-waste printed circuit board using binary and ternary deep eutectic solvents, *Jurnal Kimia Sains dan Aplikasi* 27 (2) (2024) 64–71.
- [192] R. Hossain, R.K. Nekouei, I. Mansuri, V. Sahajwalla, Sustainable recovery of Cu and Sn from problematic global waste: exploring value from waste printed circuit boards, *ACS Sustainable Chemistry & Engineering* 7 (1) (2018) 1006–1017.
- [193] R. Wang, G. Fan, C. Zhang, Process and systematic study of gold recovery from flexible printed circuit boards (FPCBs) using a choline chloride-ethylene glycol system, *Waste Manag.* 178 (2024) 351–361.
- [194] Q. Zhao, S. Ma, W. Ho, Y. Wang, J.Y.T. Ho, K. Shih, Simple and environmentally friendly metal recovery from waste printed circuit boards by using deep eutectic solvents, *J. Clean. Prod.* 421 (2023) 138508.
- [195] K. Liu, et al., Negative-carbon recycling of copper from waste as secondary resources using deep eutectic solvents, *J. Hazard. Mater.* 465 (2024) 133258.
- [196] H.A. Sabzkoobi, G. Koliopoulos, Green zero-waste metal extraction and recycling from printed circuit boards, *Materials Proceedings* 5 (1) (2021) 39.
- [197] S. Saffaj, D. Mantovani, G. Koliopoulos, Sustainable leaching of Cu, Ni, and Au from waste printed circuit boards using choline chloride-based deep eutectic solvents, *Metals* 15 (1) (2025) 82.
- [198] A. Łukomska, et al., Recovery of metals from electronic waste-printed circuit boards by ionic liquids, DESs and organophosphorous-based acid extraction, *Molecules* 27 (15) (2022) 4984.
- [199] R.M. Rivera, G. Zante, J.M. Hartley, K.S. Ryder, A.P. Abbott, Catalytic dissolution of metals from printed circuit boards using a calcium chloride-based deep eutectic solvent, *Green Chem.* 24 (7) (2022) 3023–3034.
- [200] U. Domańska, A. Wiśniewska, Z. Dąbrowski, Recovery of strategic metals from waste printed circuit boards with deep eutectic solvents and ionic liquids, *Processes* 12 (3) (2024) 530.
- [201] I.M. Pateli, A.P. Abbott, K. Binnemans, N.R. Rodríguez, Recovery of yttrium and europium from spent fluorescent lamps using pure levulinic acid and the deep eutectic solvent levulinic acid–choline chloride, *RSC Adv.* 10 (48) (2020) 28879–28890.
- [202] C. Shuping, et al., Selective leaching and recovery of rare earth from NdFeB waste through a superior selective and stable deep eutectic solvent, *Sep. Purif. Technol.* 353 (2025) 128498.
- [203] G. Yu, S. Ni, Y. Gao, D. Mo, Z. Zeng, X. Sun, Recovery of rare earth metal oxides from NdFeB magnet leachate by hydrophobic deep eutectic solvent extraction, oxalate stripping and calcination, *Hydrometallurgy* 223 (2024) 106209.
- [204] N.A. Milevskii, I.V. Zinov'eva, A.V. Kozhevnikova, Y.A. Zakhodyaeva, A. A. Voshkin, Sm/co magnetic materials: a recycling strategy using modifiable hydrophobic deep eutectic solvents based on triocetylphosphine oxide, *Int. J. Mol. Sci.* 24 (18) (2023) 14032.
- [205] S. Riano, et al., Separation of rare earths and other valuable metals from deep-eutectic solvents: a new alternative for the recycling of used NdFeB magnets, *RSC Adv.* 7 (51) (2017) 32100–32113.
- [206] S. Heo, R. Kim, H.-S. Yoon, C.-J. Kim, K.W. Chung, S. Lee, Effect of pretreatment methods on the selective leaching of rare earth elements from NdFeB permanent magnets using deep eutectic solvents, *Hydrometallurgy* 226 (2024) 106284.
- [207] A.T. Varghese, C.G. Malar, M. Seenuvasan, V. Jayapradha, Neodymium recovery from permanent magnets: a breakthrough approach using deep eutectic solvents and Mica, *J. Mol. Liq.* 394 (2024) 123690.
- [208] Q. Yang, Y. Li, B. Li, P. Duan, Z. Ren, Z. Zhou, Selective leaching and recovery of neodymium from NdFeB carbonyl residues, *Sep. Purif. Technol.* 329 (2024) 125137.
- [209] S.B. Mk, N. Katchala, T.S. Natarajan, S. Suresh, S. Kancharla, Nd (III) and Dy (III) extraction from discarded NdFeB magnets using TOPO-based hydrophobic eutectic solvents, *J. Mol. Liq.* 402 (2024) 124697.
- [210] S. Ni, J. Su, H. Zhang, Z. Zeng, H. Zhi, X. Sun, A cleaner strategy for comprehensive recovery of waste SmCo magnets based on deep eutectic solvents, *Chem. Eng. J.* 412 (2021) 128602.
- [211] T. Hanada, N. Schaeffer, M. Katoh, J.A. Coutinho, M. Goto, Improved separation of rare earth elements using hydrophobic deep eutectic solvents: liquid–liquid extraction to selective dissolution, *Green Chem.* 26 (18) (2024) 9671–9675.
- [212] C. Lemoine, et al., Circular recycling concept for silver recovery from photovoltaic cells in ethaline deep eutectic solvent, *RSC Adv.* 14 (40) (2024) 29174–29183.
- [213] C. Zhang, et al., Recovery of silver from crystal silicon solar panels in self-synthesized choline chloride-urea solvents system, *Waste Manag.* 150 (2022) 280–289.
- [214] E. Ebrahimi, H. Safari, M. Rezaei, A. Rezaei, H. Abdollahi, An environmentally friendly method for extraction of cobalt and molybdenum from spent catalysts using deep eutectic solvents (DESS), *Environ. Sci. Pollut. Res.* 30 (39) (2023) 90243–90255.
- [215] Z. Abbas, S.M. Jung, Green and selective recovery process of Mo, V, and Ni from spent hydrosulfurization catalysts via novel ionic liquids and deep eutectic solvents technology, *Sep. Purif. Technol.* 346 (2024) 127450.
- [216] O. Lanaridi, et al., Benign recovery of platinum group metals from spent automotive catalysts using choline-based deep eutectic solvents, *Green Chem. Lett. Rev.* 15 (2) (2022) 405–415.
- [217] M. Kamisono, T. Hanada, M. Goto, Platinum group metal recycling from spent automotive catalysts using reusable hydrophobic deep eutectic solvent, *ACS Sustainable Resource Management* 1 (5) (2024) 1021–1028.
- [218] M. Zhang, et al., Highly efficient selective extraction of Mo with novel hydrophobic deep eutectic solvents, *J. Air Waste Manage. Assoc.* 71 (12) (2021) 1492–1501.
- [219] Z. Qiao, et al., Recovery of bismuth and other metals from blast furnace dust by leaching with oxalic acid-based deep eutectic solvent and precipitation, *Hydrometallurgy* 220 (2023) 106091.
- [220] M. Kul, K.O. Oskay, M. Doğan, N. Aslan, Efficient Zinc Removal from Blast Furnace Sludge Using Choline Chloride–Malonic Acid: Towards Sustainable Recycling of Iron Extraction Waste, *JOM* 75 (9) (2023) 3882–3889.
- [221] A. Bakkar, V. Neubert, Recycling of cupola furnace dust: extraction and electrodeposition of zinc in deep eutectic solvents, *J. Alloys Compd.* 771 (2019) 424–432.
- [222] S. Wang, et al., Recycling of zinc oxide dust using ChCl-urea deep eutectic solvent with nitrilotriacetic acid as complexing agents, *Miner. Eng.* 175 (2022) 107295.
- [223] P. Züerner, G. Frisch, Leaching and selective extraction of indium and tin from zinc flue dust using an oxalic acid-based deep eutectic solvent, *ACS sustainable chemistry & engineering* 7 (5) (2019) 5300–5308.
- [224] K. Surauzhanov, A. Ul'tarakova, Extraction of vanadium from the dust of ore-thermal melting? F ilmenite concentrate using deep eutectic solvents, *Chem. Eng. Trans.* 88 (2021) 1015–1020.
- [225] A. Bakkar, Recycling of electric arc furnace dust through dissolution in deep eutectic ionic liquids and electrowinning, *J. Hazard. Mater.* 280 (2014) 191–199.
- [226] F. Niu, S. He, J. Zhang, C. Wen, Study on ultrasonically-enhanced deep eutectic solvents leaching of zinc from zinc-containing metallurgical dust sludge, *Metals* 12 (11) (2022) 1856.
- [227] X.-L. Zhu, et al., Selective recovery of zinc from zinc oxide dust using choline chloride based deep eutectic solvents, *Trans. Nonferrous Met. Soc. Chin.* 29 (10) (2019) 2222–2228.
- [228] M. Sadigh, M. Gharabaghi, S.Z.S. Tonkaboni, Mechanism, kinetic, and thermodynamic studies on the leaching of Mo, Fe, and Zn oxides from ferromolybdenum furnace dusts using choline chloride/fructose deep eutectic solvent (DES), *J. Mol. Liq.* 425 (2025) 127266.
- [229] Y. Wang, et al., Study on the extraction and separation of precious metals from wastewater using a hydrophobic deep eutectic solvent, *J. Environ. Chem. Eng.* 11 (5) (2023) 111102.

- [230] M. Shaibuna, S.P. Padinhattath, R.L. Gardas, Efficient removal of multiple heavy metal ions from wastewater using task-specific hydrophobic deep eutectic solvents: a circular approach, *J. Mol. Liq.* 416 (2024) 126487.
- [231] L.F. Guidugli, T. Reza, Extraction of rare earth elements from water by trioctylphosphine oxide-based hydrophobic deep eutectic solvents, *Ind. Eng. Chem. Res.* 63 (28) (2024) 12582–12590.
- [232] R.S. Lamarca, et al., Sustainable cadmium extraction from sewage sludge samples: a novel approach with hydrophobic deep eutectic solvents and ultrasound-assisted extraction (HDES-UAE) prior to ICP-MS analysis, *J. Mol. Liq.* 398 (2024) 124264.
- [233] D.J. van Osch, D. Parmentier, C.H. Dietz, A. van den Bruinhorst, R. Tuinier, M. C. Kroon, Removal of alkali and transition metal ions from water with hydrophobic deep eutectic solvents, *Chem. Commun.* 52 (80) (2016) 11987–11990.
- [234] A. Sadeghi, G.R. Pazuki, Separation of heavy metals using aqueous two-phase systems based on choline chloride/urea and betaine/urea deep eutectic solvents, *Sci. Rep.* 15 (1) (2025) 3948.
- [235] T.N. Huynh, et al., An eco-friendly method for the efficient and straightforward removal of hexavalent chromium from aqueous media using hydrophobic deep eutectic solvents, *J. Mol. Liq.* 422 (2025) 126954.
- [236] E. Yilmaz, M. Soylak, Ultrasound assisted-deep eutectic solvent based on emulsification liquid phase microextraction combined with microsample injection flame atomic absorption spectrometry for valence speciation of chromium (III/VI) in environmental samples, *Talanta* 160 (2016) 680–685.
- [237] A.G. Yiksek, A. Elik, N. Altunay, Rapid and safe determination of vitamin B1 in dairy products, fruits, nuts and vitamin tablets: combination of natural deep eutectic solvents, experimental design and artificial intelligence, *J. Food Compos. Anal.* 131 (2024) 106222.
- [238] A. Elik, B. Doğan, A. Demirbaş, H.U. Haq, N. Altunay, Investigation of use of hydrophilic/hydrophobic NADESs for selective extraction of as (III) and Sb (III) ions in vegetable samples: air assisted liquid phase microextraction and chemometric optimization, *Food Chem.* 451 (2024) 139538.
- [239] P.D. Ola, M. Matsumoto, Use of deep eutectic solvent as extractant for separation of Fe (III) and Mn (II) from aqueous solution, *Sep. Sci. Technol.* 54 (5) (2019) 759–765.
- [240] J. Ali, et al., Separation and preconcentration of trivalent chromium in environmental waters by using deep eutectic solvent with ultrasound-assisted based dispersive liquid-liquid microextraction method, *J. Mol. Liq.* 291 (2019) 111299.
- [241] R. Liu, et al., Extraction of platinum (IV) by hydrophobic deep eutectic solvents based on trioctylphosphine oxide, *Hydrometallurgy* 199 (2021) 105521.
- [242] Z. Erbas, M. Soylak, E. Yilmaz, M. Dogan, Deep eutectic solvent based liquid phase microextraction of nickel at trace level as its diethyldithiocarbamate chelate from environmental samples, *Microchem. J.* 145 (2019) 745–750.
- [243] F. Olea, et al., Hydrophobic eutectic solvents for the selective solvent extraction of molybdenum (VI) and rhenium (VII) from a synthetic pregnant leach solution, *J. Mol. Liq.* 385 (2023) 122415.
- [244] K. Xue, et al., Lithium extraction from aqueous medium using hydrophobic deep eutectic solvents, *J. Environ. Chem. Eng.* 11 (5) (2023) 110490.
- [245] H. Dong, et al., Hydrophobic deep eutectic solvents as the green media for highly efficient extraction of Cr (VI) over a broad pH range and low oil-water ratio, *Sep. Purif. Technol.* 334 (2024) 126104.
- [246] O. Mokhodoeva, V. Maksimova, A. Shishov, V. Shkinev, Separation of platinum group metals using deep eutectic solvents based on quaternary ammonium salts, *Sep. Purif. Technol.* 305 (2023) 122427.
- [247] N. Tang, et al., Environmentally benign hydrophobic deep eutectic solvents for palladium (II) extraction from hydrochloric acid solution, *J. Taiwan Inst. Chem. Eng.* 121 (2021) 92–100.
- [248] L. Zhang, J. Li, L. Ji, L. Li, Separation of lithium from alkaline solutions with hydrophobic deep eutectic solvents based on β -diketone, *J. Mol. Liq.* 344 (2021) 117729.
- [249] A. Mehmood, M. Sharif, A. Hussain, M. Suleman, M.B. Hayat, “extraction of rare earth elements from coal ash by chemical leaching using deep eutectic solvents,” *Journal of King Abdulaziz University*, *Eng. Sci.* 34 (2) (2024).
- [250] R. Karan, T. Sreenivas, M.A. Kumar, D. Singh, Recovery of rare earth elements from coal flyash using deep eutectic solvents as leachants and precipitating as oxalate or fluoride, *Hydrometallurgy* 214 (2022) 105952.
- [251] M.A. Topçu, S.A. Çeltek, A. Rüsen, Green leaching and predictive model for copper recovery from waste smelting slag with choline chloride-based deep eutectic solvent, *Chin. J. Chem. Eng.* 75 (2024) 14–24.
- [252] B. Salgado, D. Endara, C.F. Aragón-Tobar, E. de la Torre, L. Ullauri, Recovery of residual Lead from automotive battery recycling slag using deep eutectic solvents, *Molecules* 29 (02) (2024).
- [253] V.E. Huntington, F. Coulon, S.T. Wagland, Assessing metal extraction from metalliferous waste: a study using deep eutectic solvents and chelating agents vs. ethylenediaminetetraacetic acid, *J. Environ. Manage.* 363 (2024) 121350.
- [254] N.R. Rodriguez, L. Machiels, B. Onghena, J. Spooren, K. Binnemans, Selective recovery of zinc from goethite residue in the zinc industry using deep-eutectic solvents, *RSC Adv.* 10 (12) (2020) 7328–7335.
- [255] C.F. Aragón-Tobar, D. Endara, E. de la Torre, Dissolution of metals (Cu, Fe, Pb, and Zn) from different metal-bearing species (sulfides, oxides, and sulfates) using three deep eutectic solvents based on choline chloride, *Molecules* 29 (2) (2024) 290.
- [256] N. Mohammad, et al., Recovery of Cu from waste smelting slag through extraction with novel deep eutectic solvents (DESSs), *Can. Metall. Quart.* (2024) 1–8.
- [257] S. Drogobuzhskaya, M. Frolova, A. Shishov, N. Tsvetov, Comparison of extraction abilities of deep eutectic solvents and aqueous acid solutions for extraction of rare earths and transition metals, *Journal of Rare Earths* 42 (6) (2024) 1157–1164.
- [258] K. Moreno, X. Díaz, D. Endara, F. Sánchez, C.F. Aragón-Tobar, Zinc and Lead Leaching from Sphalerite–Galena Concentrate Using Deep Eutectic Solvents Based on Choline Chloride: Effect of Roasting and Iodine as Oxidizing Agent, *Molecules* 29 (16) (2024) 3742.
- [259] M. Sudová, M. Sisol, M. Kanuchova, M. Marcin, J. Kurty, Environmentally friendly leaching of antimony from mining residues using deep eutectic solvents: optimization and sustainable extraction strategies, *Processes* 12 (3) (2024) 555.
- [260] G. Shakiba, R. Saneie, H. Abdollahi, E. Ebrahimi, A. Rezaei, M. Mohammadkhani, Application of deep eutectic solvents (DESSs) as a green lixiviant for extraction of rare earth elements from caustic-treated monazite concentrate, *J. Environ. Chem. Eng.* 11 (5) (2023) 110777.
- [261] T. Sakamoto, T. Hanada, H. Sato, M. Kamisono, M. Goto, Hydrophobic deep eutectic solvents for the direct leaching of nickel laterite ores: selectivity and reusability investigations, *Sep. Purif. Technol.* 331 (2024) 125619.
- [262] F. Shahrezaei, S. Karimi, B. Behnadjady, Experimental and dynamic molecular study of zinc extraction from sphalerite concentrate in ternary deep eutectic solvent, *Colloids Surf. A Physicochem. Eng. Asp.* 695 (2024) 134241.
- [263] A. Entezari-Zarandi, F. Larachi, Selective dissolution of rare-earth element carbonates in deep eutectic solvents, *J. Rare Earths* 37 (5) (2019) 528–533.
- [264] Q. Yan, C. Liu, X. Zhang, L. Lei, C. Xiao, Selective dissolution and separation of rare earths using guanidine-based deep eutectic solvents, *ACS Sustainable Chemistry & Engineering* 9 (25) (2021) 8507–8514.
- [265] S. Ni, Y. Gao, G. Yu, S. Zhang, Z. Zeng, X. Sun, Tailored ternary hydrophobic deep eutectic solvents for synergistic separation of yttrium from heavy rare earth elements, *Green Chem.* 24 (18) (2022) 7148–7161.
- [266] J. Almeida, et al., Electrodialytic removal of tungsten and arsenic from secondary mine resources—deep eutectic solvents enhancement, *Sci. Total Environ.* 710 (2020) 136364.
- [267] X. Li, W. Li, Y. Gao, G. Tian, Effect of mechanical activation on the leaching process of rare earth metal yttrium in deep eutectic solvents, *Appl. Sci.* 12 (16) (2022) 8253.
- [268] S. Barakan, V. Aghazadeh, Rhenium extraction from pressure oxidative leaching solution of molybdenite concentrate using hydrophobic deep eutectic solvents, *J. Mol. Liq.* 376 (2023) 121481.
- [269] X. Li, W. Monnens, Z. Li, J. Franssaer, K. Binnemans, Solvometallurgical process for extraction of copper from chalcopyrite and other sulfidic ore minerals, *Green Chem.* 22 (2) (2020) 417–426.
- [270] S.M. Ghadami, A. Babakhani, G. Barati Darband, H. Shalchian, R. Behmadi, Solvometallurgical properties of choline chloride-based deep eutectic solvents for copper extraction from chalcopyrite: optimization and analysis, *Mining* 5 (1) (2025) 8.
- [271] X. Jin, et al., Separation of indium and tin from ITO powders with short-chain dicarboxylic acid-CHCl deep eutectic solvents: indium tin leaching and splitting mechanism, *Process Saf. Environ. Prot.* 185 (2024) 1268–1276.
- [272] Z. Liu, F. Yang, Z. Sun, Q. Chi, Y. Li, Efficient selective leaching of ytterbium and lutetium oxides by new hydrophobic deep eutectic solvents, *Sep. Purif. Technol.* 343 (2024) 127097.
- [273] Y. Shi, D. Xiong, Y. Zhao, T. Li, K. Zhang, J. Fan, Highly efficient extraction/separation of Cr (VI) by a new family of hydrophobic deep eutectic solvents, *Chemosphere* 241 (2020) 125082.
- [274] J. Werner, Ligandless, deep eutectic solvent-based ultrasound-assisted dispersive liquid-liquid microextraction with solidification of the aqueous phase for preconcentration of lead, cadmium, cobalt and nickel in water samples, *J. Sep. Sci.* 43 (7) (2020) 1297–1305.
- [275] N. Altunay, A. Elik, R. Gürkan, Innovative and practical deep eutectic solvent based vortex assisted microextraction procedure for separation and preconcentration of low levels of arsenic and antimony from sample matrix prior to analysis by hydride generation-atomic absorption spectrometry, *Food Chem.* 293 (2019) 378–386.
- [276] A.H. Panhwar, M. Tuzen, N. Deligonul, T.G. Kazi, Ultrasonic assisted deep eutectic solvent liquid-liquid microextraction using azadipyrromethene dye as complexing agent for assessment of chromium species in environmental samples by electrothermal atomic absorption spectrometry, *Appl. Organomet. Chem.* 32 (5) (2018) e4319.
- [277] M. Tavakoli, M.R. Jamali, A. Nezhadali, Ultrasound-assisted dispersive liquid-liquid microextraction (DLLME) based on solidification of floating organic drop using a deep eutectic solvent for simultaneous preconcentration and determination of nickel and cobalt in food and water samples, *Anal. Lett.* 54 (18) (2021) 2863–2873.
- [278] A.H. Panhwar, M. Tuzen, T.G. Kazi, Ultrasonic assisted dispersive liquid-liquid microextraction method based on deep eutectic solvent for speciation, preconcentration and determination of selenium species (IV) and (VI) in water and food samples, *Talanta* 175 (2017) 352–358.
- [279] S.N. Rashid, et al., Deep eutectic solvents for the removal of lead contaminants in mangrove soil, *J. Environ. Chem. Eng.* 10 (2) (2022) 107264.
- [280] H. Liu, J. Lv, Y. Yang, Recyclable water-modified deep eutectic solvents for removal of multiple heavy metals from soil, *Chemosphere* 350 (2024) 141141.
- [281] S. Mukhopadhyay, et al., Ammonium-based deep eutectic solvents as novel soil washing agent for lead removal, *Chem. Eng. J.* 294 (2016) 316–322.
- [282] S. Mukhopadhyay, S. Mukherjee, A. Hayyan, M. Hayyan, M.A. Hashim, B. S. Gupta, Enhanced removal of lead from contaminated soil by polyl-based deep eutectic solvents and saponin, *J. Contam. Hydrol.* 194 (2016) 17–23.

- [283] K. Huang, X. Wang, W. Yuan, J. Xie, J. Wang, J. Li, Remediation of lead-contaminated soil by washing with choline chloride-based deep eutectic solvents, *Process Safety and Environmental Protection* 160 (2022) 650–660.
- [284] N. Osowska, K. Paduszyński, M. Matczuk, L. Ruzik, New solvents for metal extraction–NADES. Prediction and optimization of efficient extraction of selected metals by ICP-MS/MS, *J. Anal. At. Spectrom* 36 (5) (2021) 946–953.
- [285] A. Elik, N. Altunay, Preparation and investigation of green and cheap deep eutectic solvents for determination of essential and toxic metal ions at trace levels in different coffee products, *Sustain. Chem. Pharm.* 39 (2024) 101602.
- [286] C. Zhang, et al., Metal removal from heavy metal-enriched plants by deep eutectic solvents and its mechanism investigation, *Sep. Purif. Technol.* 357 (2025) 130189.
- [287] A.N. Shikov, E.D. Obluchinskaya, E.V. Flisyuk, I.I. Terninko, Y.E. Generalova, O. N. Pozharitskaya, The impact of natural deep eutectic solvents and extraction method on the co-extraction of trace metals from *Fucus vesiculosus*, *Mar. Drugs* 20 (5) (2022) 324.
- [288] H.U. Haq, et al., Deep eutectic solvents based assay for extraction and determination of zinc in fish and eel samples using FAAS, *J. Mol. Liq.* 333 (2021) 115930.
- [289] T. Borahan, T. Unutkan, N.B. Turan, F. Turak, S. Bakırdere, Determination of lead in milk samples using vortex assisted deep eutectic solvent based liquid phase microextraction-slotted quartz tube-flame atomic absorption spectrometry system, *Food Chem.* 299 (2019) 125065.
- [290] M.B. Arain, E. Yilmaz, M. Soylak, Deep eutectic solvent based ultrasonic assisted liquid phase microextraction for the FAAS determination of cobalt, *J. Mol. Liq.* 224 (2016) 538–543.
- [291] A. Shishov, S. Savinov, N. Volodina, I. Gurev, A. Bulatov, Deep eutectic solvent-based extraction of metals from oil samples for elemental analysis by ICP-OES, *Microchem. J.* 179 (2022) 107456.
- [292] A. Shishov, A. Gerasimov, A. Bulatov, Deep eutectic solvents based on carboxylic acids for metals separation from plant samples: elemental analysis by ICP-OES, *Food Chem.* 366 (2022) 130634.
- [293] E. Yilmaz, M. Soylak, Ultrasound assisted-deep eutectic solvent extraction of iron from sheep, bovine and chicken liver samples, *Talanta* 136 (2015) 170–173.
- [294] K. Ghanemi, M.-A. Navidi, M. Fallah-Mehrjardi, A. Dadolahi-Sohrab, Ultra-fast microwave-assisted digestion in choline chloride–oxalic acid deep eutectic solvent for determining Cu, Fe, Ni and Zn in marine biological samples, *Anal. Methods* 6 (6) (2014) 1774–1781.
- [295] R. Akramipour, M.R. Golpayegani, S. Gheini, N. Fattahi, Speciation of organic/inorganic mercury and total mercury in blood samples using vortex assisted dispersive liquid-liquid microextraction based on the freezing of deep eutectic solvent followed by GFAAS, *Talanta* 186 (2018) 17–23.
- [296] X. Yang, Y.Y. Zang, S. Yang, Z.G. Chen, Green and efficient removal of heavy metals from *Porphyra haitanensis* using natural deep eutectic solvents, *J. Sci. Food Agric.* 101 (7) (2021) 2930–2939.
- [297] M. Karimi, A.M.H. Shabani, S. Dadfarnia, Deep eutectic solvent-mediated extraction for ligand-less preconcentration of lead and cadmium from environmental samples using magnetic nanoparticles, *Microchim. Acta* 183 (2016) 563–571.
- [298] Y. Ji, M. Zhao, A. Li, L. Zhao, Hydrophobic deep eutectic solvent-based ultrasonic-assisted dispersive liquid-liquid microextraction for preconcentration and determination of trace cadmium and arsenic in wine samples, *Microchem. J.* 164 (2021) 105974.
- [299] E. Bağda, H. Altundağ, M. Soylak, Highly simple deep eutectic solvent extraction of manganese in vegetable samples prior to its ICP-OES analysis, *Biol. Trace Elem. Res.* 179 (2017) 334–339.
- [300] E. Habibi, K. Ghanemi, M. Fallah-Mehrjardi, A. Dadolahi-Sohrab, A novel digestion method based on a choline chloride–oxalic acid deep eutectic solvent for determining Cu, Fe, and Zn in fish samples, *Anal. Chim. Acta* 762 (2013) 61–67.
- [301] S.M. Sorouraddin, M.A. Farajzadeh, H. Dastoori, T. Okhravi, Development of an air-assisted liquid-liquid microextraction method based on a ternary solidified deep eutectic solvent in extraction and preconcentration of Cd (II) and Zn (II) ions, *Int. J. Environ. Anal. Chem.* 101 (11) (2021) 1567–1580.
- [302] S.M. Sorouraddin, M.A. Farajzadeh, H. Dastoori, Development of a dispersive liquid-liquid microextraction method based on a ternary deep eutectic solvent as chelating agent and extraction solvent for preconcentration of heavy metals from milk samples, *Talanta* 208 (2020) 120485.
- [303] A.H. Panhwar, M. Tuzen, T.G. Kazi, Deep eutectic solvent based advance microextraction method for determination of aluminum in water and food samples: multivariate study, *Talanta* 178 (2018) 588–593.
- [304] A. Petračić, A. Sander, M. Cvetnić, A novel approach for the removal of trace elements from waste fats and oils, *Sep. Sci. Technol.* 55 (18) (2020) 3487–3501.
- [305] M. Karimi, S. Dadfarnia, A.M.H. Shabani, F. Tamaddon, D. Azadi, Deep eutectic liquid organic salt as a new solvent for liquid-phase microextraction and its application in ligandless extraction and preconcentration of lead and cadmium in edible oils, *Talanta* 144 (2015) 648–654.
- [306] K. Binnemans, P.T. Jones, Ionic liquids and deep-eutectic solvents in extractive metallurgy: mismatch between academic research and industrial applicability, *Journal of Sustainable Metallurgy* 9 (2) (2023) 423–438.